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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-0052	1	SCHEM, MLB-4GB, J44	SCH	CRITICAL	
820-3536	1	PCBF, MLB-4GB, J44	PCB	CRITICAL	



BOM Groups

BOM GROUP	BOM OPTIONS
J44_COMMON	ALTERNATE, COMMON, J44_COMMON1, J44_COMMON2, J44_COMMON3, J44_COMMON4, J44_PROGPARTS
J44_COMMON1	TBTHV:P15V, SKIP_5V3V3:AUDIBLE, SPI:DUAL_IO
J44_COMMON2	EDP, EDP_LS_CAP, CAMERA_3V3:S0, CAM_WAKE:NO, CAM_XTAL:NO, MEM_ODT:PU, VCORE_FETS
J44_COMMON3	XDP, LPCPLUS, BKL T:PROD, CPUTHRM:ALRT, LOADRC:NO, OTHERRC:NO, DDRRC:NO, TBTRC:NO, BMONRC:NO
J44_PROGPARTS	SMC_PROG:PVT, BOOTROM:PVT, TBTRM:PVT, TPAD_PSOC:PROG
ENGISNS	LOADISNS, OTHERISNS, DDRISNS, TBTISNS, BMONISNS

Programmables (All Builds)

TBT

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S3918	1	EPROM,FALCON RIDGE (V13.7) J44	U2890	CRITICAL	TBTROM:PVT

SMC

341S3922	1	IC,SMC-B1,EXT(V2.16F39),PVT,J44	U5000	CRITICAL	SMC_PROG:PVT
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Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4596	1	HSWULT,SR18A,PRQ,C0,2.4,28W,2+3,3M,BGA	U0500	CRITICAL	CPU_HSW:2.4G
337S4597	1	HSWULT,SR189,PRQ,C0,2.6,28W,2+3,3M,BGA	U0500	CRITICAL	CPU_HSW:2.6G
337S4598	1	HSWULT,SR188,PRQ,C0,2.8,28W,2+3,4M,BGA	U0500	CRITICAL	CPU_HSW:2.8G
338S1247	1	IC,TBT,FR-4C,A0,PRQ,C10,SR13C,FCBGA288	U2800	CRITICAL	
338S1186	1	IC,BCM15708A2,S2 PCIE CAMERA PROCESSOR	U3900	CRITICAL	
376S1194	2	MOSFET,N-CH,30V,15.3A,12M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:VSHY
376S1193	2	MOSFET,N-CH,30V,22A,6.0M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:VSHY
376S0964	2	MOSFET,N-CH,25V,30A,9.6M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:REN
376S1104	2	MOSFET,N-CH,25V,30A,6.1M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:REN

EFI ROM

341S3924	1	IC,EFI ROM (V0116),PVT,J44	U6100	CRITICAL	BOOTROM:PVT
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PSOC

341S3862	1	IC,TRKPD/KYBD PSOC,CU ONLY(V224) J44	U4801	CRITICAL	TPAD_PSOC:PROG
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Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1053	376S0604		ALL	
128S0311	128S0329		ALL	
138S0739	138S0706		ALL	
197S0481	197S0480		ALL	
152S0461	152S1645		ALL	
376S1080	376S0820		ALL	
155S0667	155S0583		ALL	
138S0725	138S0724		ALL	
376S1032	376S0855		ALL	
376S1129	376S0855		ALL	
376S1089	376S1128		ALL	
353S3452	353S1286		ALL	
376S1180	376S0761		ALL	
128S0364	128S0264		ALL	
107S0254	107S0241		ALL	
138S0843	138S0674		ALL	
138S0803	138S0639		ALL	
138S0846	138S0811		ALL	
197S0542	197S0544		ALL	
197S0545	197S0544		ALL	
152S1876	152S1804		ALL	
107S0255	107S0240		ALL	
107S0250	107S0248		ALL	
127S0164	127S0162		ALL	
353S4070	353S4069		ALL	
353S4068	353S4069		ALL	
353S3814	353S3812		ALL	
311S0649	311S0541		ALL	
128S0436	128S0392		ALL	

Diodes alt to Fairchild
NEC alt to Sanyo
Samsung alt to Murata
Epson alt to NDK
Cyntec alt to Vishay
Diodes alt to On Semi
Panasonic alt to TDK
Samsung alt to Murata
Toshiba alt for Diodes Dual
NXP Alt for Diodes Dual
NXP Alt for Diodes Single
Maxim alt to Microchip
Renesas alt to Vishay
Sanyo 2nd Factory alt
Cyntec alt to TFT
Samsung alt to Murata (BKLT)
Samsung alt to Murata (BKLT)
Samsung alt to Murata (BKLT)
NDK alt to TXC
Epson alt to TXC
TDK alt to Toko
Cyntec alt to TFT
Cyntec alt to TFT
Rohm alt to Vishay
Pericom alt to TI DP Mux U9750
NXP alt to TI DP Mux U9750
TI alt to NXP
ONsemi alt to Toshiba
Kemet alt to Sanyo

SYNC MASTER=J44

SYNC DATE=09/29/2011

PAGE TITLE

BOM Configuration

 Apple Inc.

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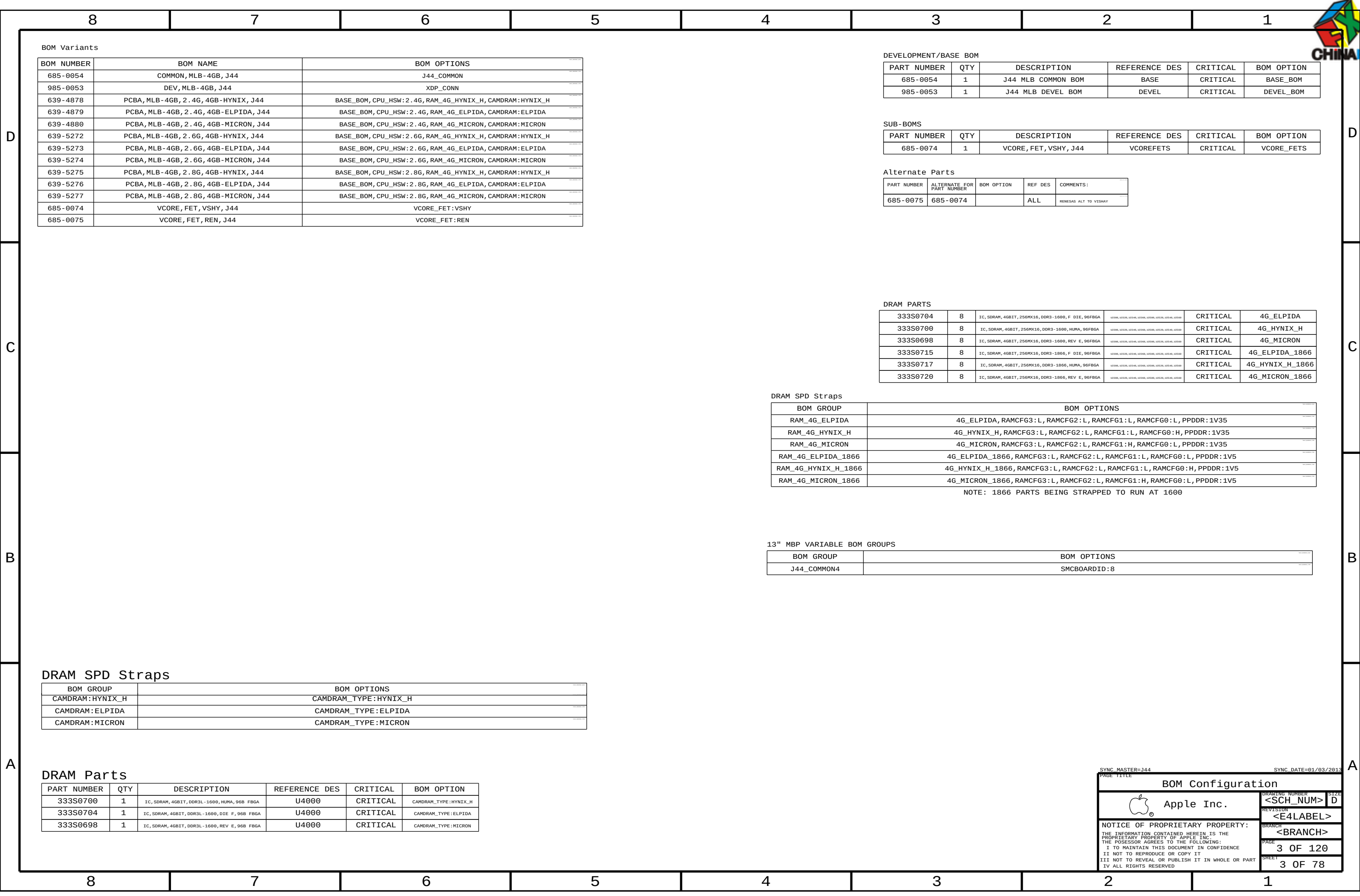
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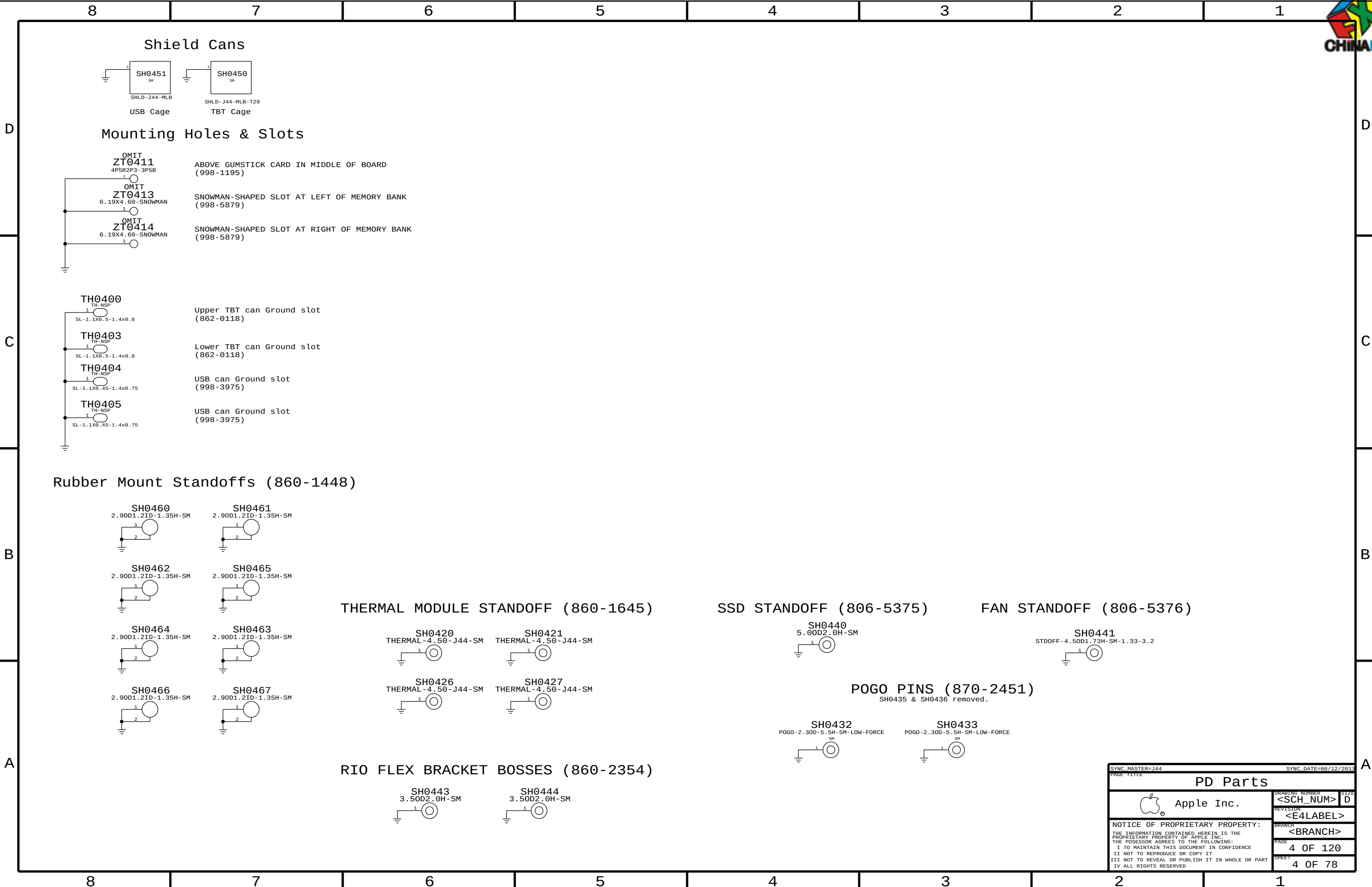
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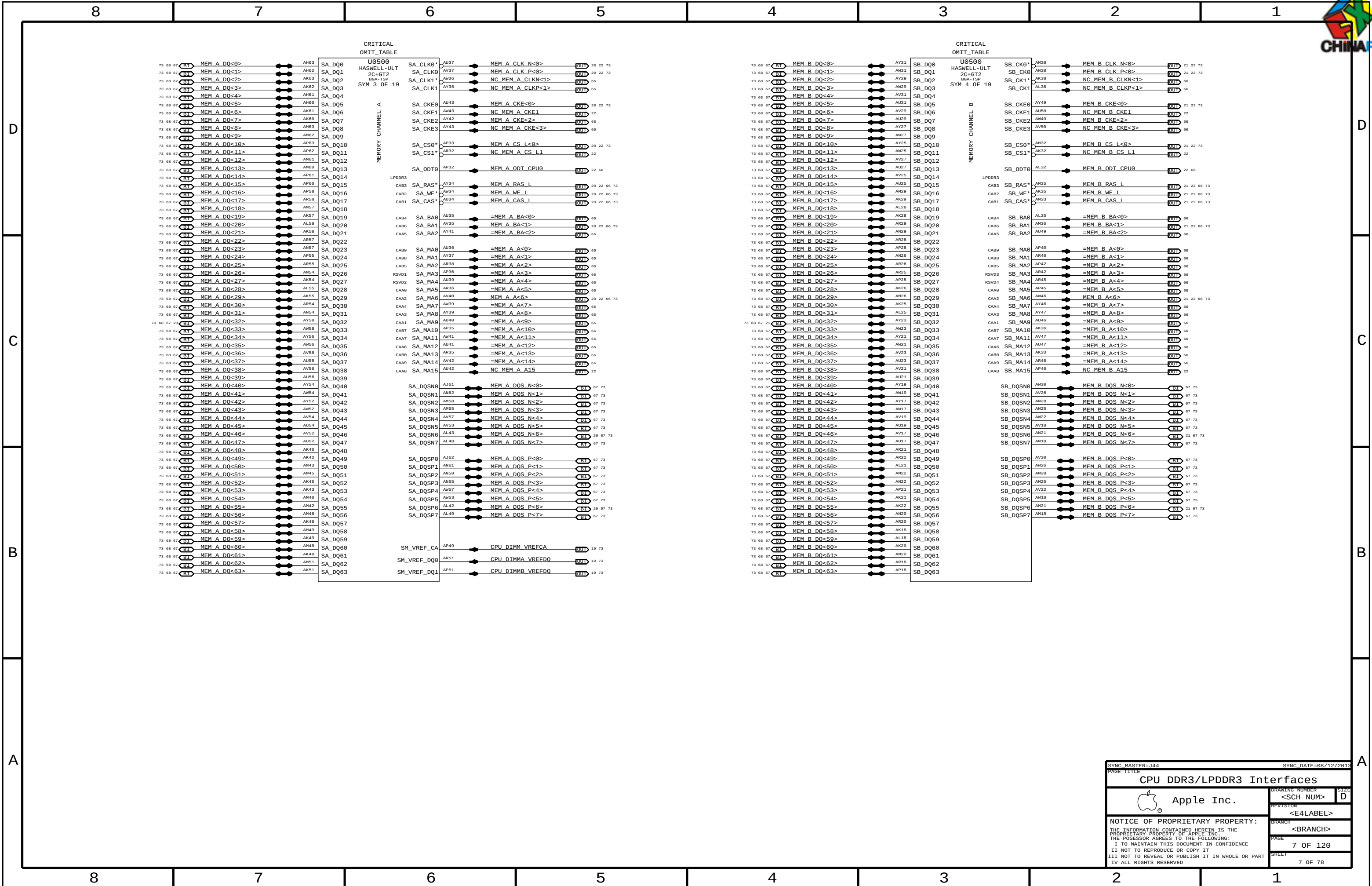


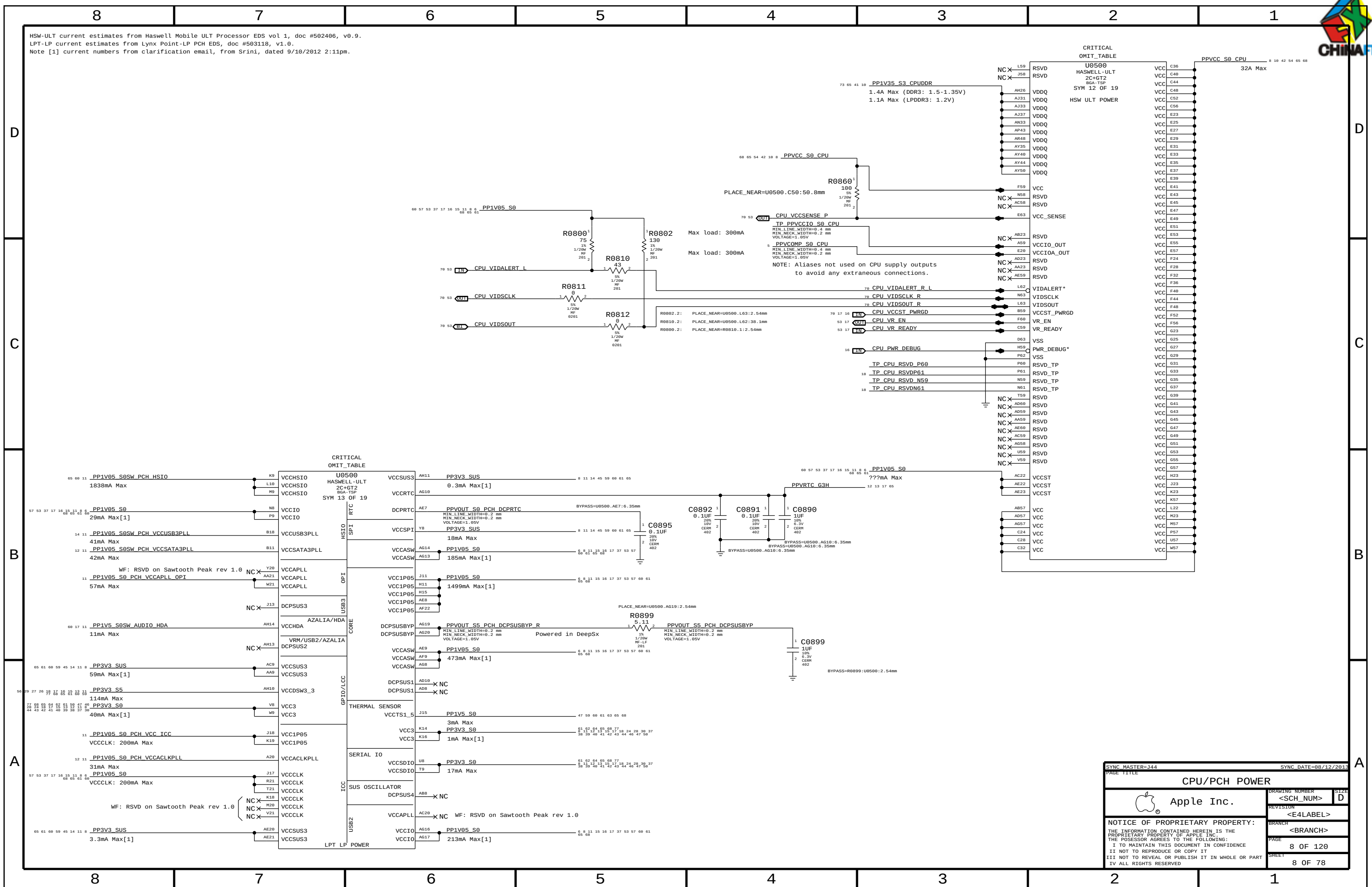
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		SHEET	4 OF 78

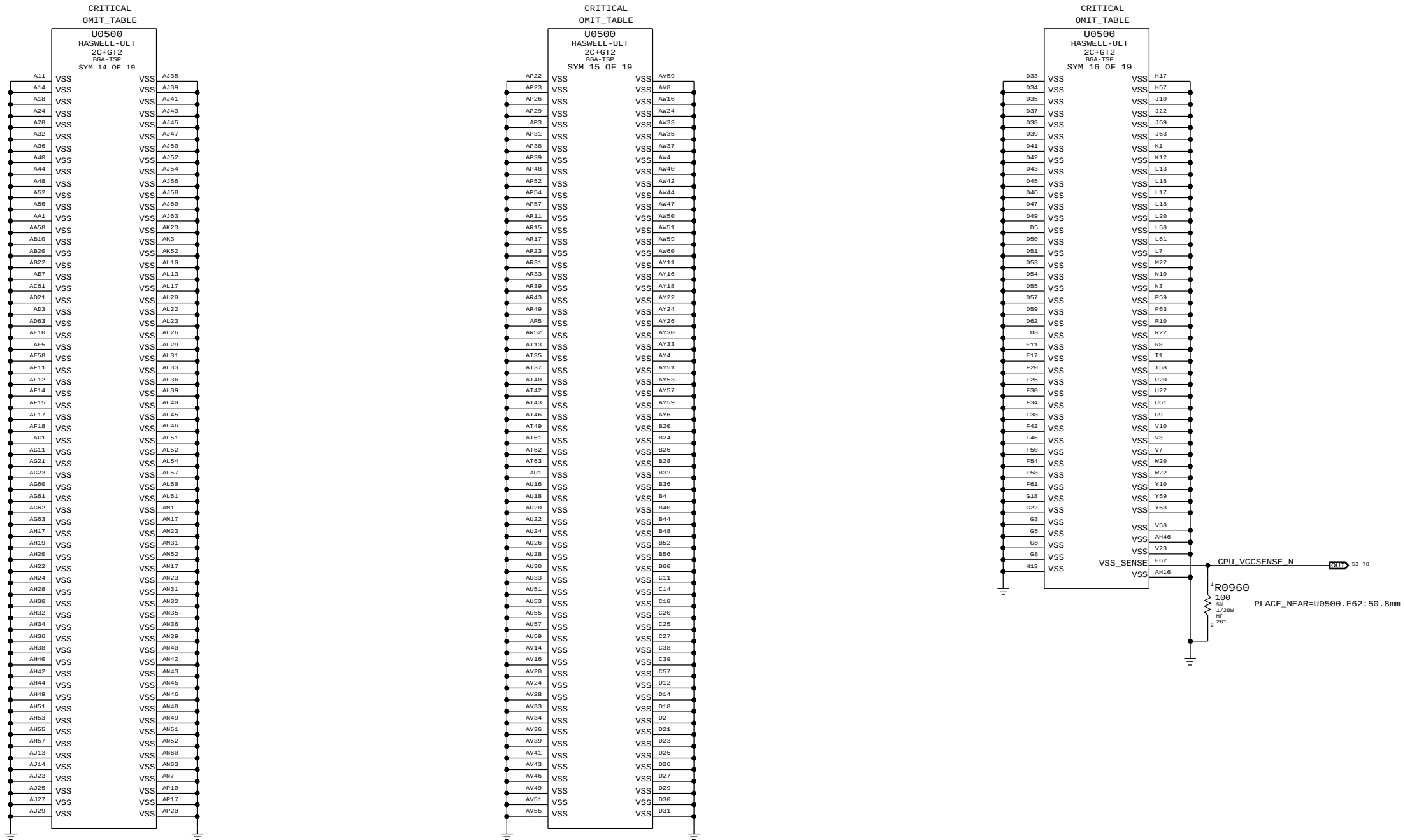




CPU CFG<4> 6 16 70
EDP
1 R0634
1K
5%
1/20W
MF
2 201







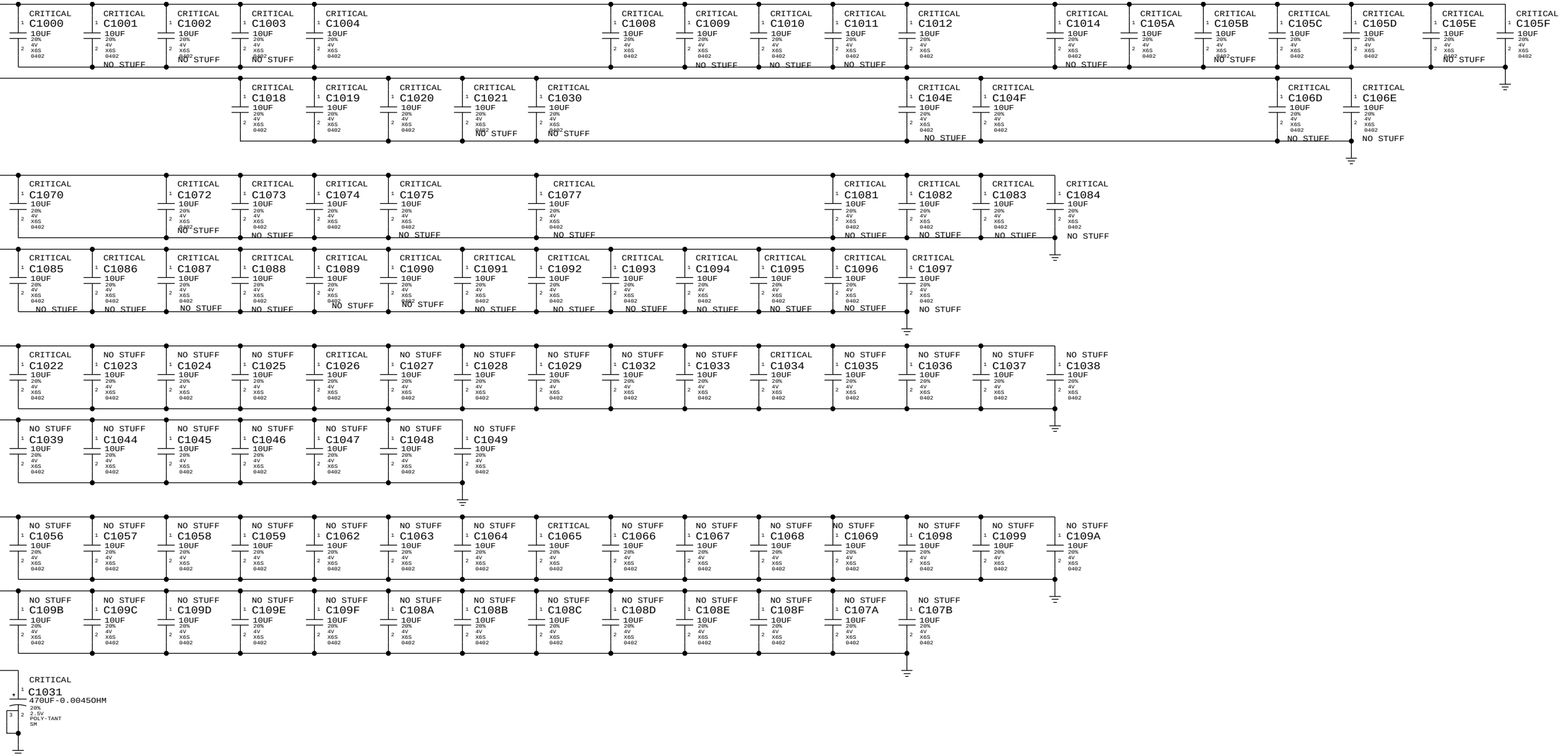


All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 0.9 unless stated otherwise

CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff
Apple implementation : 18x 22uF 0603 stuff, 80x 22uF 0603 nostuff

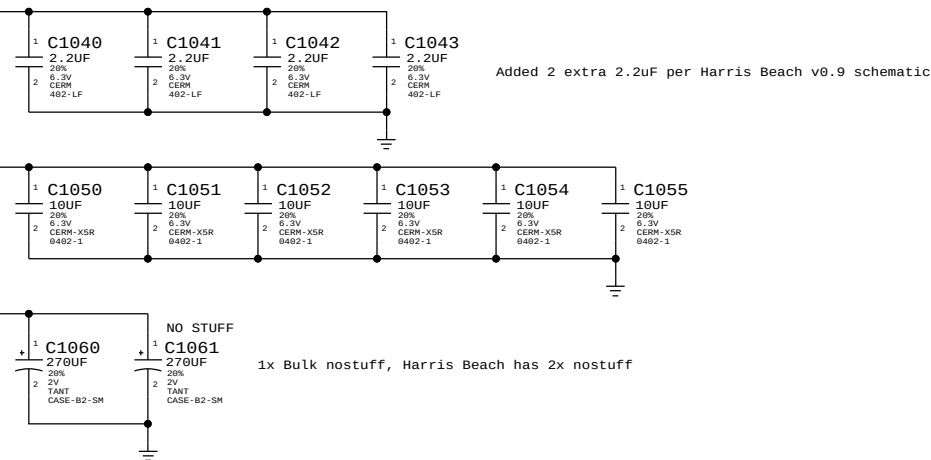
68 65 54 42 8 PPVCC_S0_CPU



CPU VDDQ DECOUPLING

Intel recommendation (Table 5-4): 2x 2.2uF 0402, 6x 10uF 0603
Apple implementation : 2x 2.2uF 0402, 6x 10uF 0402

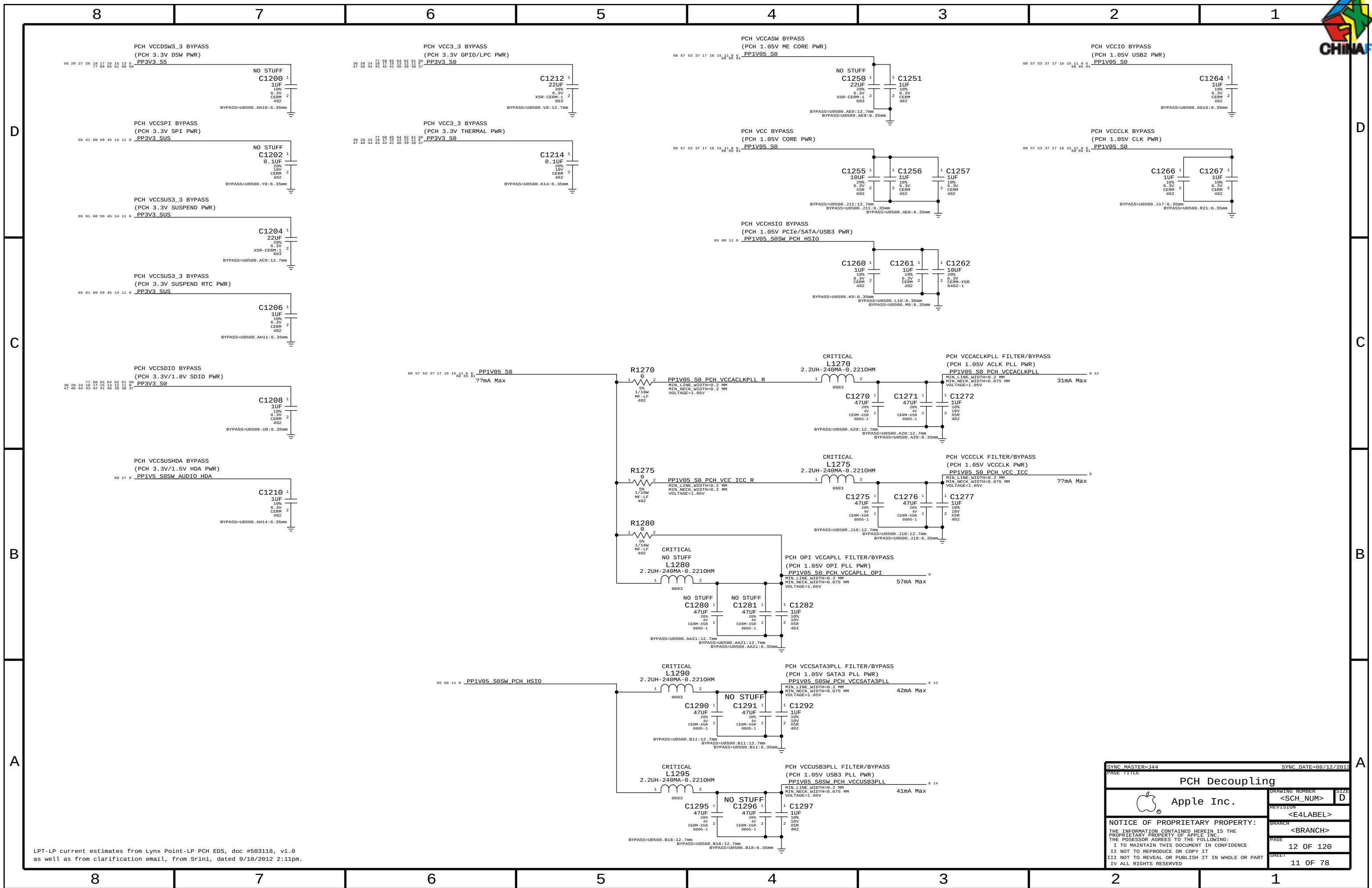
73 65 41 8 PP1V35_S3_CPUDDR



CPU VCC Decoupling

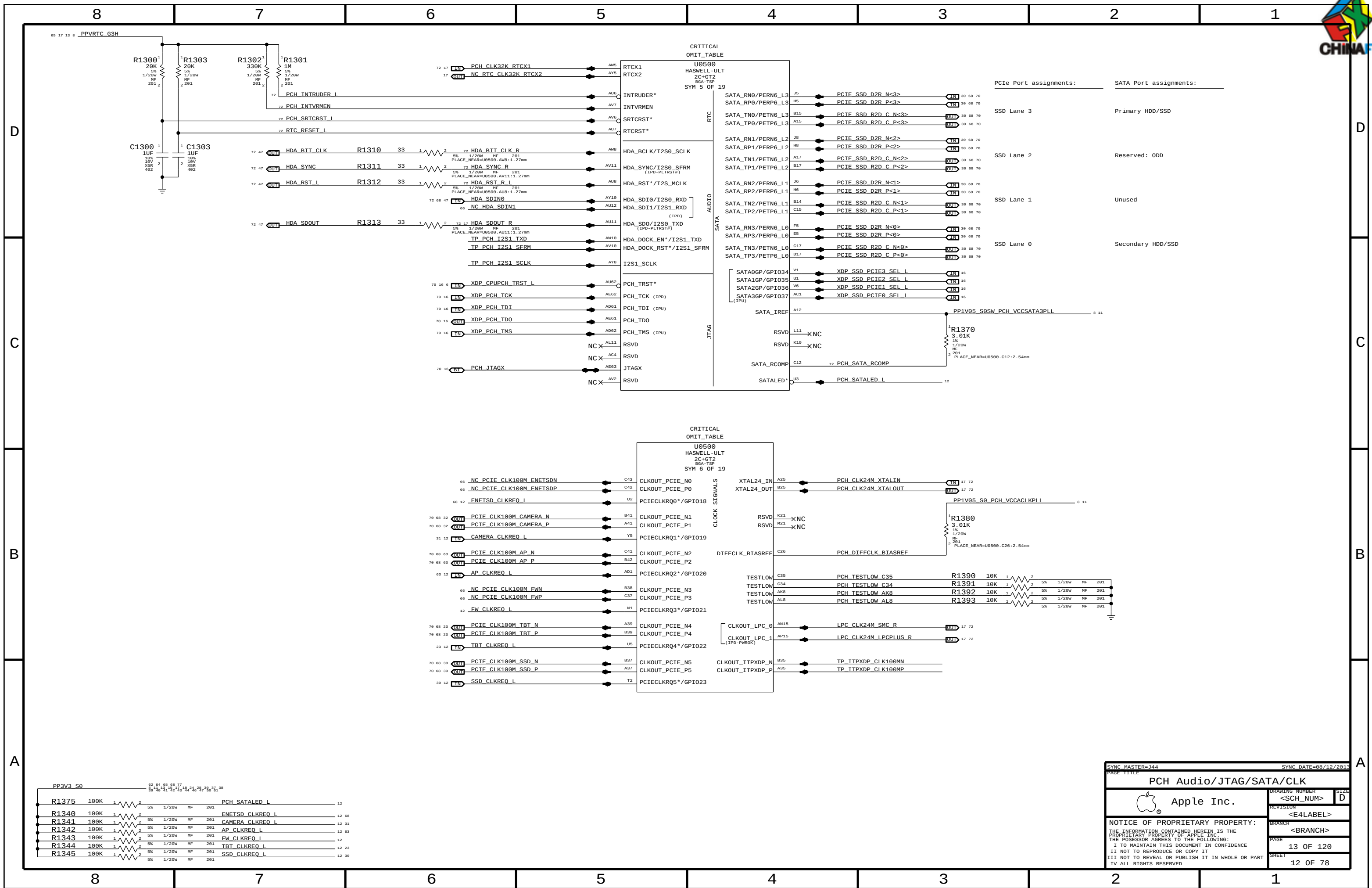
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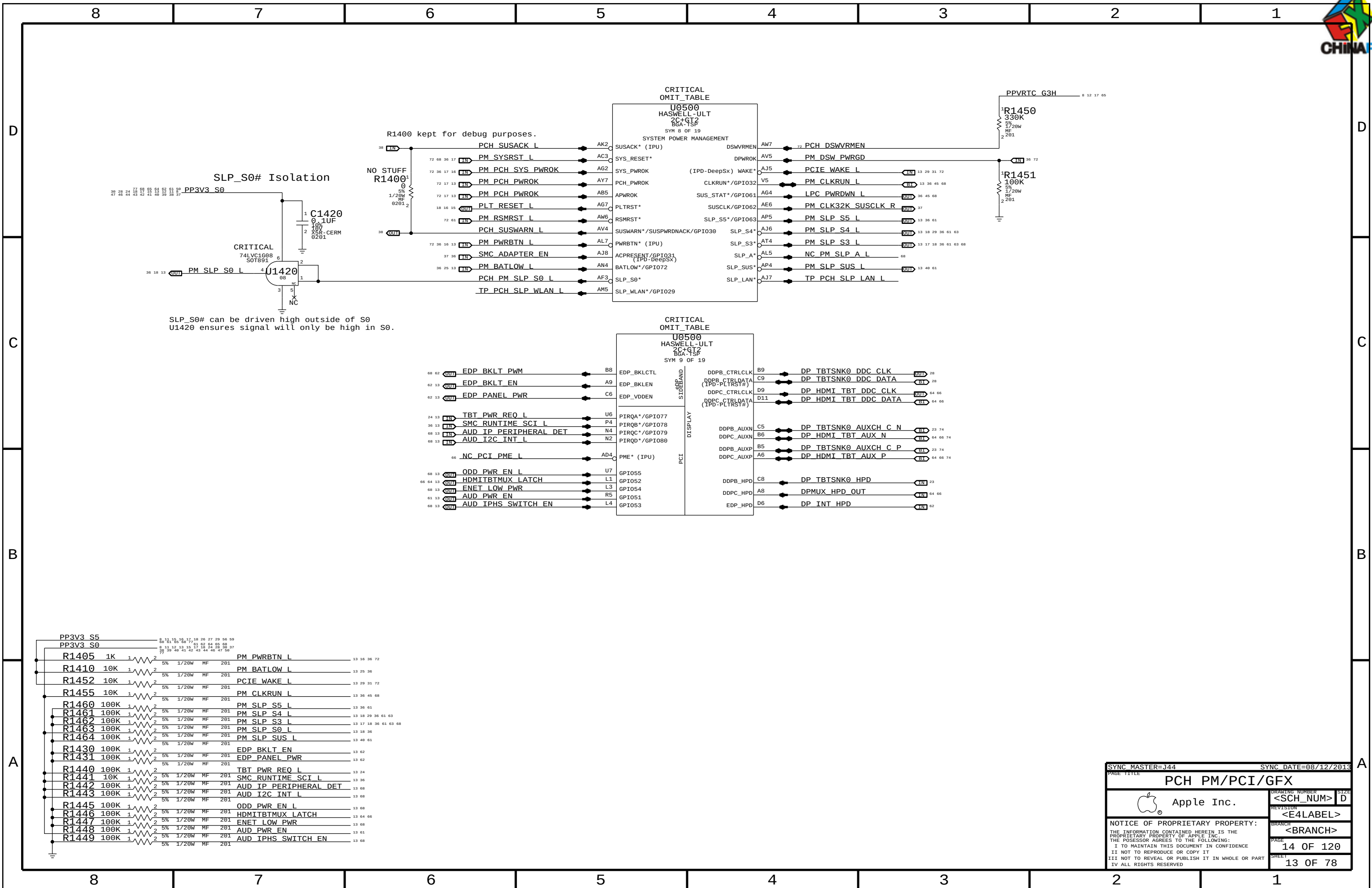


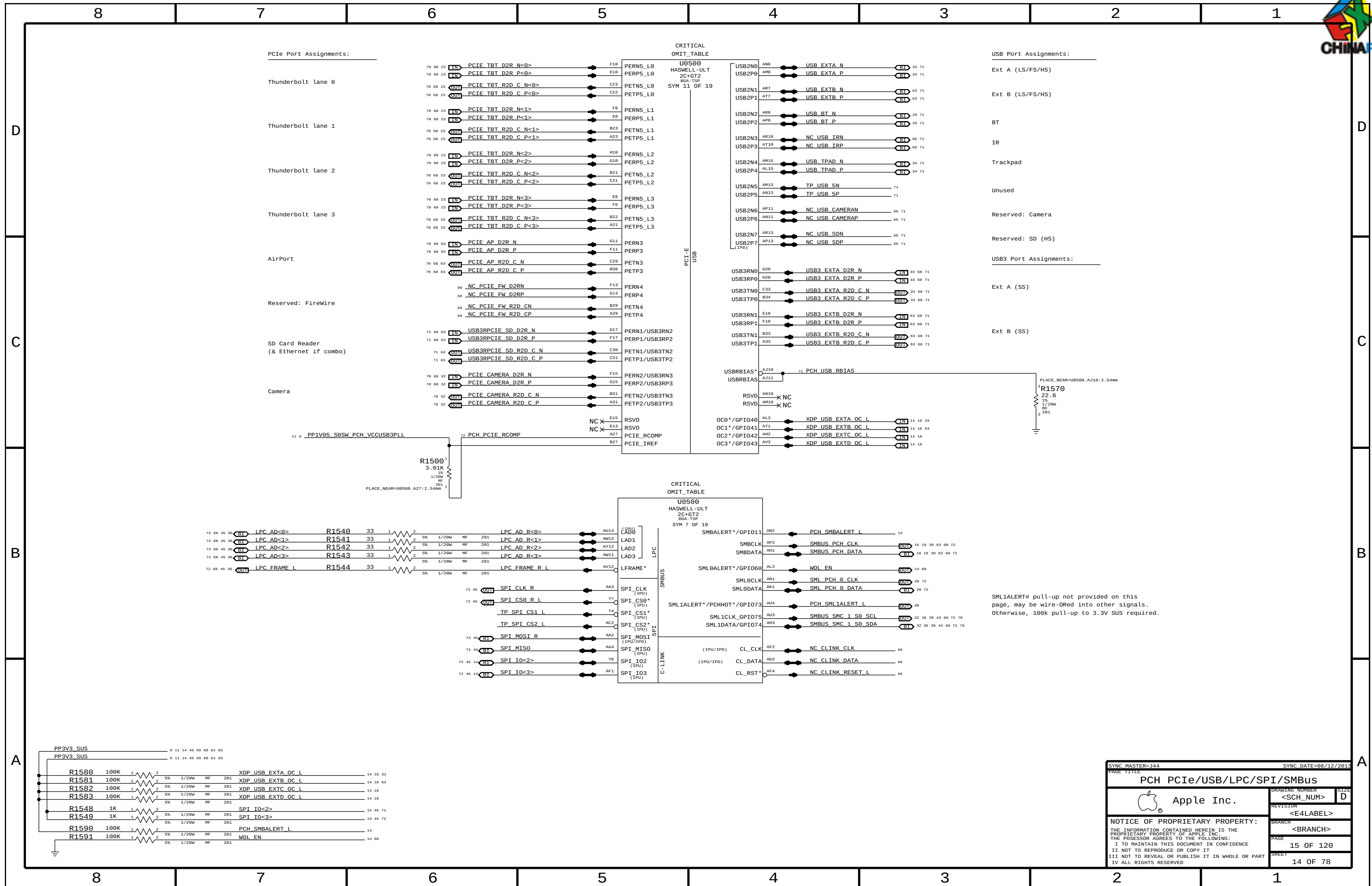
LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0
as well as from clarification email, from Ssrini, dated 9/10/2012 2:11pm.

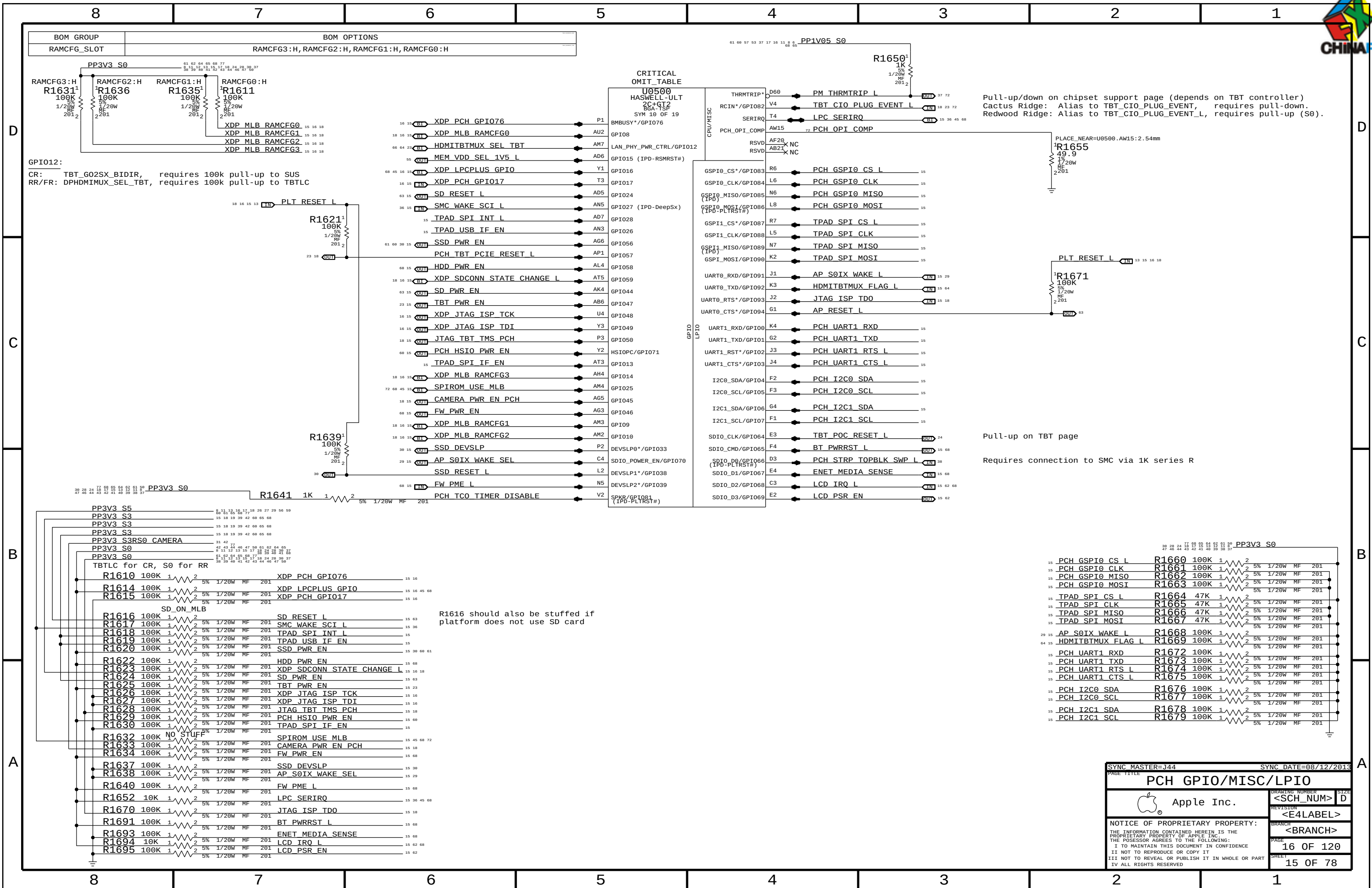
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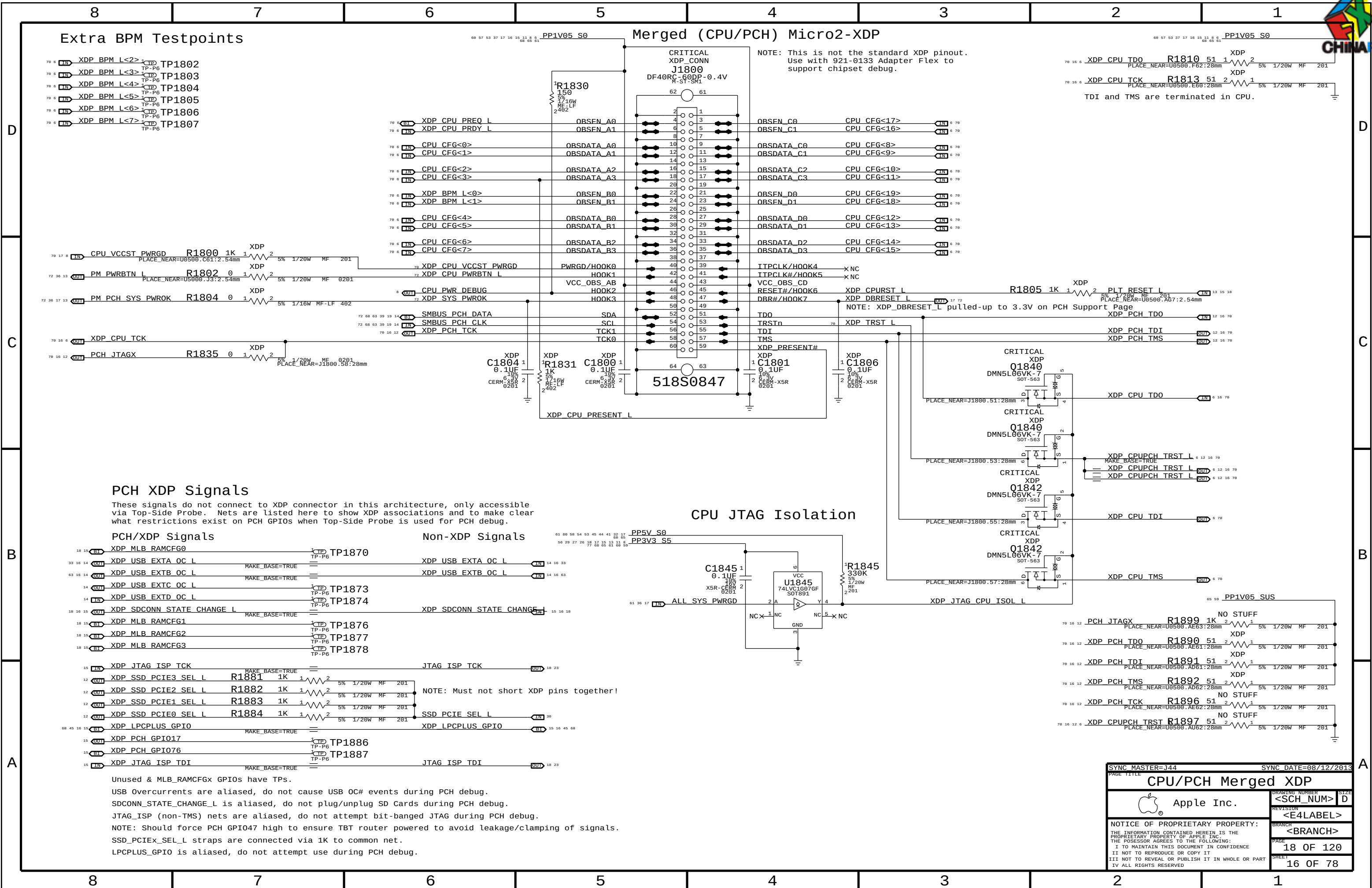


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		SHEET	12 OF 78











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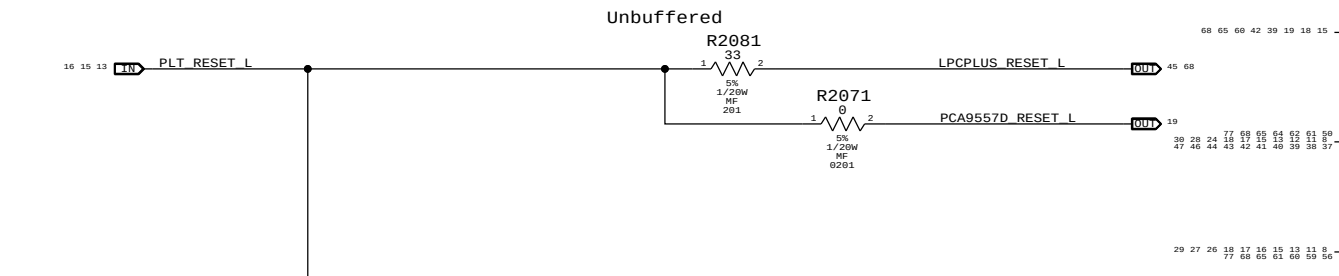
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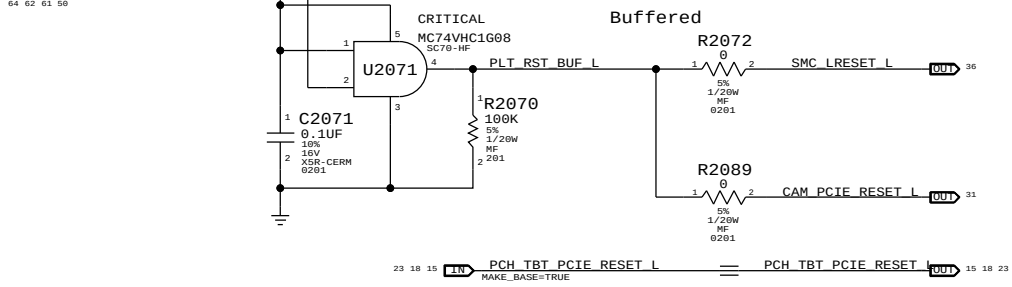
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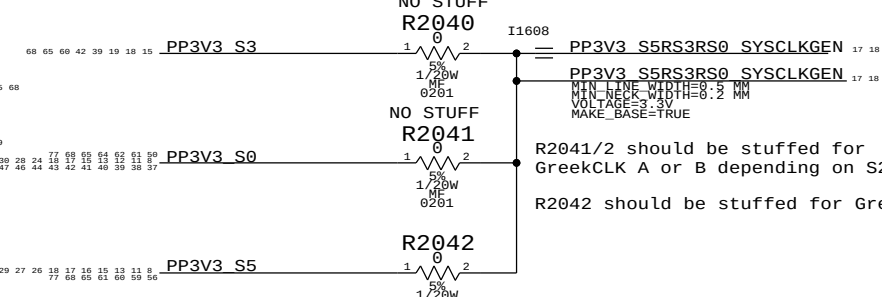
Platform Reset Connections



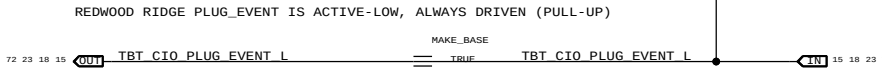
Scrub for Layout Optimization



GreenCLK 25MHz Power

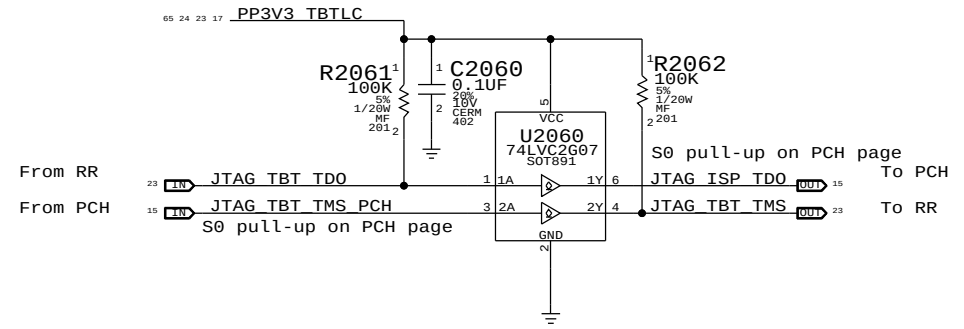


THUNDERBOLT PULL-UP



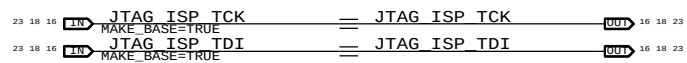
Redwood Ridge JTAG Isolation

TBTLIC can be on when S0 is off, and vice-versa
Isolation ensures no leakage to RR or PCH

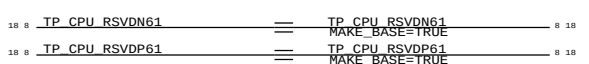


NOTE: Solution shown is for LPT-LP. Other PCH's may require isolation on TCK and TDI as well for PCH glitch-prevention.

NOTE: This reference schematic assumes PCH JTAG GPIOs are only used for Thunderbolt. If other ASIC JTAG signals are wired into these GPIOs different isolation techniques will likely be necessary. Multi-router designs also require different circuitry.

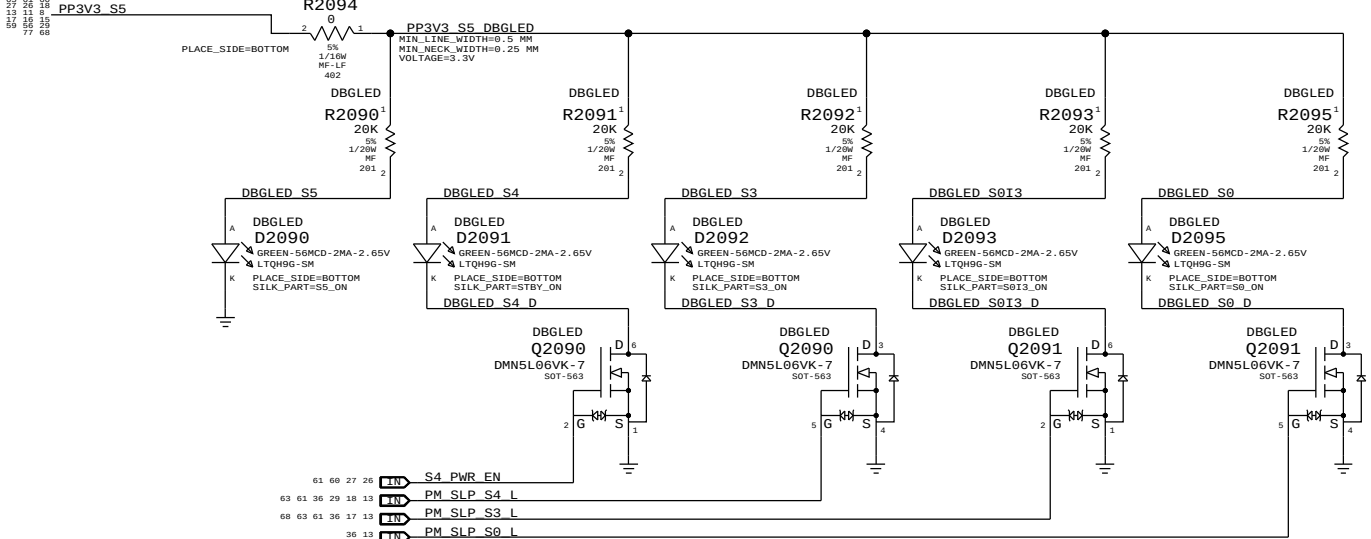


Pin N61 needs a TP for Power to perform iFDIM test
Renaming the pins N61 and P61 to remove automatic diffpari property

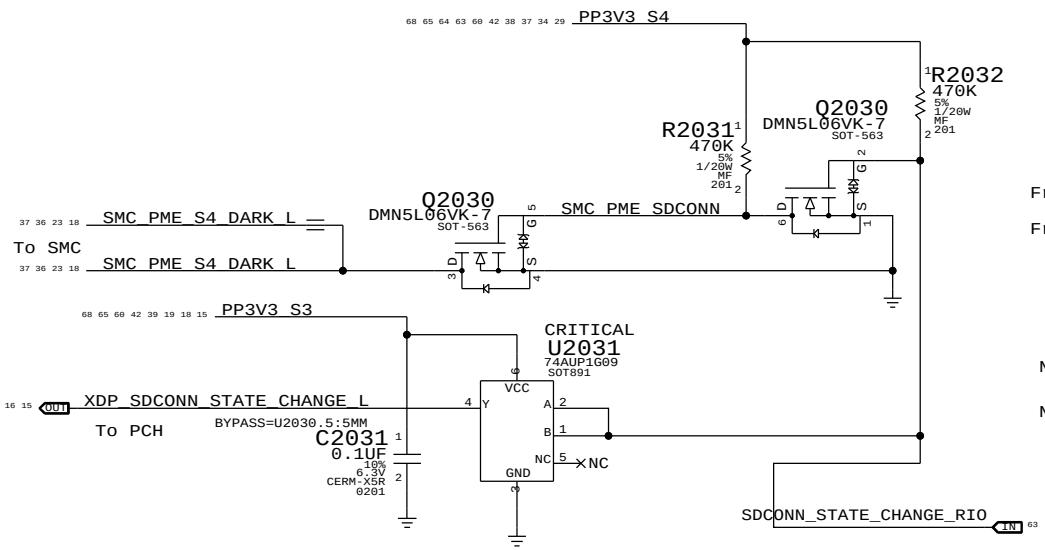


Power State Debug LEDs

(For development only)

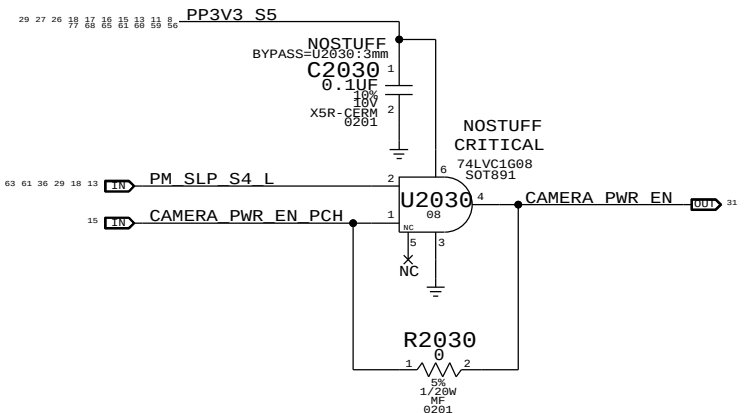
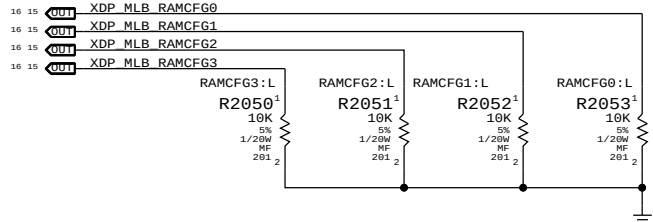


SDCONN_STATE_CHANGE Isolation



RAM Configuration Straps

Pull-downs for chip-down RAM systems



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Project Chipset Support			
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Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

- =I2C_VREFDAC5_SCL
- =I2C_VREFDAC5_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

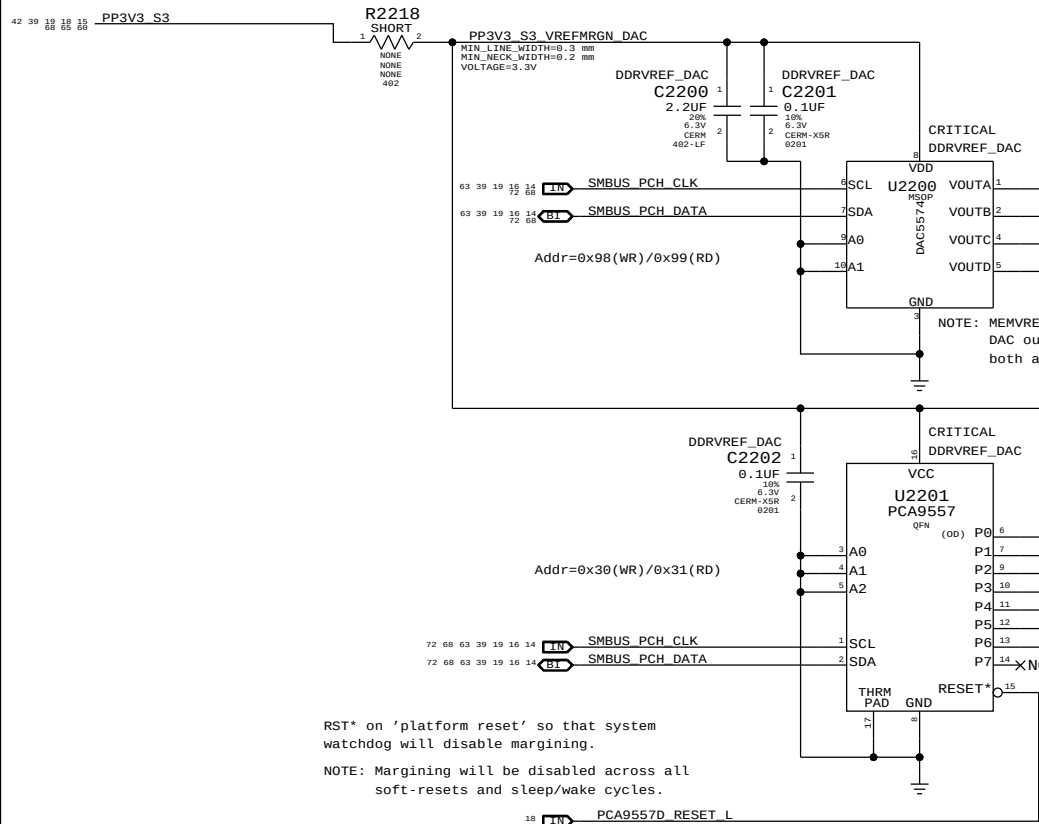
FETs for CPU isolation during DAC margining

NOTE: CPU DAC output step sizes:

DDR3	(1.5V)	7.70mV	per step
DDR3L	(1.35V)	6.99mV	per step
LPDDR3	(1.2V)	?.??mV	per step

NOTE: CPU has single output for VREFA. Split into two signals for independent DAC margining support. When DAC margining VREFA ensure VREFMRGN_CPU_EN is low to remove short due to CPU.

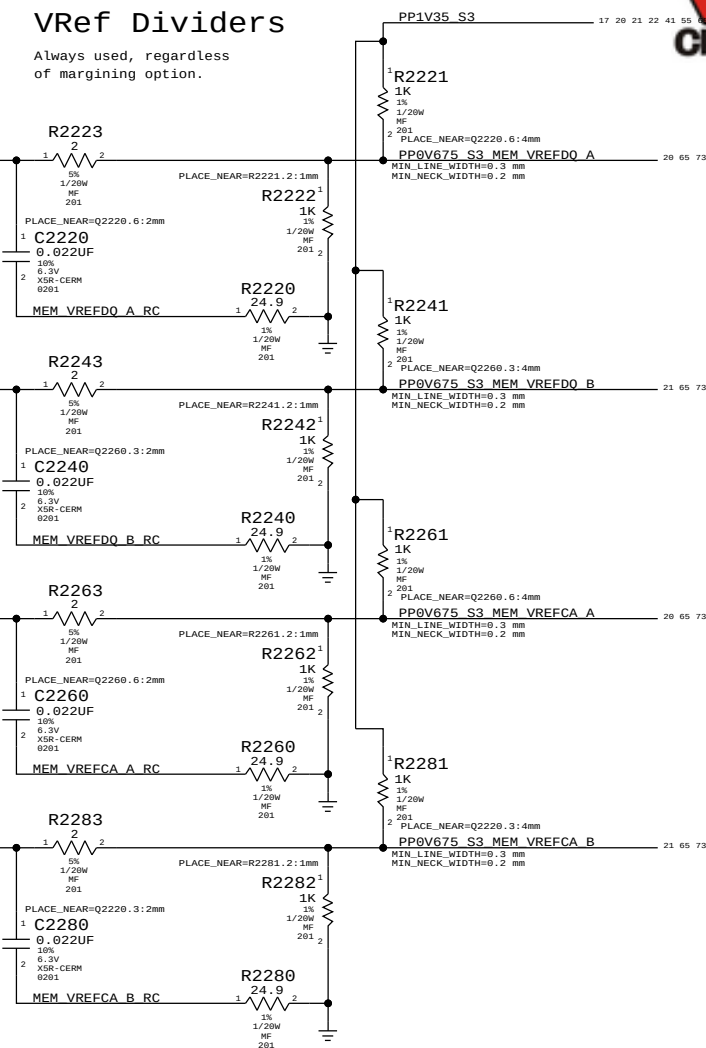
DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	LPDDR3 (1.2V)		DDR3L (1.35V)		LPDDR3 (1.2V) DDR3L (1.35V)
Nominal value	0.600V (DAC: 0x2E.5)		0.675V (DAC: 0x34)		1.200V (DAC: 0x5D) 1.343V (DAC: 0x68)
Margined target:	0.300V - 0.900V (+/- 300mV)		0.337V - 1.013V (+/- 337.5mV)		0.800V - 1.600V (+/- 400mV) 0.972V - 1.714V (+/- 371mV)
DAC range:	0.000V - 1.199V (0x00 - 0x5D)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 2.397V (0x00 - 0xBA)
VRef current:	+73uA - -73uA (- = sourced)		+82uA - -82uA (- = sourced)		+21uA - -21uA (- = sourced) +25uA - -25uA (- = sourced)
DAC step size:	6.36mV / step @ output		6.36mV / step @ output		4.28mV / step @ output 3.53mV / step @ output

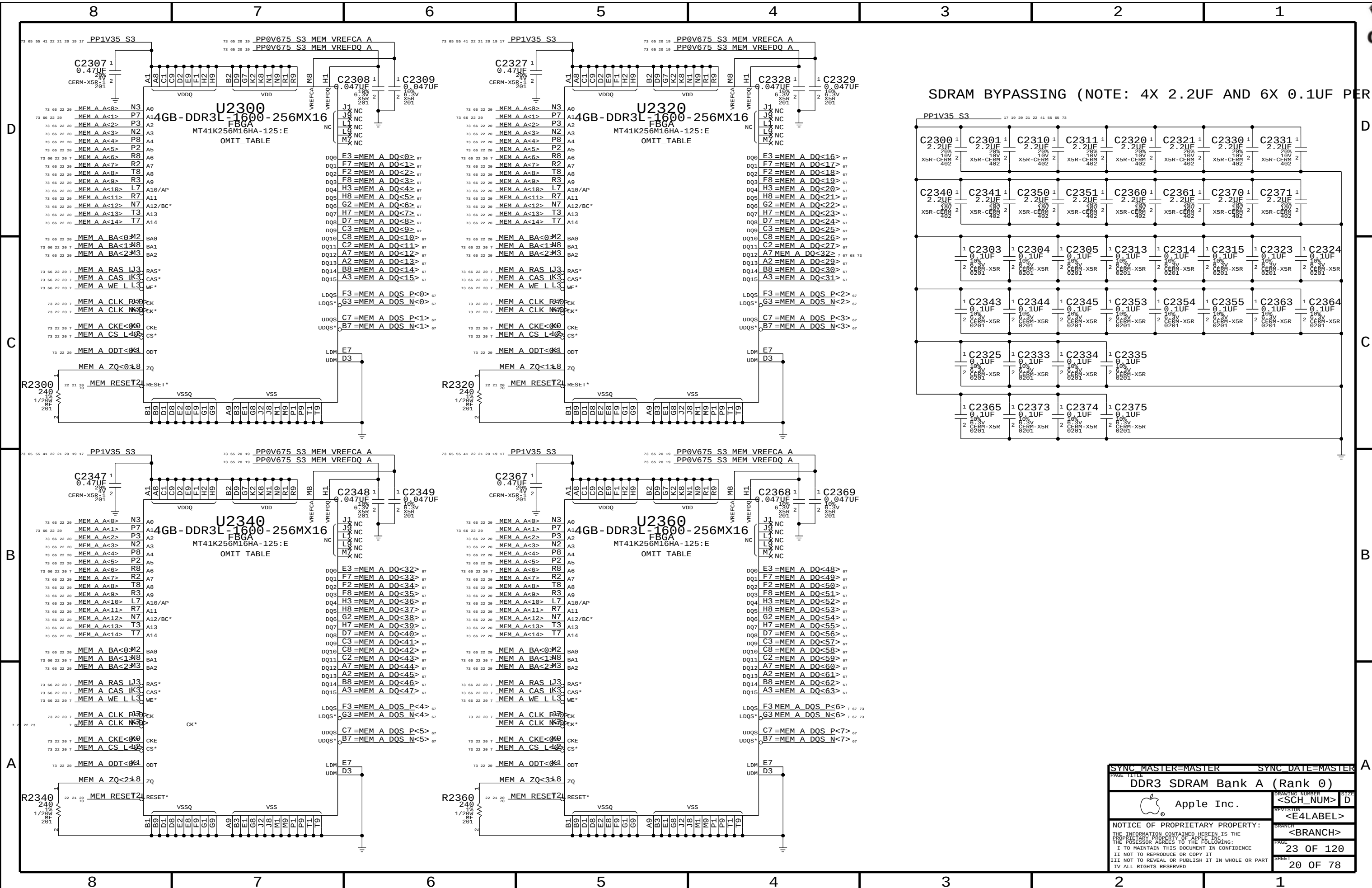
NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

Always used, regardless
of margining option.



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SDRAM BYPASSING (NOTE: 4X 2.2UF AND 6X 0.1UF PER CHIP)

SYNC MASTER=MASTER

SYNC DATE=MASTER

PAGE TITLE

DDR3 SDRAM Bank A (Rank 0)

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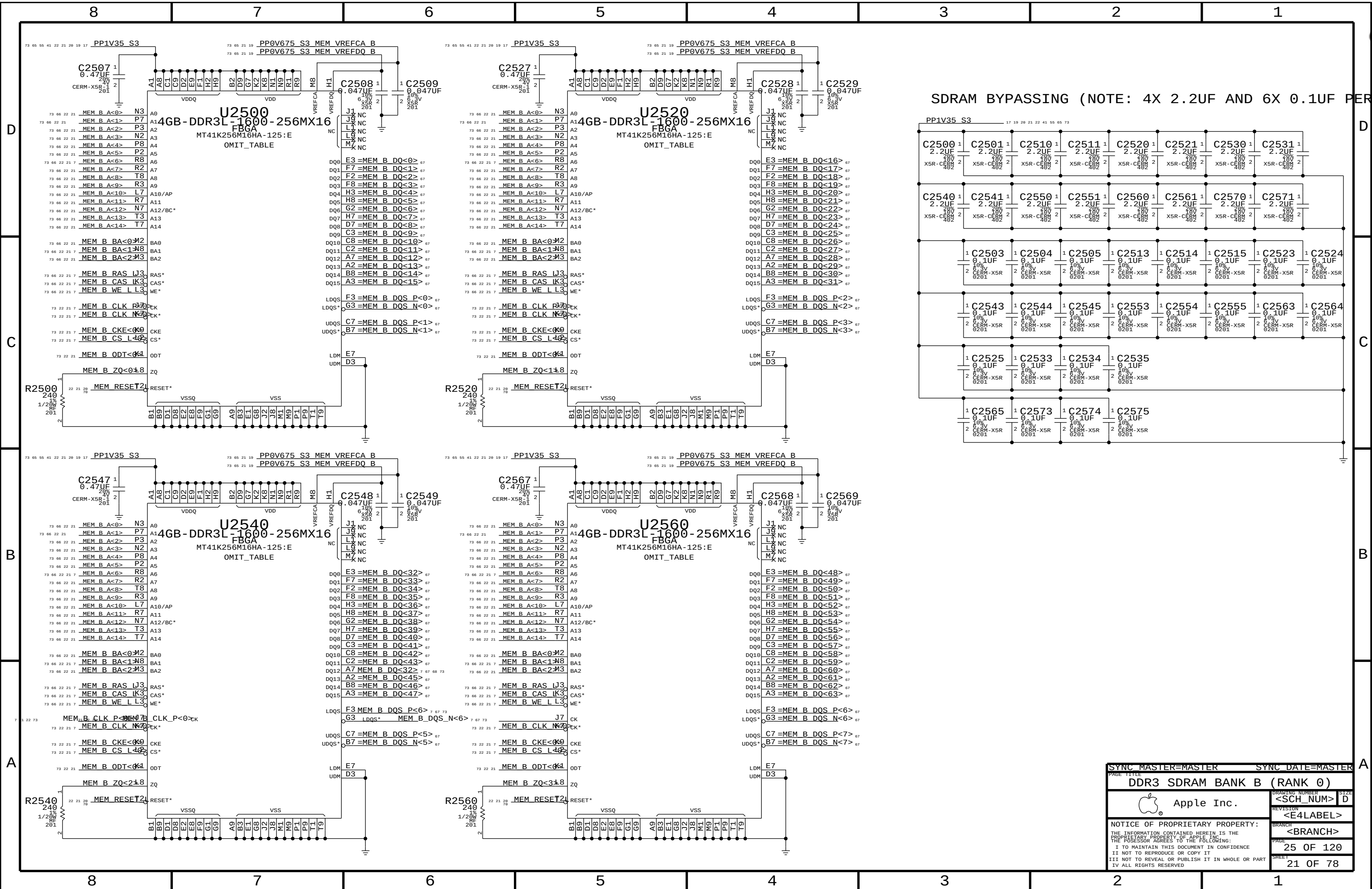
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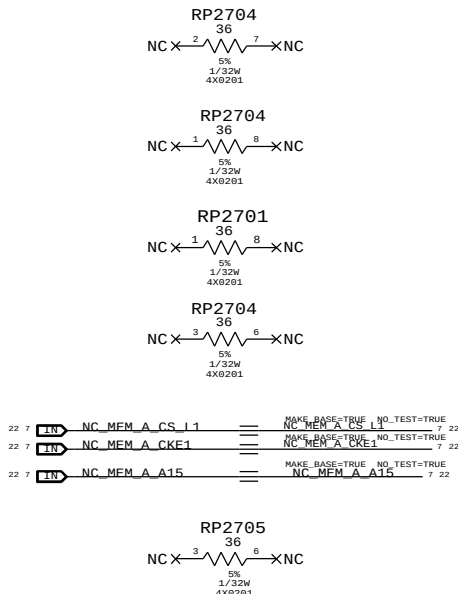
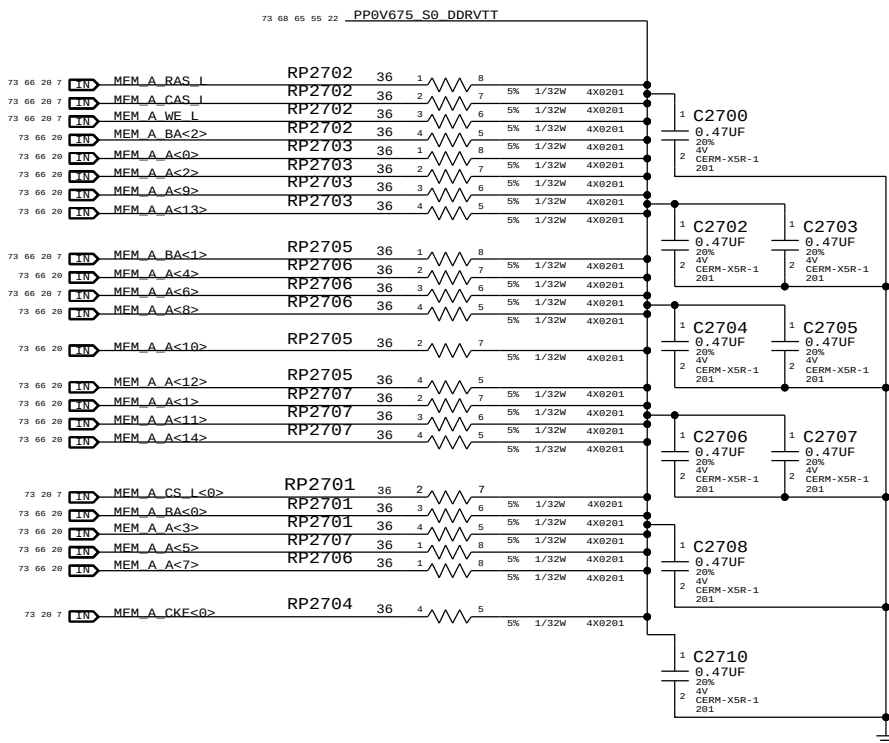
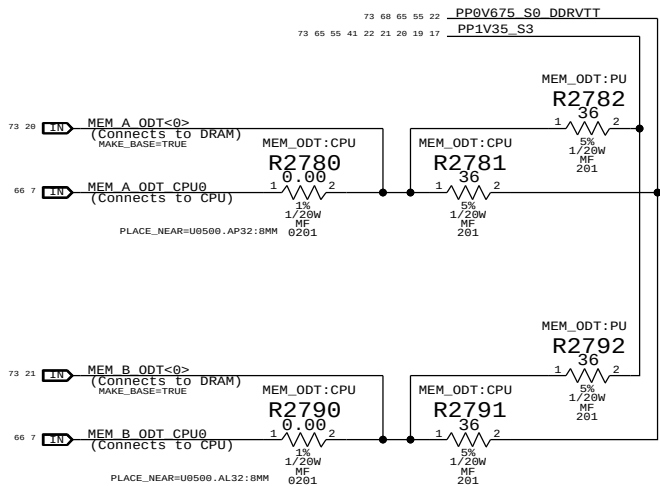
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Memory CMD/CTL Termination - Channel A

MEMORY RPACK SPARES

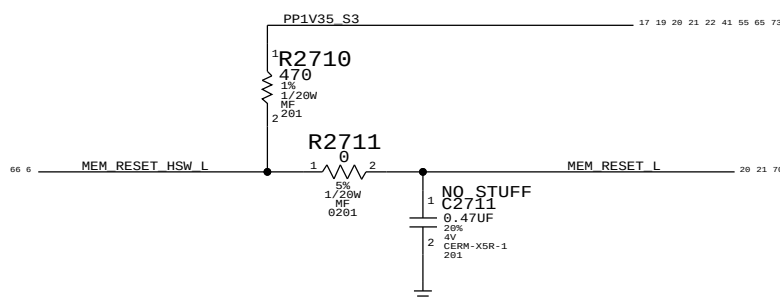
Memory ODT Option

MEM_ODT:CPU drives ODT from CPU, terminated to 0.675V VTT.
MEM_ODT:PU disconnect ODT from CPU, ODT pins on DRAM pulled up to 1.35V VDDQ.



Memory Reset Pull Up

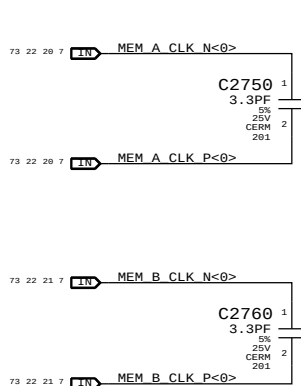
Reset is an open drain in Haswell ULT and needs pull up



Memory CMD/CTL Termination - Channel B

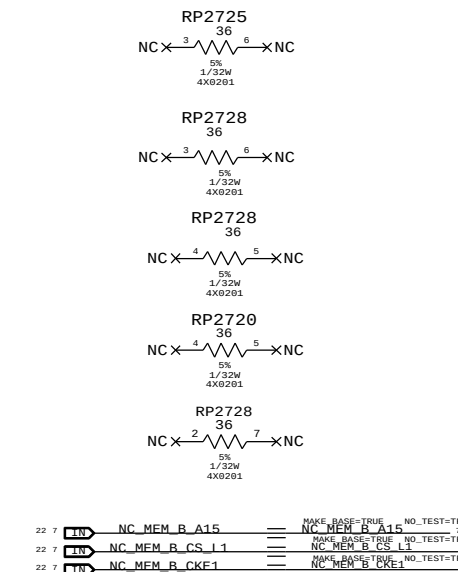
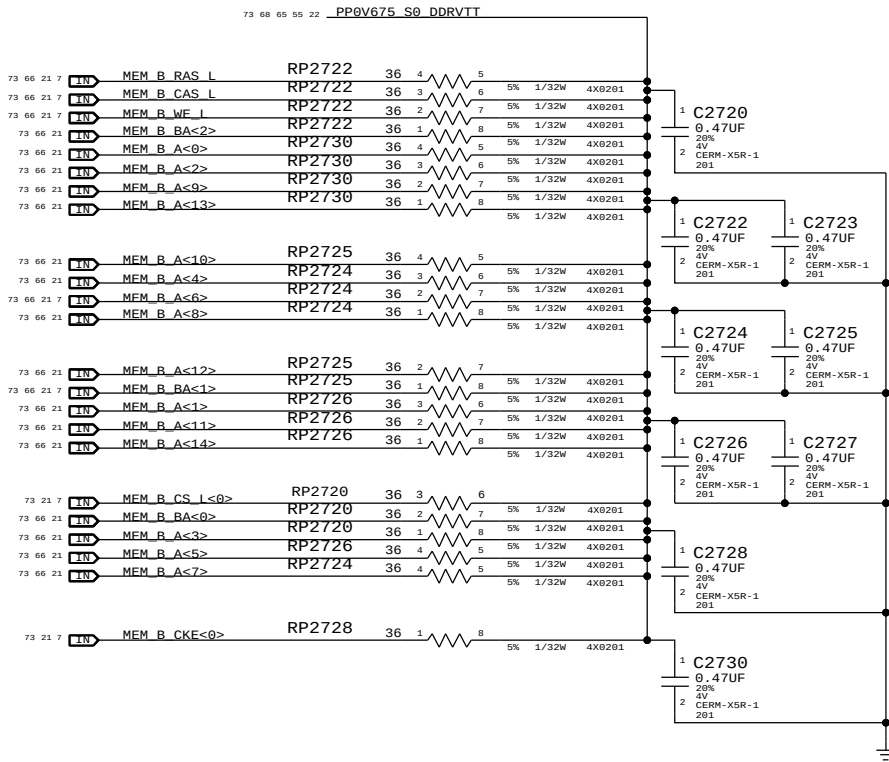
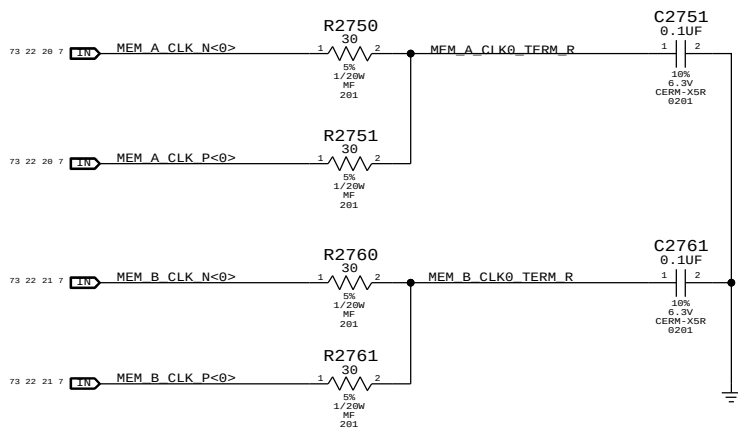
Memory Clock Near-End Termination

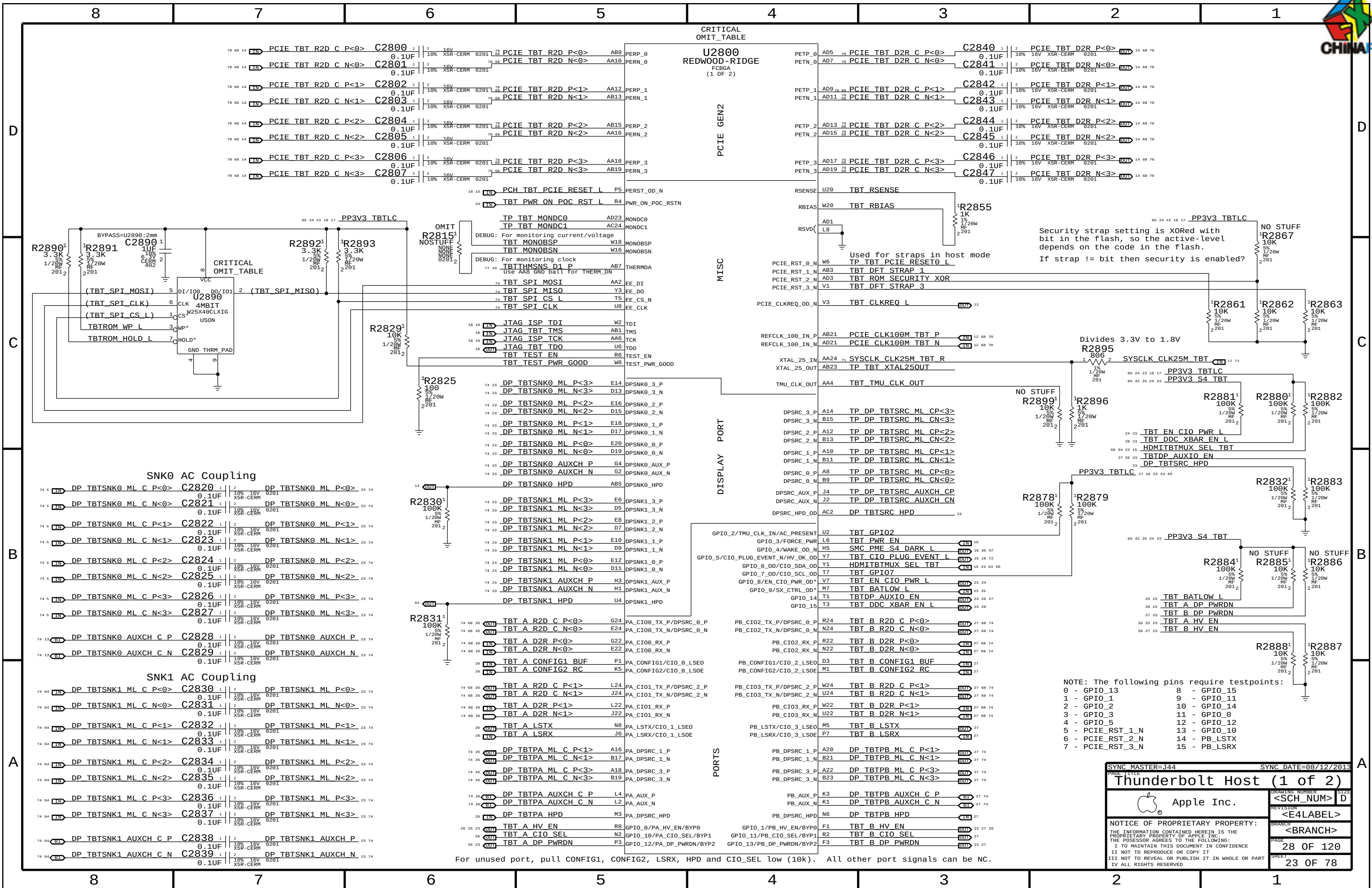
Place Source C termination before first DRAM



Memory Clock Far-End Termination

Place RC end termination after last DRAM





SYNC MASTER=J44

SYNC DATE=08/12/2013

Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER<SCH NUM>D

REVISION<E4LABEL>

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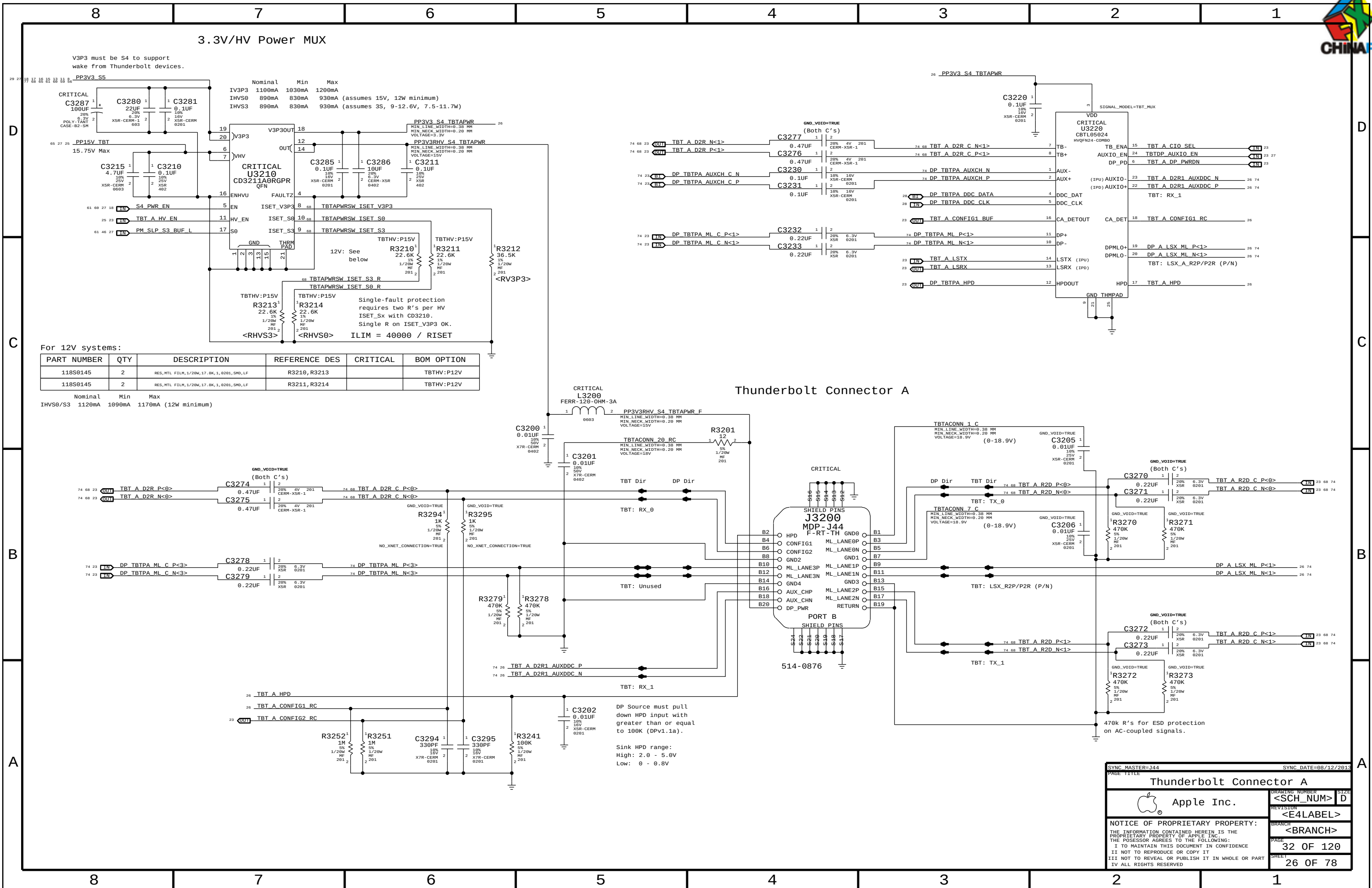
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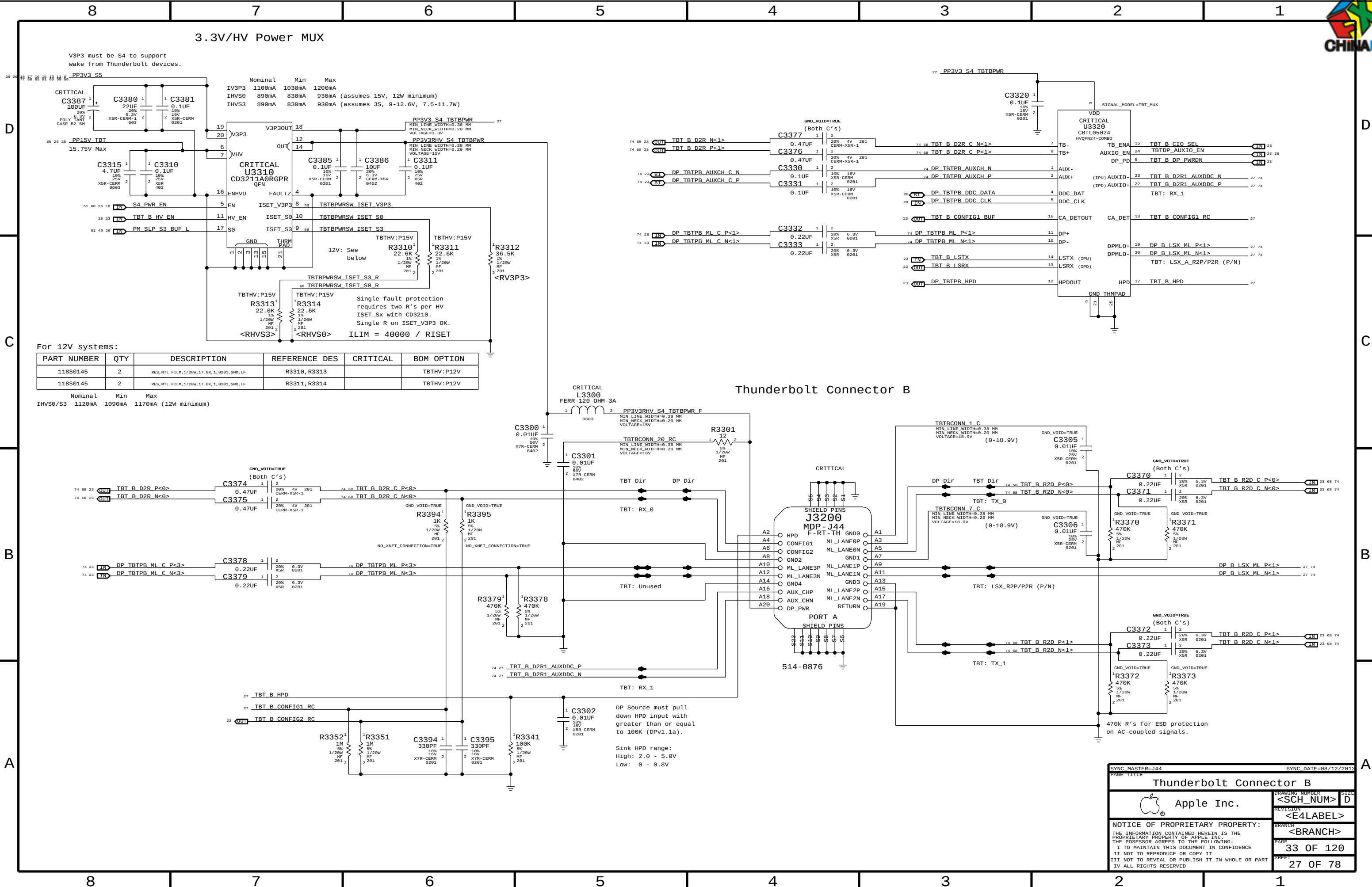
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Thunderbolt Host (2 of 2)			
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Apple Inc.		REVISION <E4LABEL>	
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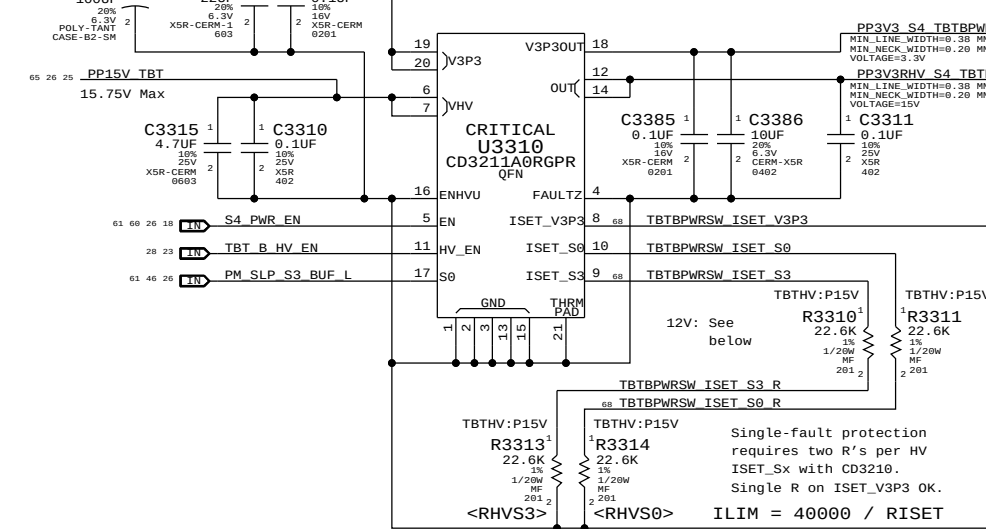
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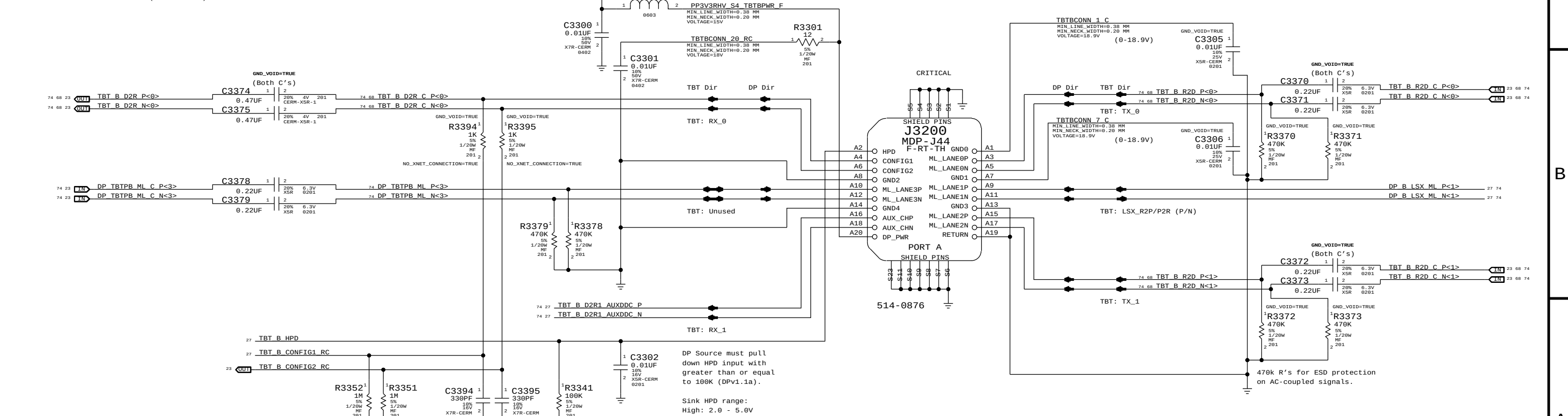
V3P3 must be S4 to support wake from Thunderbolt devices.

	Nominal	Min	Max
IV3P3	1100mA	1030mA	1200mA
IHVS0	890mA	830mA	930mA (assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA (assumes 3S, 9-12.6V, 7.5-11.7W)



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES, MTL FILM, 1/20W, 17.8K, 1, 0201, SMD, LF	R3310, R3313		TBTHV:P12V
118S0145	2	RES, MTL FILM, 1/20W, 17.8K, 1, 0201, SMD, LF	R3311, R3314		TBTHV:P12V

	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)



8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

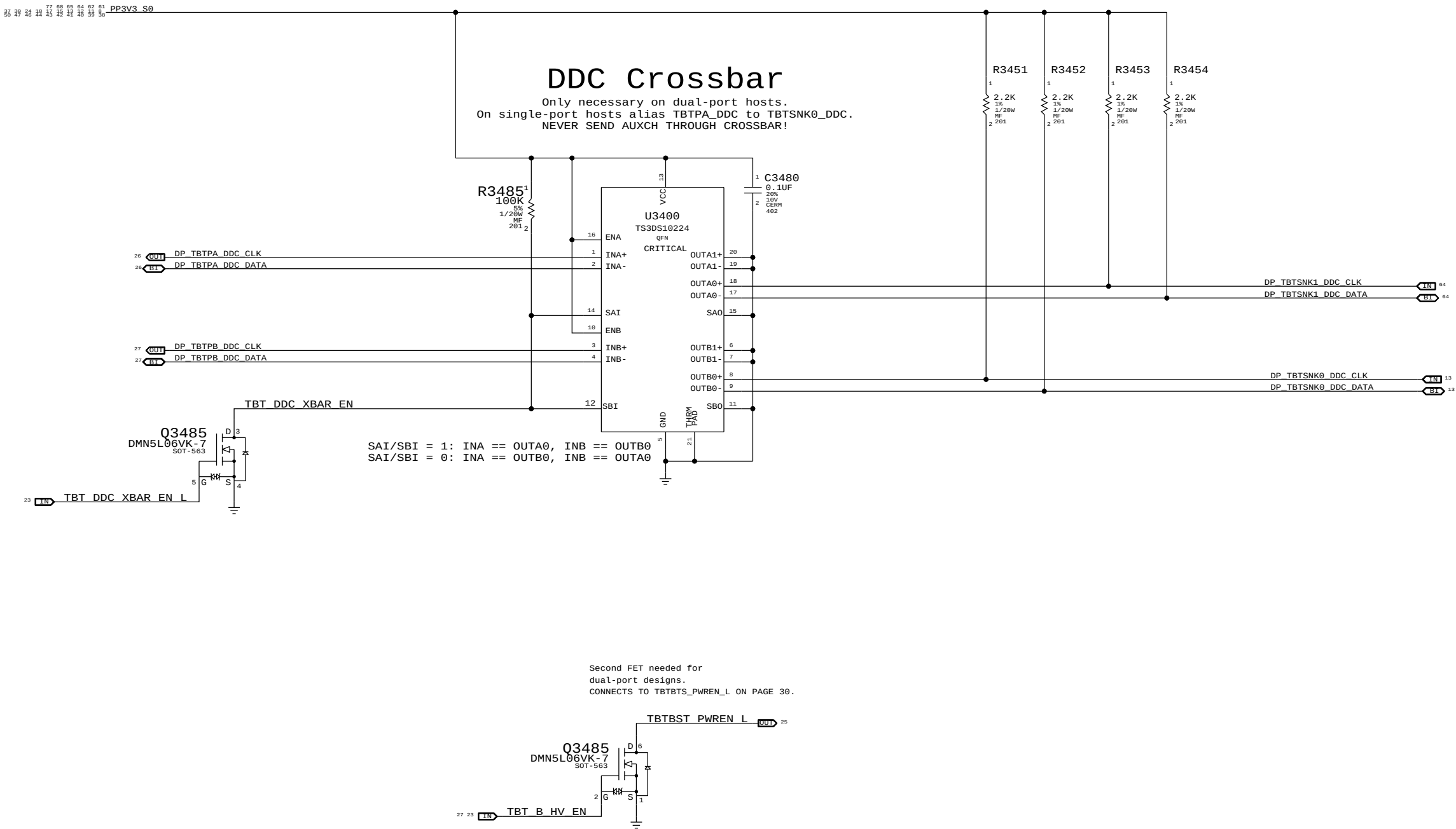
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PAGE TITLE
Thunderbolt Connector B
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PAGE
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DDC Pull-Ups

2.2k pull-ups are required by PCH
to indicate active display interface.
DP++ spec violation, should remove!

NOTE: Only DDC_DATA is sensed, so DDC_CLK
pull-ups are unstuffed.





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56 27 26 18 17 16 15 13 11 8
77 68 65 61 60 59

PP3V3 S5

1 R3561 100K 5% 1/20W MF 201

2 C3560 0.1uF 63V CERM-X5R 0201

U3560 NC7SB3157P6XG SC70 VER-3

VCC CRITICAL

5

1

2

4

GND

2

SEL OUTPUT

L H

PCIE_WAKE_L (B0)

AP_S0IX_WAKE_L (B1)

6 AP_S0IX_WAKE_SEL 15

3 PCIE_WAKE_L 13 31 72

1 AP_S0IX_WAKE_L 15

NOSTUFF

R3560

1 0 2

5% 1/20W MF 0201

[illegible]

SYNC MASTER-J44		SYNC DATE=08/12/2013	
PAGE TITLE			
WIRELESS SUPPORT			
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		PAGE 35 OF 120	
		SHEET 29 OF 78	

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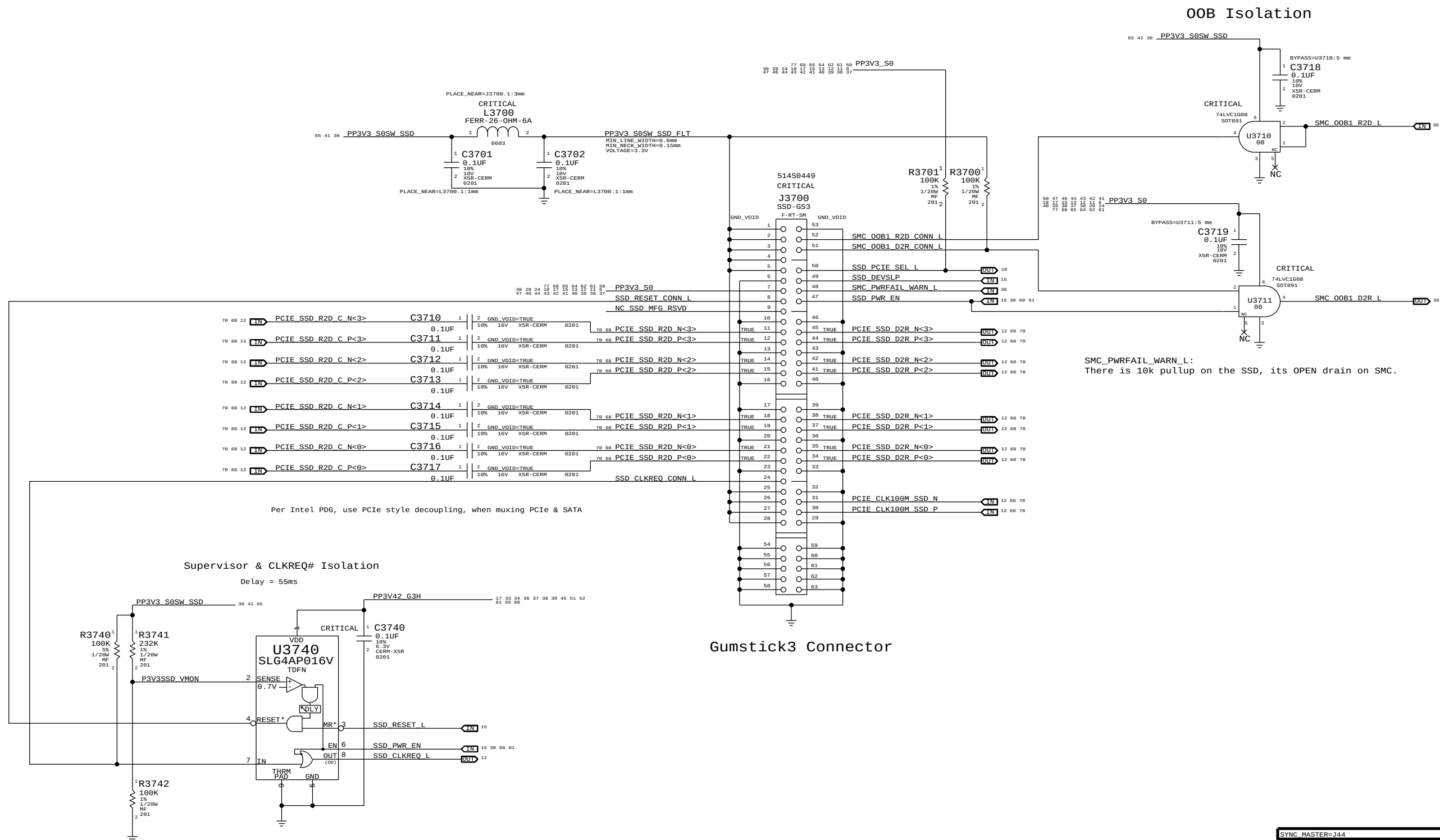
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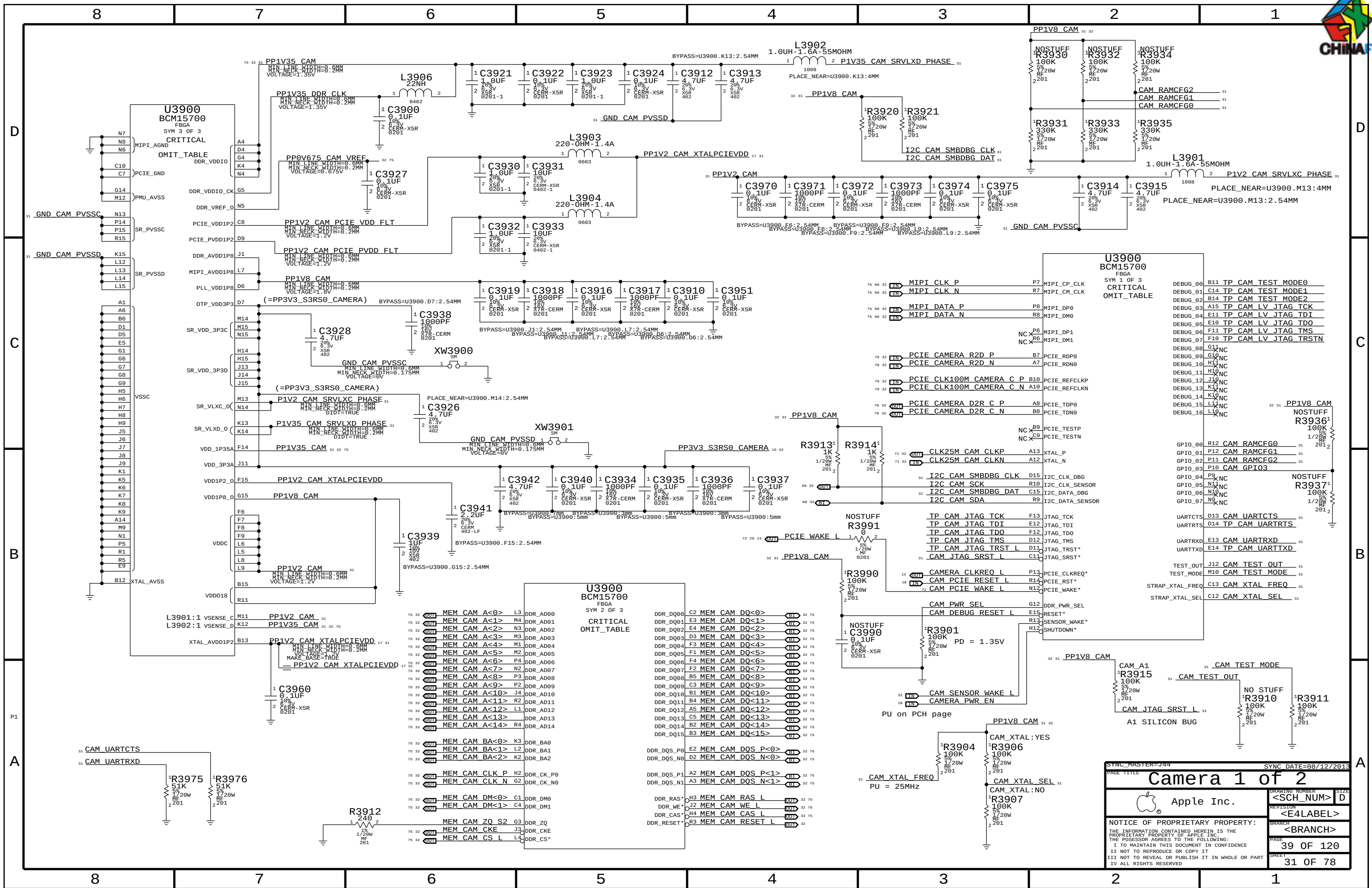
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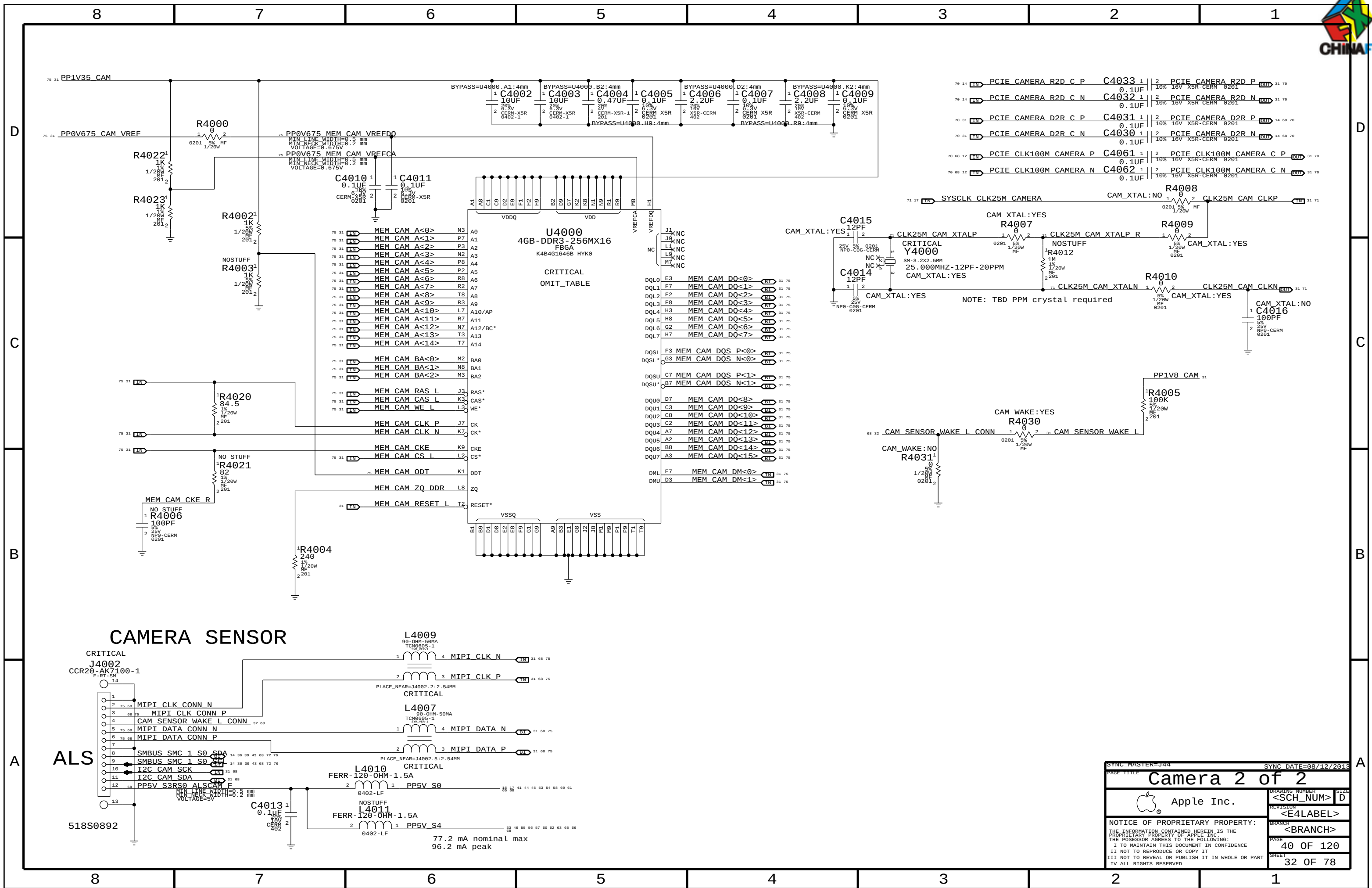
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SYNC MASTER=J44		SYNC DATE=08/12/2013	
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		SHEET	30 OF 78

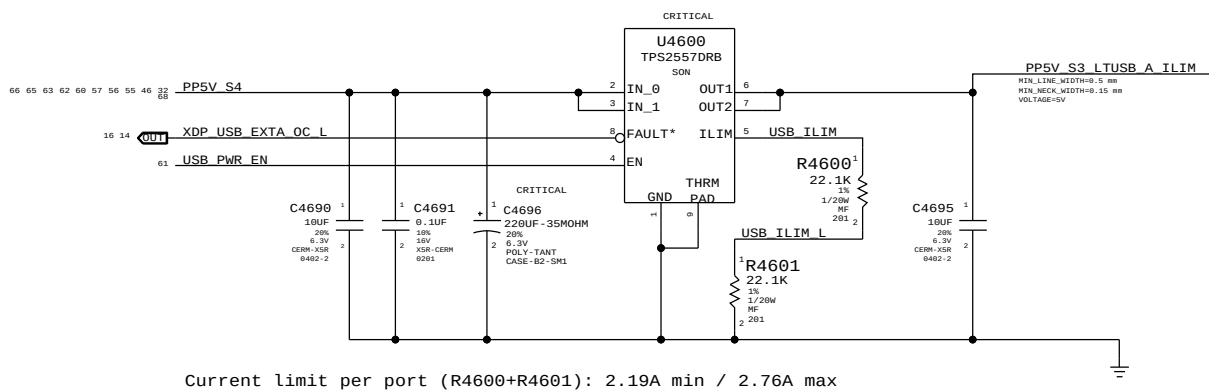






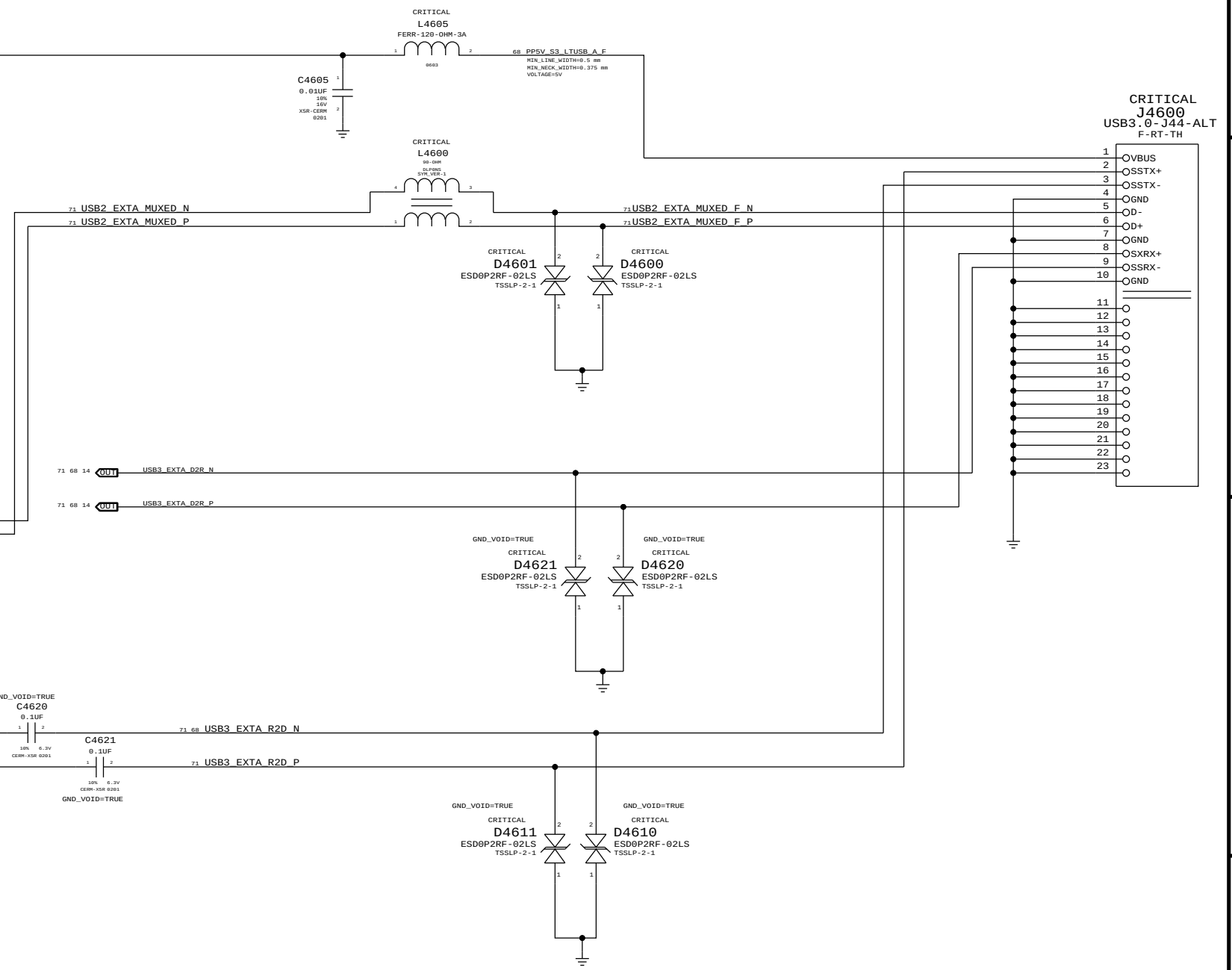
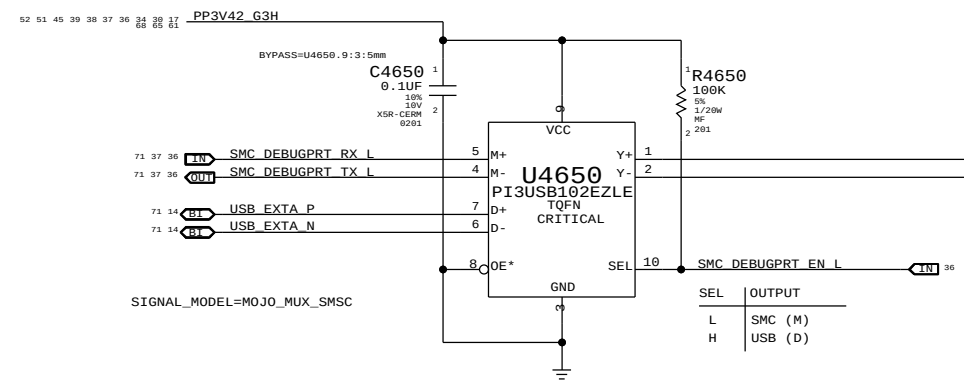
RIGHT USB PORT A

USB Port Power Switch



Mojo SMC Debug Mux

THE PI3USB102E CAN CLAMP VOLTAGE IN THE INTERNAL USB PINS



SYNC MASTER=J44

SYNC DATE=08/12/2013

External A USB3 Connector

Apple Inc.

DRAWING NUMBER

<SCH_NUM>

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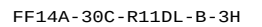
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SHEET

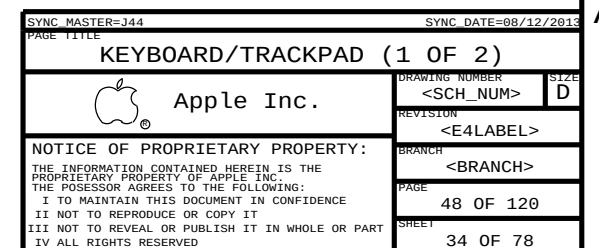
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J4800
FF14-18C-R11DL
E=RT-SM

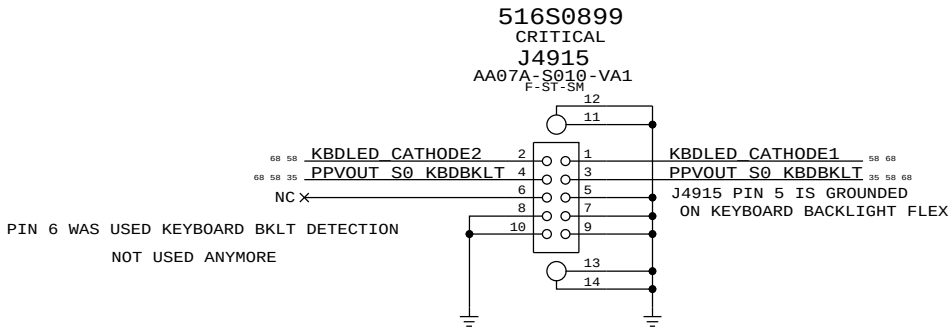


```
No IPD on OE input pin PP3V3_S4 (symbol error).
```





Keyboard Backlight Connector



SYNC_MASTER=J44		SYNC_DATE=08/12/2013	
PAGE TITLE			
KEYBOARD/TRACKPAD		(2 OF 2)	
 Apple Inc.		DRAWING NUMBER	SIZE
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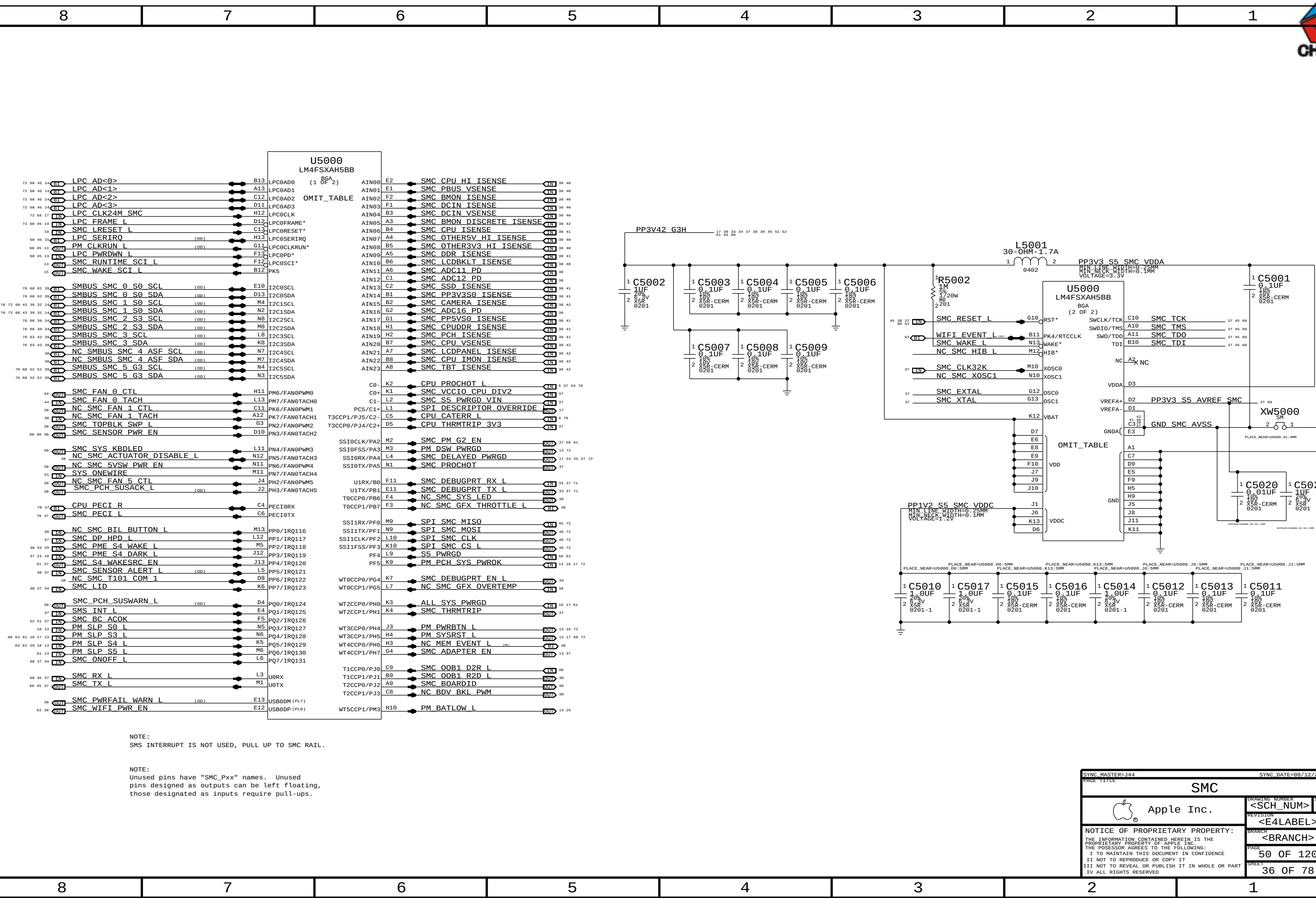
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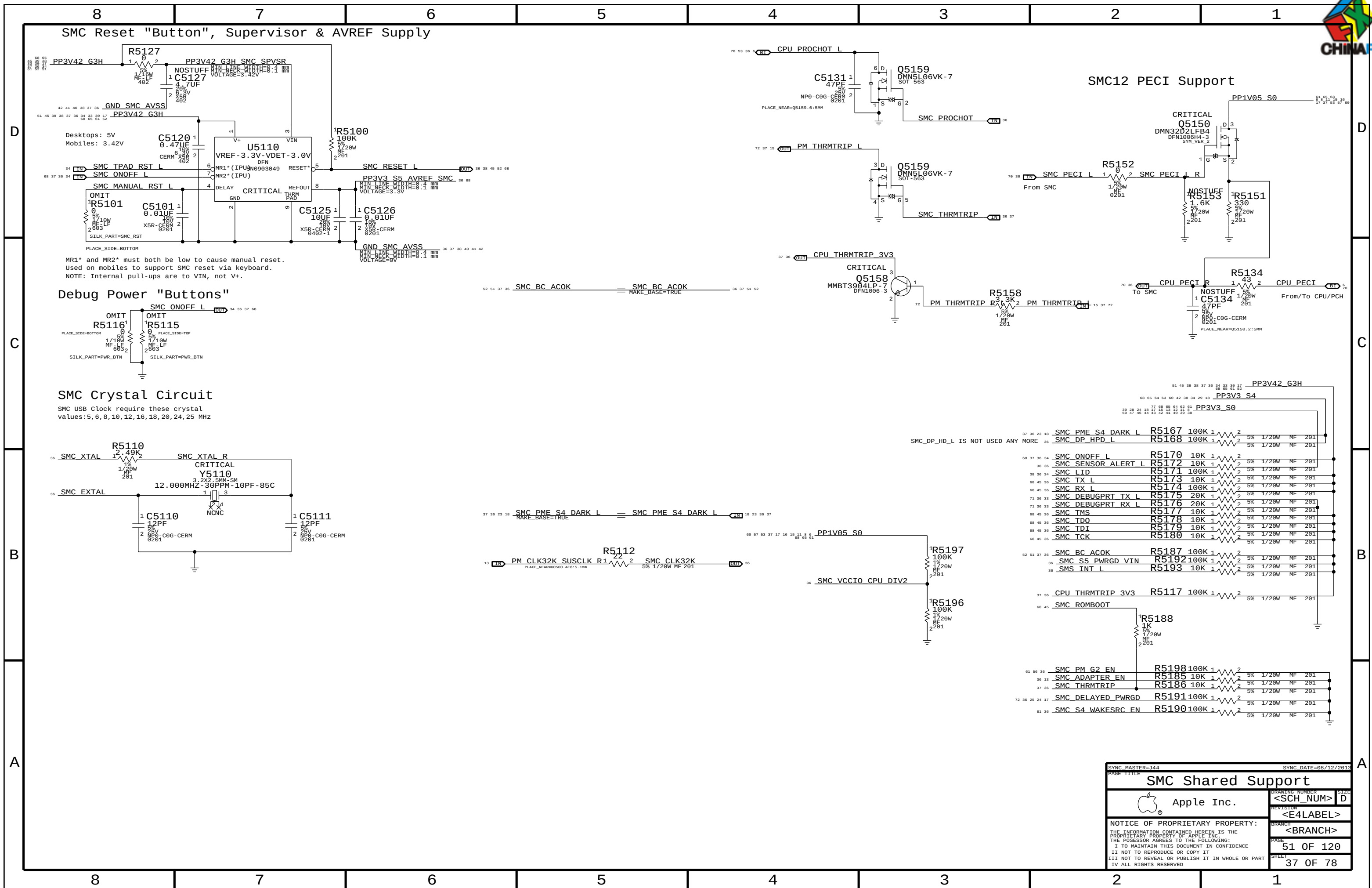
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NOTE:
SMS INTERRUPT IS NOT USED, PULL UP TO SMC RAIL.

NOTE:
Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

SYNC MASTER=J44		SYNC DATE=08/12/2013	
PAGE TITLE			
SMC			
Apple Inc.		DRAWING NUMBER	SIZE
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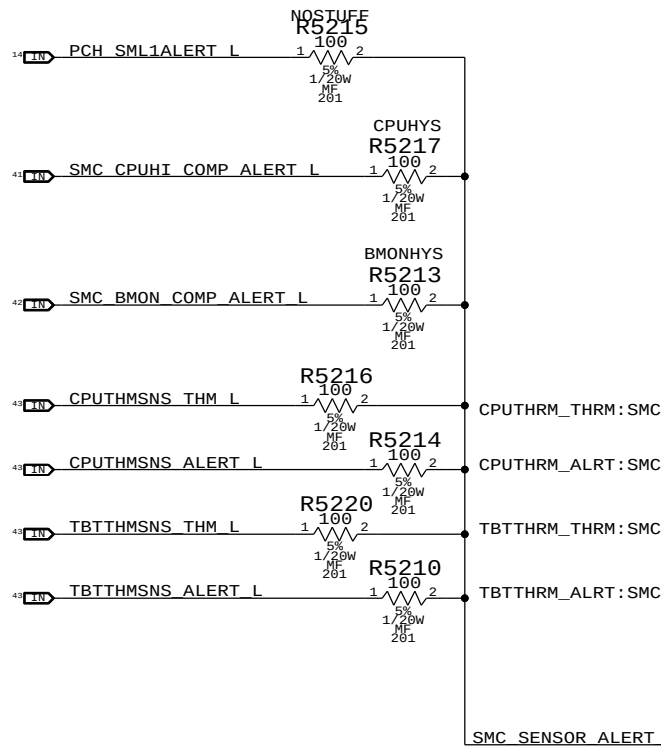
SMC12 ADC Assignments

36	OUT	SMC CPU HI ISENSE	=	SMC CPU HI ISENSE	IN	36	38	40
36	OUT	SMC PBUS VSENSE	=	SMC PBUS VSENSE	IN	36	38	40
36	OUT	SMC BMON ISENSE	=	SMC BMON ISENSE	IN	36	38	40
36	OUT	SMC DCIN ISENSE	=	SMC DCIN ISENSE	IN	36	38	40
36	OUT	SMC DCIN VSENSE	=	SMC DCIN VSENSE	IN	36	38	40
36	OUT	SMC BMON DISCRETE ISENSE	=	SMC BMON DISCRETE ISENSE	IN	36	38	42
36	OUT	SMC CPU ISENSE	=	SMC CPU ISENSE	IN	36	38	41
36	OUT	SMC OTHER5V HI ISENSE	=	SMC OTHER5V HI ISENSE	IN	36	38	40
36	OUT	SMC OTHER3V3 HI ISENSE	=	SMC OTHER3V3 HI ISENSE	IN	36	38	40
36	OUT	SMC DDR ISENSE	=	SMC DDR ISENSE	IN	36	38	41
36	OUT	SMC LCDBKLT ISENSE	=	SMC LCDBKLT ISENSE	IN	36	38	40
36	OUT	SMC ADC11 PD	=	SMC ADC11 PD	IN	36	38	38
36	OUT	SMC ADC12 PD	=	SMC ADC12 PD	IN	36	38	38
36	OUT	SMC SSD ISENSE	=	SMC SSD ISENSE	IN	36	38	41
36	OUT	SMC PP3V3S0 ISENSE	=	SMC PP3V3S0 ISENSE	IN	36	38	41
36	OUT	SMC CAMERA ISENSE	=	SMC CAMERA ISENSE	IN	36	38	42
36	OUT	SMC ADC16 PD	=	SMC ADC16 PD	IN	36	38	38
36	OUT	SMC PP5VS0 ISENSE	=	SMC PP5VS0 ISENSE	IN	36	38	41
36	OUT	SMC CPUDDR ISENSE	=	SMC CPUDDR ISENSE	IN	36	38	41
36	OUT	SMC PCH ISENSE	=	SMC PCH ISENSE	IN	36	38	41
36	OUT	SMC CPU VSENSE	=	SMC CPU VSENSE	IN	36	38	42
36	OUT	SMC LCDPANEL ISENSE	=	SMC LCDPANEL ISENSE	IN	36	38	42
36	OUT	SMC CPU IMON ISENSE	=	SMC CPU IMON ISENSE	IN	36	38	42
36	OUT	SMC TBT ISENSE	=	SMC TBT ISENSE	IN	36	38	42

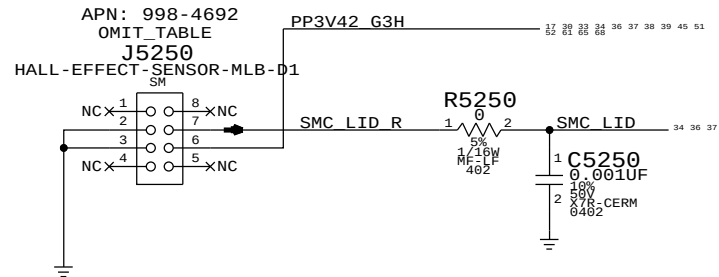
SMC12 Pin Assignments

36	NC	SMC 4 ASF SCL	=	SMC 4 ASF SCL	36	38
36	NC	SMC 4 ASF SDA	=	SMC 4 ASF SDA	36	38
36	NC	BDV BKL PWM	=	BDV BKL PWM	36	38
36	NC	SMC SYS LED	=	SMC SYS LED	36	38
36	NC	SMC GFX THROTTLE L	=	SMC GFX THROTTLE L	36	38
36	NC	SMC GFX OVERTEMP	=	SMC GFX OVERTEMP	36	38
36	NC	SMC FAN 1 CTL	=	SMC FAN 1 CTL	36	38
36	NC	SMC FAN 1 TACH	=	SMC FAN 1 TACH	36	38
36	NC	SMC 5VSW PWR EN	=	SMC 5VSW PWR EN	36	38
36	NC	SMC FAN 5 CTL	=	SMC FAN 5 CTL	36	38
36	NC	SMC BIL BUTTON L	=	SMC BIL BUTTON L	36	38
36	NC	MEM EVENT L	=	MEM EVENT L	36	38
36	NC	SMC T101 COM 1	=	SMC T101 COM 1	36	38
36	NC	SMC ACTUATOR DISABLE	=	SMC ACTUATOR DISABLE	36	38

Thermal Alerts



Hall Effect Pads



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
677-0912	1	SUBASSY,PCBA HALL EFFECT, J44	J5250	CRITICAL	

639-4502 (J44 HALL EFFECT BOARD) REPORTS TO 677-0912

Specify one of these BOM GROUPS.

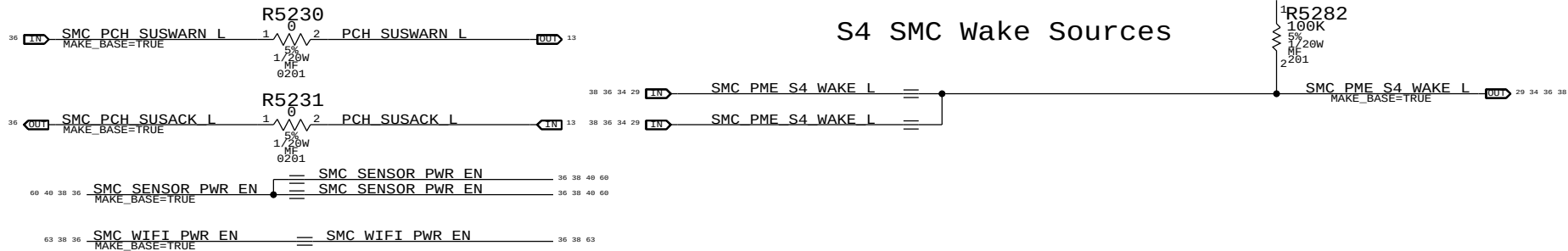
BOM GROUP	BOM OPTIONS
CPUTHRM:BOTH	CPUTHRM_THRM:SMC, CPUTHRM_ALRT:SMC
CPUTHRM:THRM	CPUTHRM_THRM:SMC, CPUTHRM_ALRT:PU
CPUTHRM:ALRT	CPUTHRM_THRM:PU, CPUTHRM_ALRT:SMC
CPUTHRM:NONE	CPUTHRM_THRM:PU, CPUTHRM_ALRT:PU

Specify one of these BOM GROUPS.

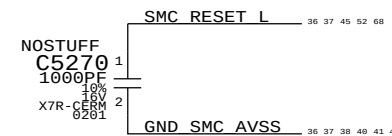
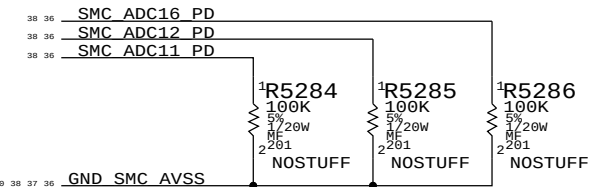
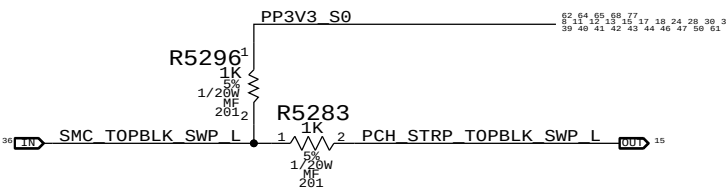
BOM GROUP	BOM OPTIONS
TBTTTHRM:BOTH	TBTTTHRM_THRM:SMC, TBTTTHRM_ALRT:SMC
TBTTTHRM:THRM	TBTTTHRM_THRM:SMC, TBTTTHRM_ALRT:PU
TBTTTHRM:ALRT	TBTTTHRM_THRM:PU, TBTTTHRM_ALRT:SMC
TBTTTHRM:NONE	TBTTTHRM_THRM:PU, TBTTTHRM_ALRT:PU

Requires EMC1412-1 or EMC1412-2 instead of EMC1412-A, new APN needs to be created.

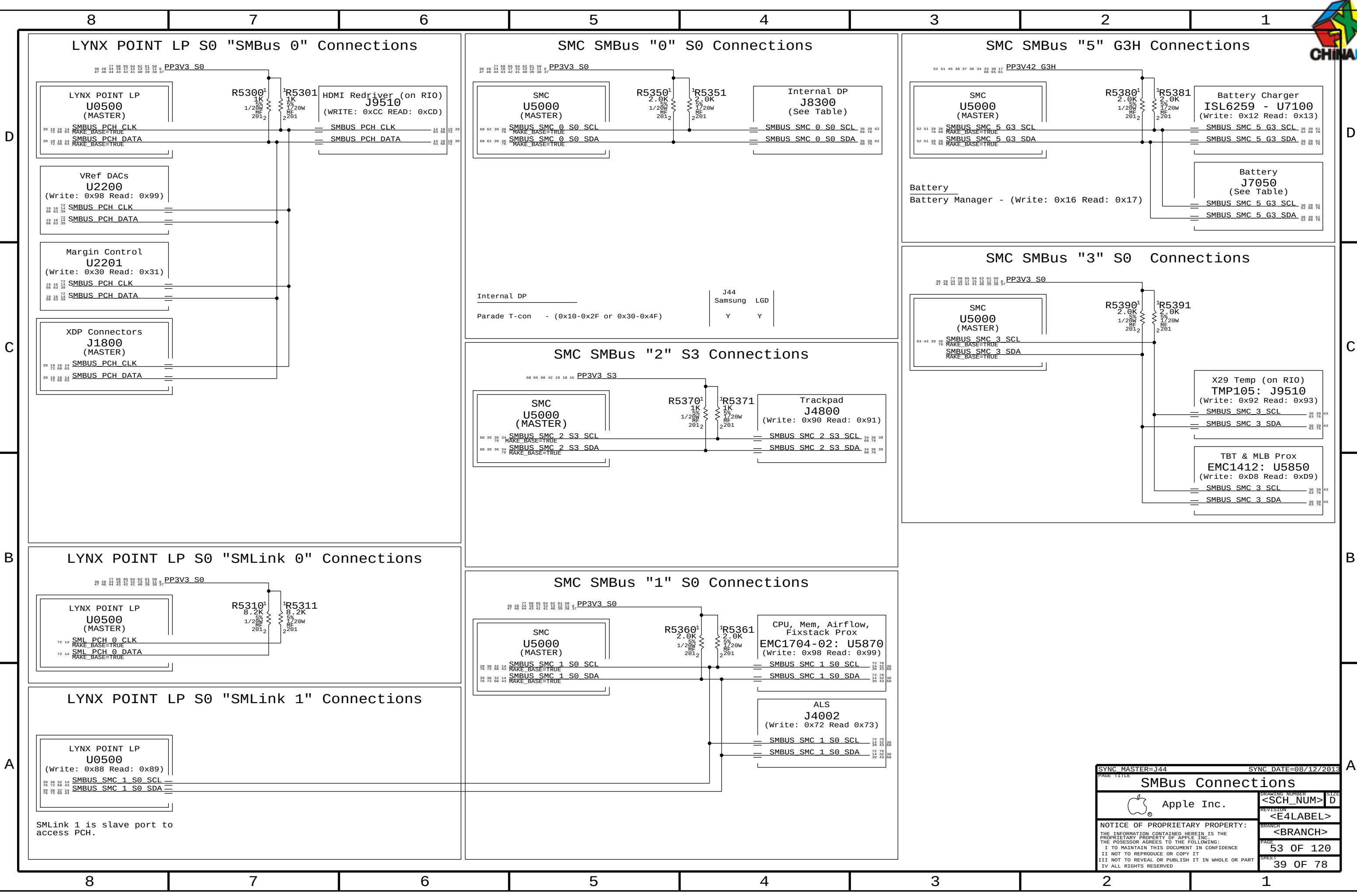
S4 SMC Wake Sources



Top Block Swap



SMC Project Support	
Apple Inc.	<SCH_NUM> D
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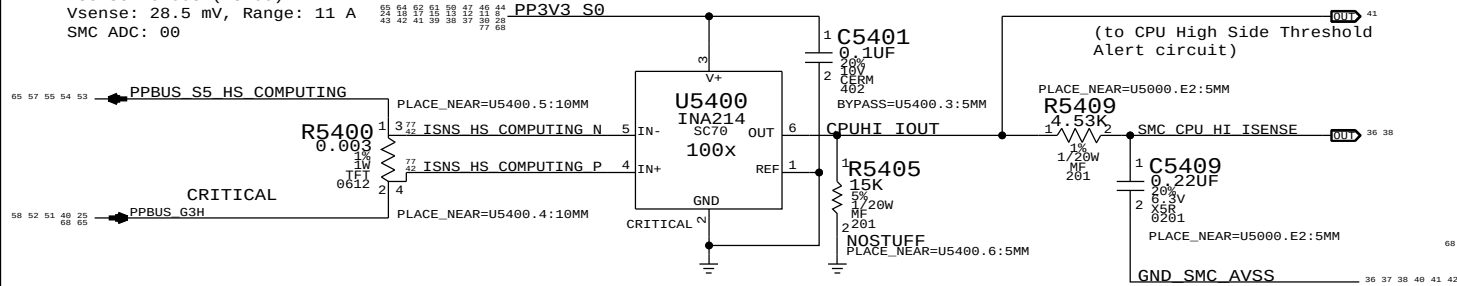


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SMBus Connections			
	Apple Inc.	DRAWING NUMBER	SIZE
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PAGE		53 OF 120	
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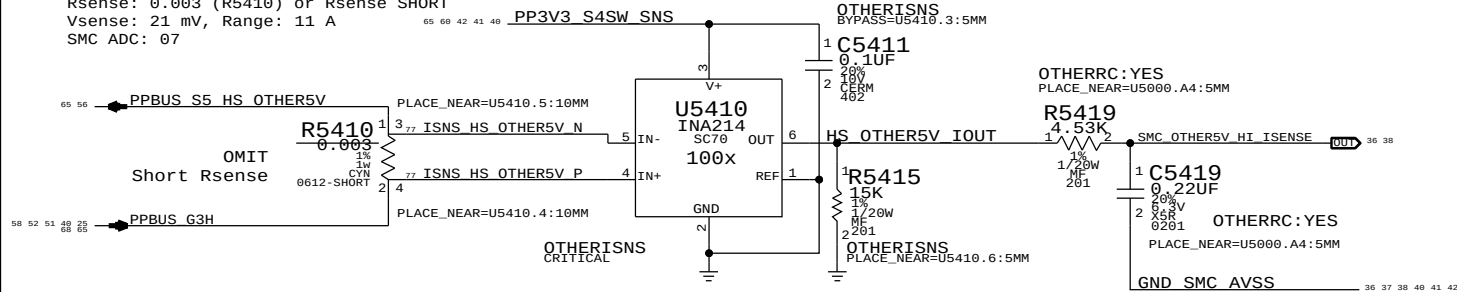
CPU High Side Current Sense (IC0R)

Gain: 100x, EDP: 9.5 A
Rsense: 0.003 (R5400)
Vsense: 28.5 mV, Range: 11 A
SMC ADC: 00



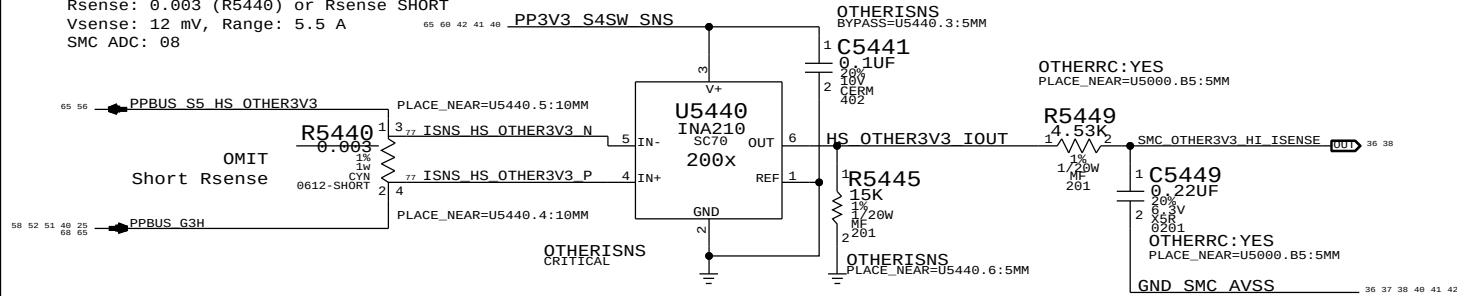
OTHER 5V High Side Current Sense (IO5R)

Gain: 100x, EDP: 7 A
Rsense: 0.003 (R5410) or Rsense SHORT
Vsense: 21 mV, Range: 11 A
SMC ADC: 07



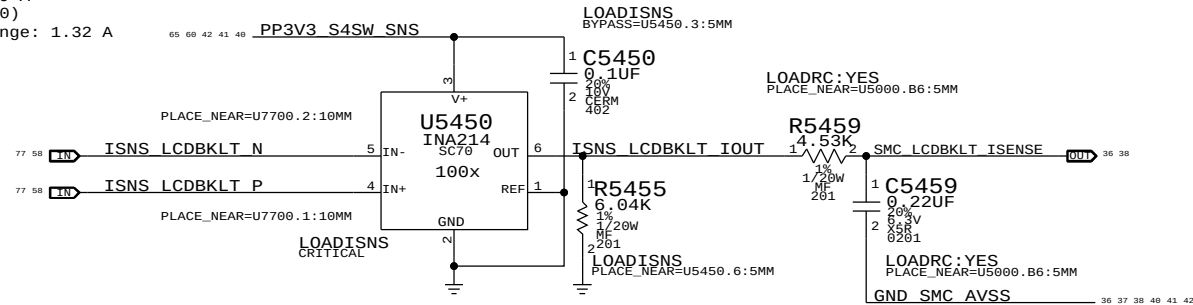
OTHER 3.3V High Side Current Sense (IO3R)

Gain: 200x, EDP: 5 A
Rsense: 0.003 (R5440) or Rsense SHORT
Vsense: 12 mV, Range: 5.5 A
SMC ADC: 08



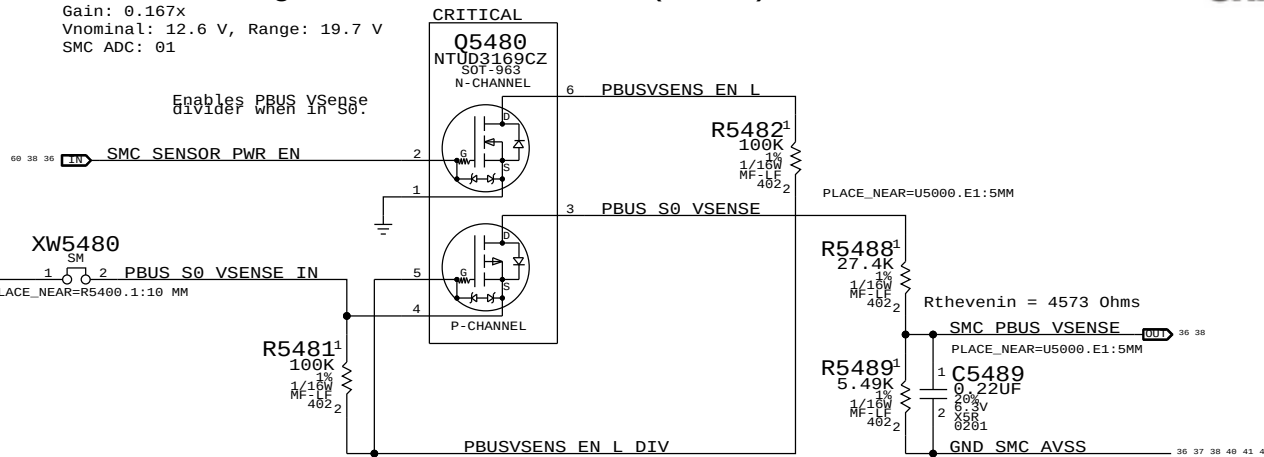
LCD Backlight Current Sense (IBLC)

Gain: 100x, EDP: 0.9 A
Rsense: 0.025 (R7700)
Vsense: 22.5 mV, Range: 1.32 A
SMC AD: 10



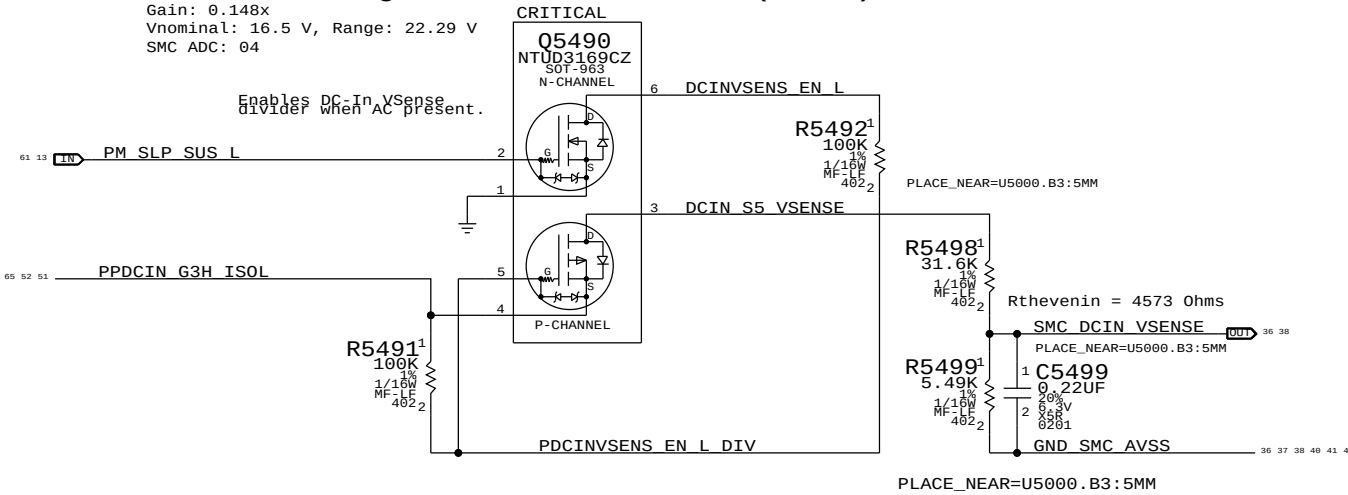
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x
Vnominal: 12.6 V, Range: 19.7 V
SMC ADC: 01



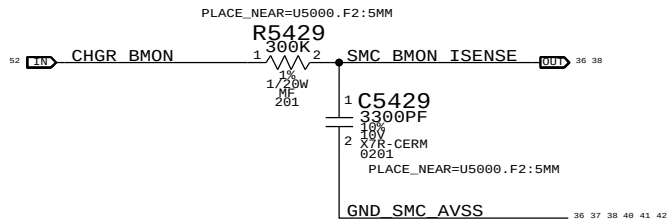
DC In Voltage Sense & Enable (VD0R)

Gain: 0.148x
Vnominal: 16.5 V, Range: 22.29 V
SMC ADC: 04



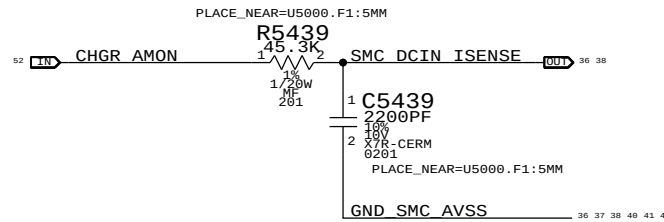
Charger (BMON) Current Sense (IPBR)

Charger Gain: 36x, EDP: 8 A
Rsense: 0.005 (R7150)
SMC ADC: 02



DC-IN (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A
Rsense: 0.020 (R7120)
SMC ADC: 03



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5419, C5449		OTHERRC: NO
117S0008	1	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5459		LOADRC: NO

SYNC_MASTER=J44		SYNC_DATE=08/12/2013	
PAGE TITLE			
Power Sensors: High Side			
		DRAWING NUMBER	SIZE
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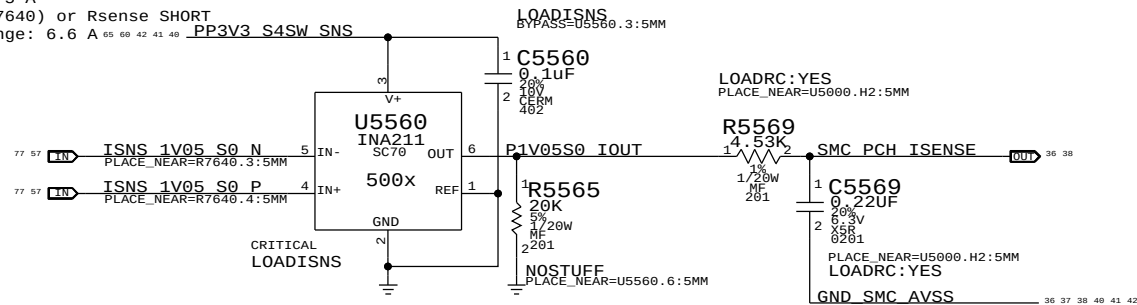
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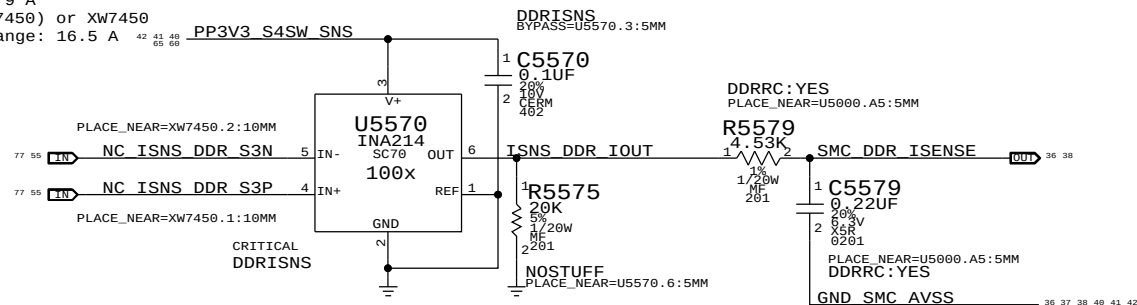
PCH 1.05V Current Sense (IC1C)

Gain: 500x, EDP: 5 A
Rsense: 0.001 (R7640) or Rsense SHORT
Vsense: 5 mV, Range: 6.6 A 65 60 42 41 40 PP3V3 S4SW SNS
SMC ADC: 19



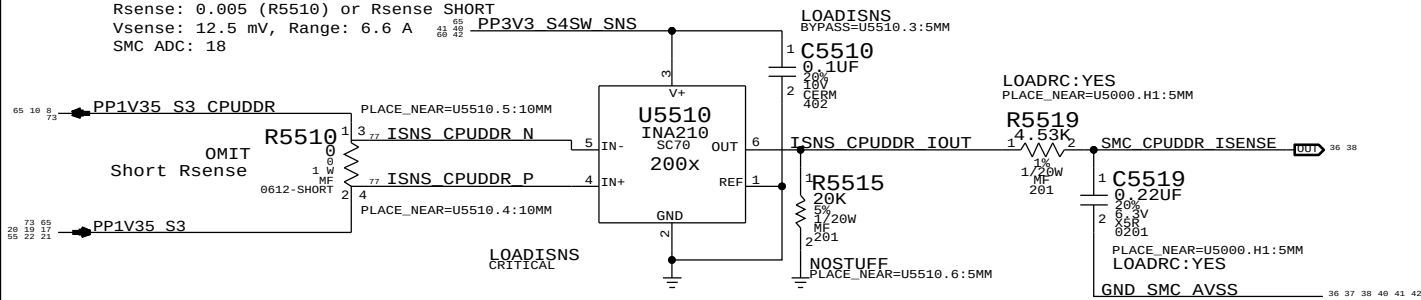
DDR 1.35V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 100x, EDP: 9 A
Rsense: 0.002 (R7450) or XW7450
Vsense: 21 mV, Range: 16.5 A 42 41 40 65 60 PP3V3 S4SW SNS
SMC ADC: 09



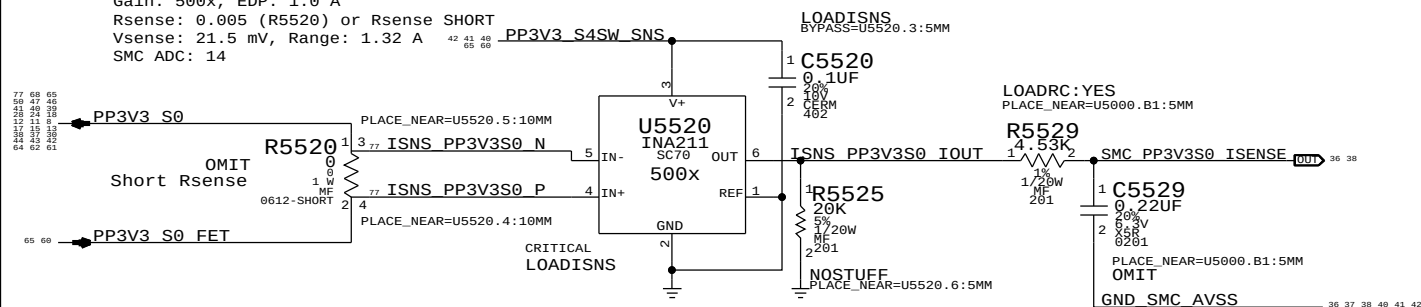
CPU DDR 1.35V S3 (CPU Only) Current Sense (IM1C)

Gain: 200x, EDP: 2.5 A
Rsense: 0.005 (R5510) or Rsense SHORT
Vsense: 12.5 mV, Range: 6.6 A 41 40 65 60 PP3V3 S4SW SNS
SMC ADC: 18



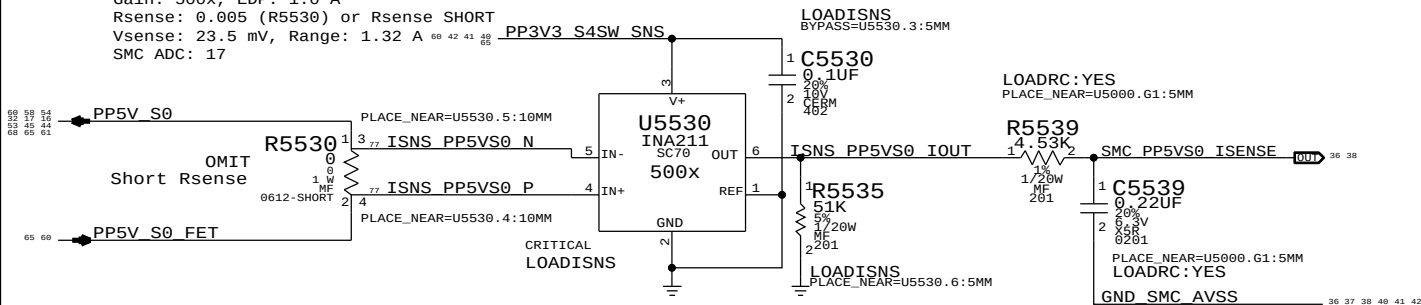
3.3V S0 Rail Current Sense (IR3C)

Gain: 500x, EDP: 1.0 A
Rsense: 0.005 (R5520) or Rsense SHORT
Vsense: 21.5 mV, Range: 1.32 A 42 41 40 65 60 PP3V3 S4SW SNS
SMC ADC: 14



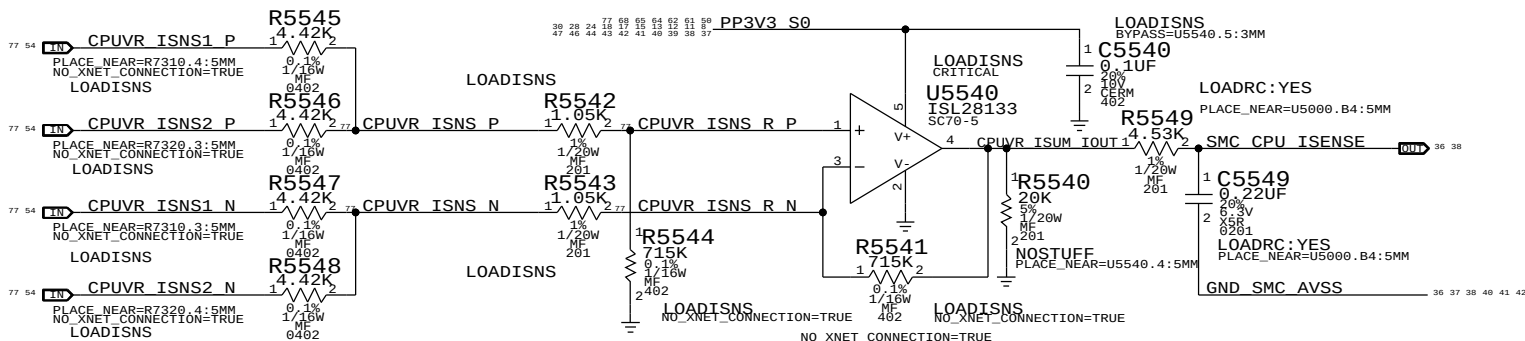
5V S0 Rail Current Sense (IR5C)

Gain: 500x, EDP: 1.0 A
Rsense: 0.005 (R5530) or Rsense SHORT
Vsense: 23.5 mV, Range: 1.32 A 60 42 41 40 PP3V3 S4SW SNS
SMC ADC: 17



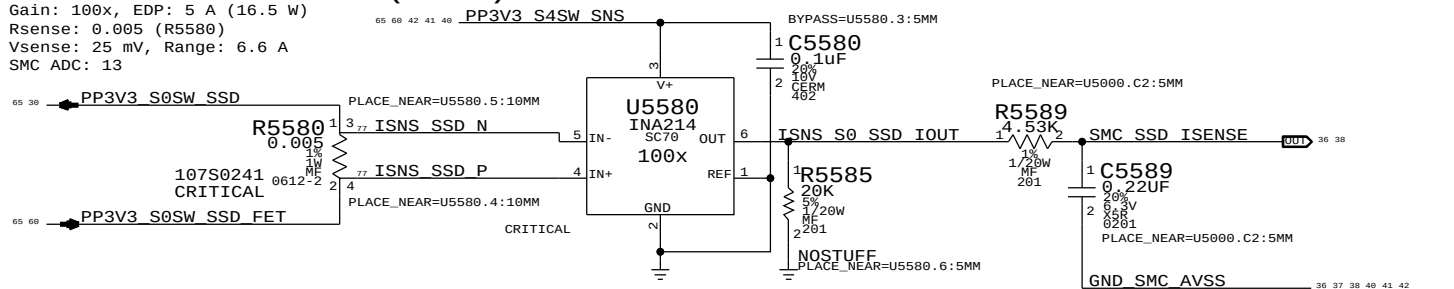
CPU Fixed Current Sense (IC0C)

Gain: 219.33x, EDP: 40 A
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375
Vsense: 15 mV, Range: 40.12 A
SMC ADC: 06



SSD Current Sense (ISDC)

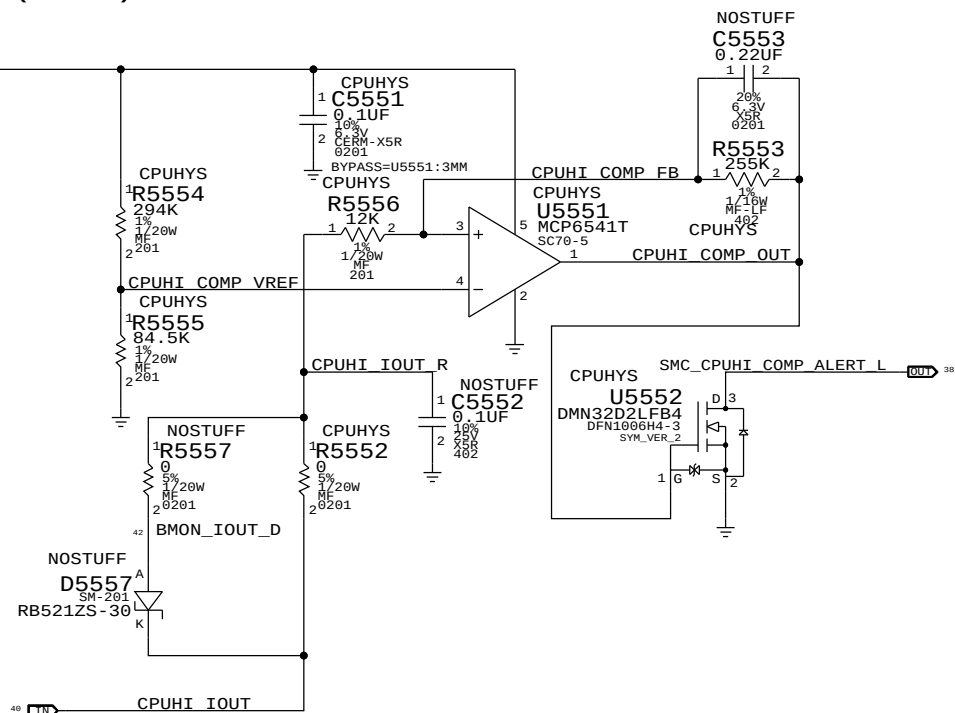
Gain: 100x, EDP: 5 A (16.5 W)
Rsense: 0.005 (R5580)
Vsense: 25 mV, Range: 6.6 A
SMC ADC: 13



CPU High Side Current (IC0R) Threshold Alert

Gain: 100x
Rsense: 0.003 (R5400)

Trip Target on CPU High current: 2.5 A
Hysteresis Circuit:
Vref = 0.737 V
Vth = 0.616 V -> 2.054 A on CPU High current
Vtl = 0.771 V -> 2.571 A on CPU High current
Hysteresis Margin = 0.518 A

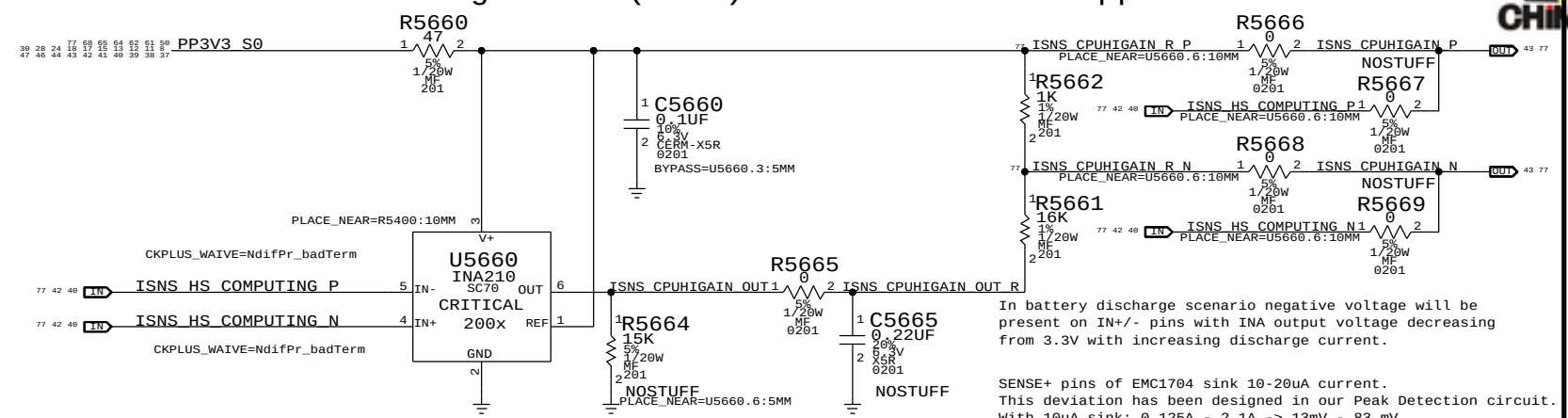


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5569, C5519		LOADRC: NO
117S0008	3	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5529, C5539, C5549		LOADRC: NO
117S0008	1	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5579		DDRRRC: NO

SYNC MASTER=J44		SYNC DATE=08/12/2013	
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Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
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		PAGE	55 OF 120
		SHEET	41 OF 78
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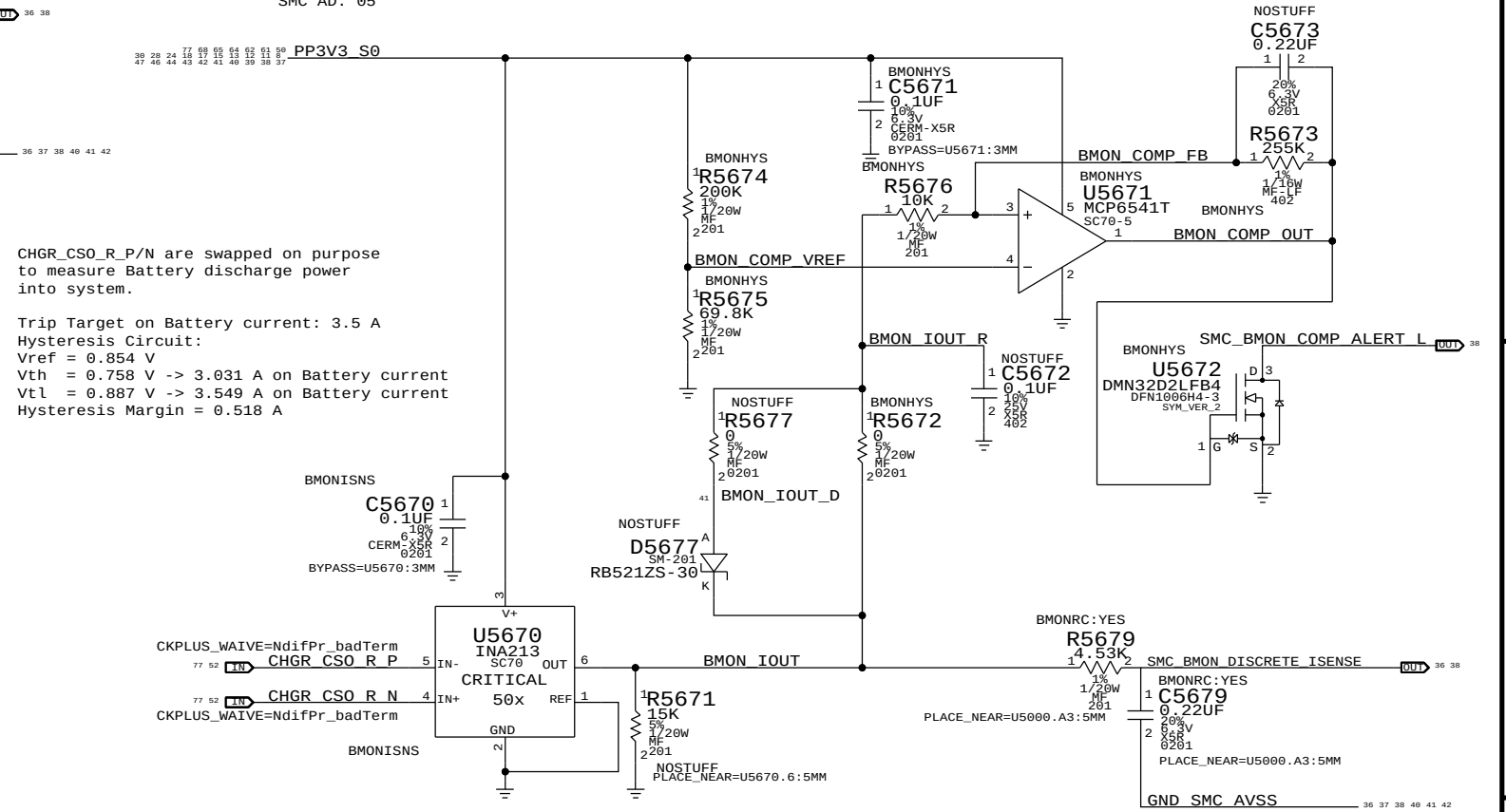


CPU High Side (IC0R) Peak Detection Support



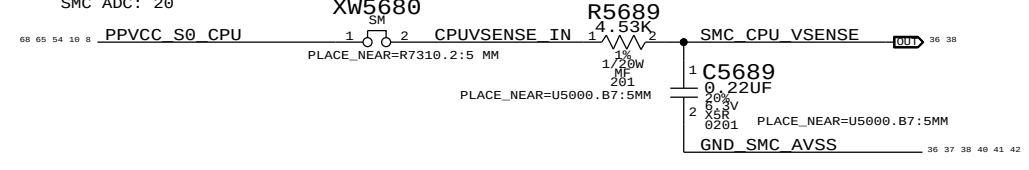
Battery BMON Discrete Current Sense (IP0R) & Threshold Alert

Gain: 50x. EDP: 8 A
Rsense: 0.005 (R7150)
Vsense: 50 mV, Range: 13.2 A
SMC AD: 05



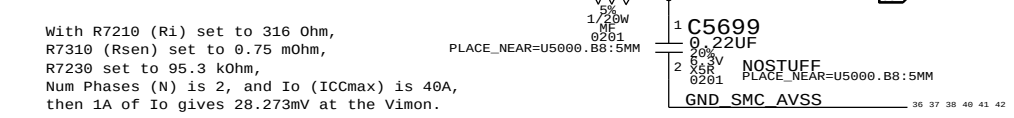
CPU Core Voltage Sense (VC0C)

SMC ADC: 20



CPU Core IMON Current Sense (IC2C)

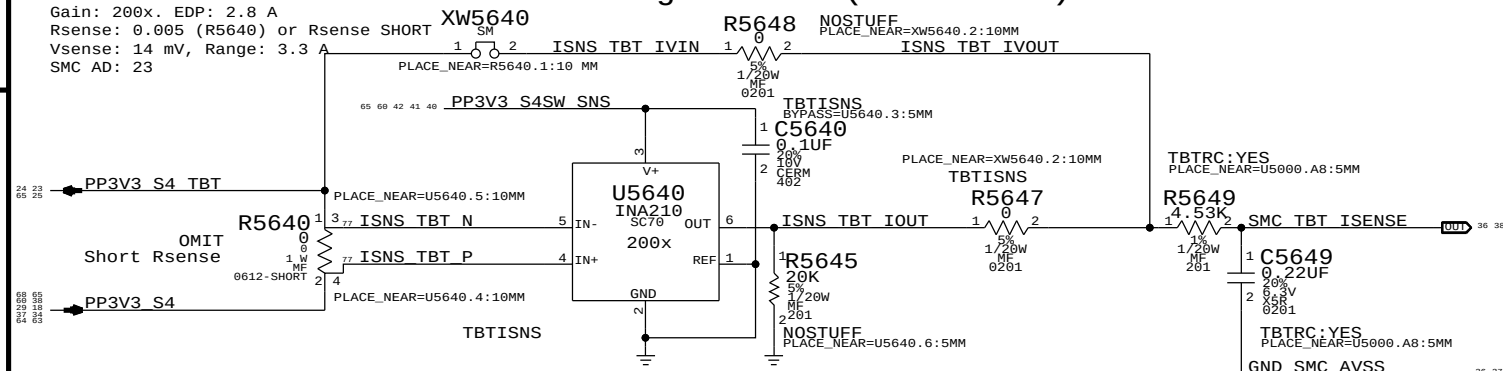
Gain: 1 A / 28.273 mV, Range: 40 A.
SMC ADC: 22
CPUVR IMON



With R7210 (Ri) set to 316 Ohm,
R7310 (Rsen) set to 0.75 mOhm,
R7230 set to 95.3 kOhm,
Num Phases (N) is 2, and Io (ICmax) is 40A,
then 1A of Io gives 28.273mV at the Vimon.

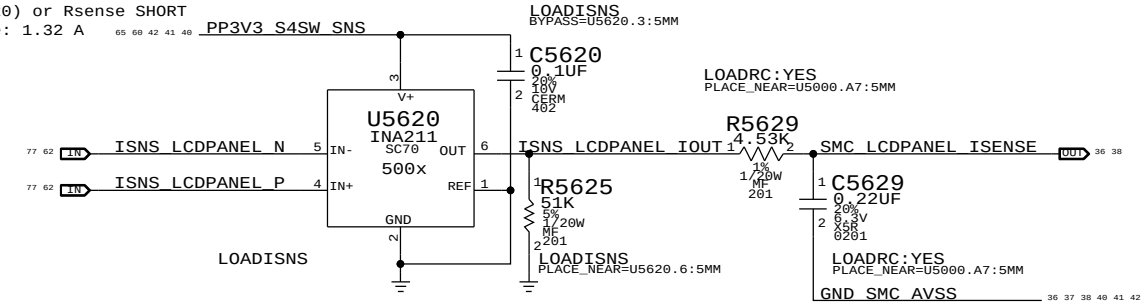
Thunderbolt TBT Current/Voltage Sense (IHSC/VHSC)

Gain: 200x. EDP: 2.8 A
Rsense: 0.005 (R5640) or Rsense SHORT
Vsense: 14 mV, Range: 3.3 A
SMC AD: 23



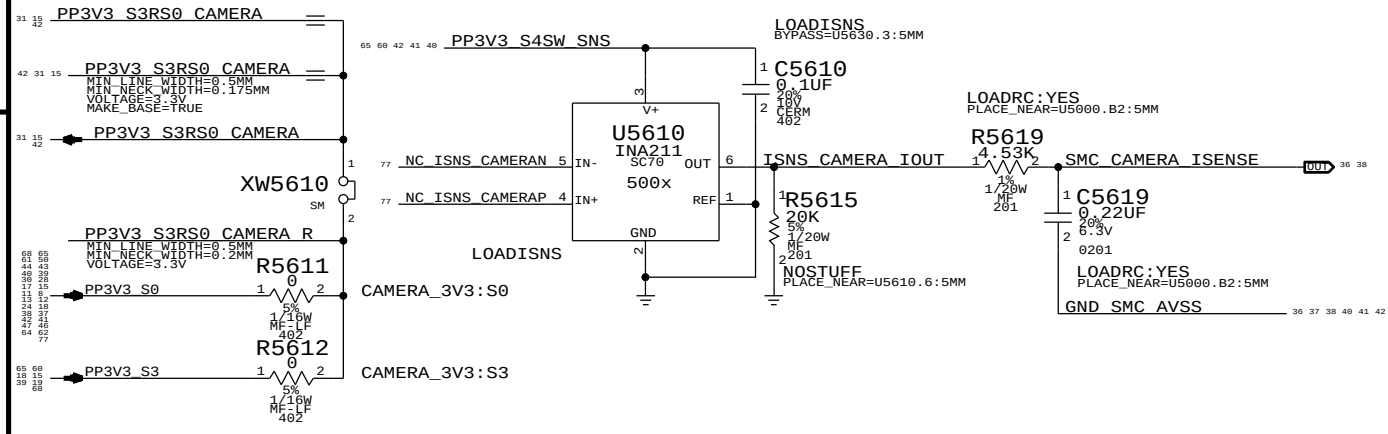
LCD Panel Current Sense (ILDC)

Gain: 500x. EDP: 1 A
RSENSE: 0.005 (R8320) or Rsense SHORT
Vsense: 5 mV, Range: 1.32 A
SMC AD: 21



Camera (S2 Controller) Current Sense (ICMC)

Gain: 500x. EDP: 0.82 A
Rsense: 0.005 (R5610) or XW5610
Vsense: 4.1 mV, Range: 1.32 A
SMC AD: 15



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5619,C5629,C5649		
117S0008	1	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5679		

SYNC MASTER=144

SYNC DATE=08/12/2013

Power Sensors: Extended

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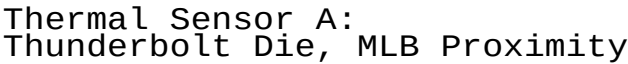
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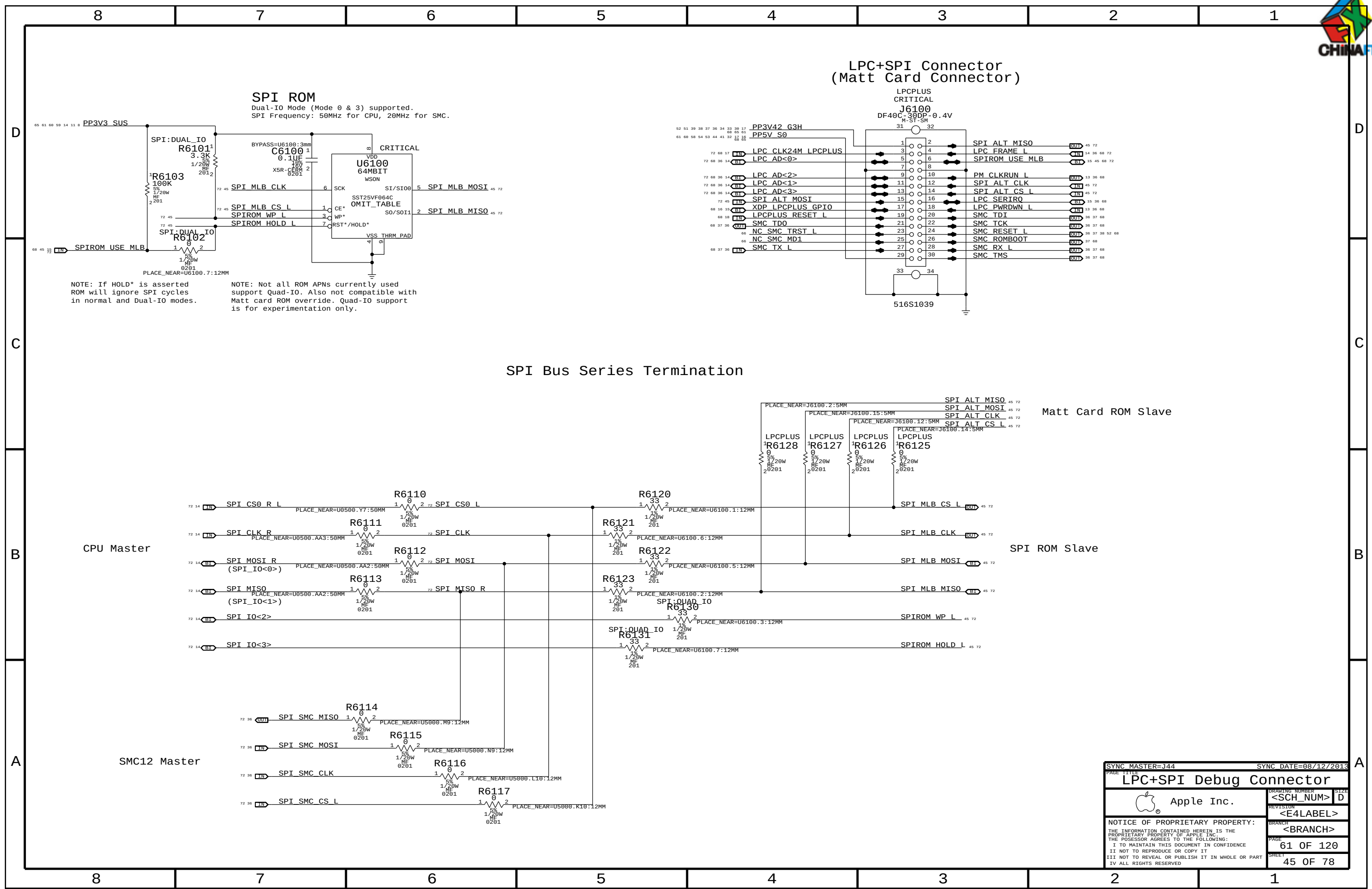


Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Airflow, Fin Stack Proximity

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Fan			
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		PAGE 60 OF 120	
		SHEET 44 OF 78	

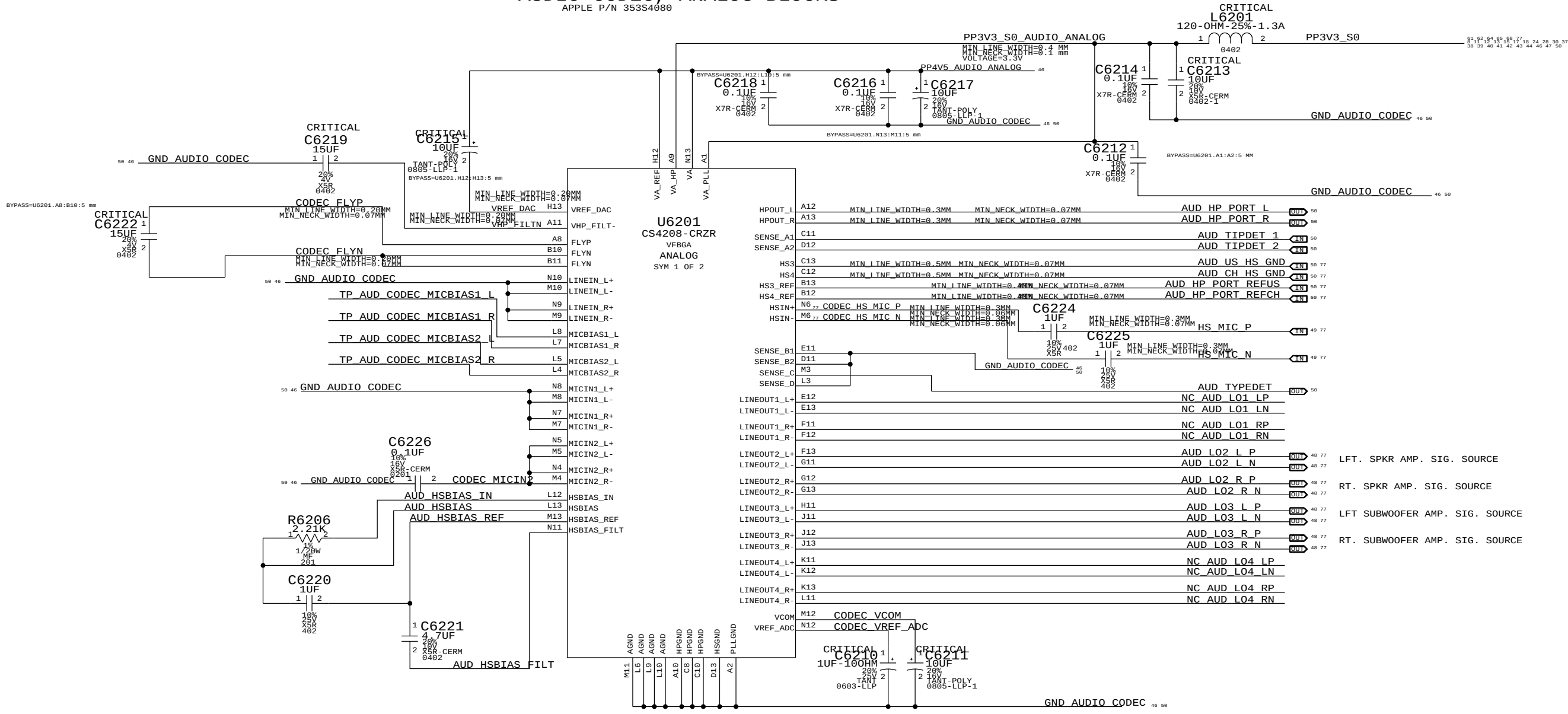


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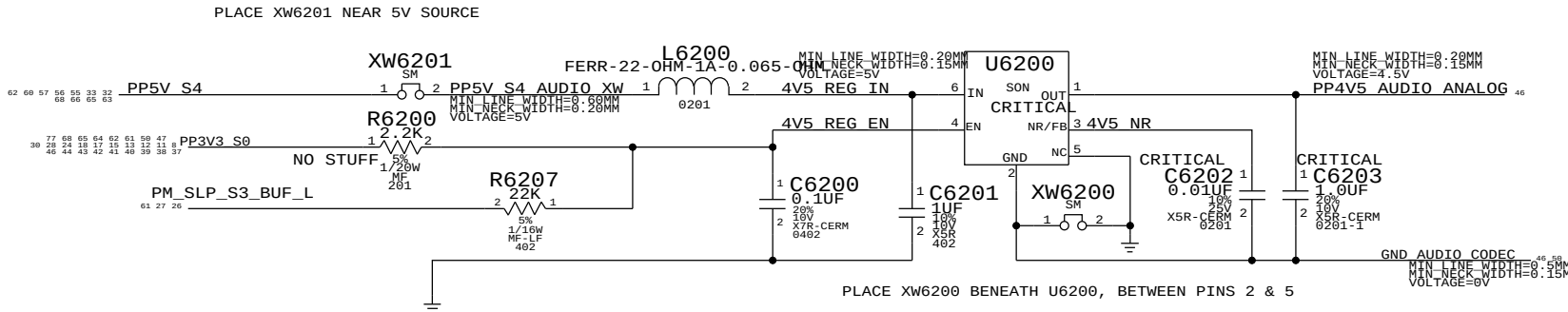
AUDIO CODEC, ANALOG BLOCKS

APPLE P/N 353S4680



4.5V POWER SUPPLY FOR CODEC

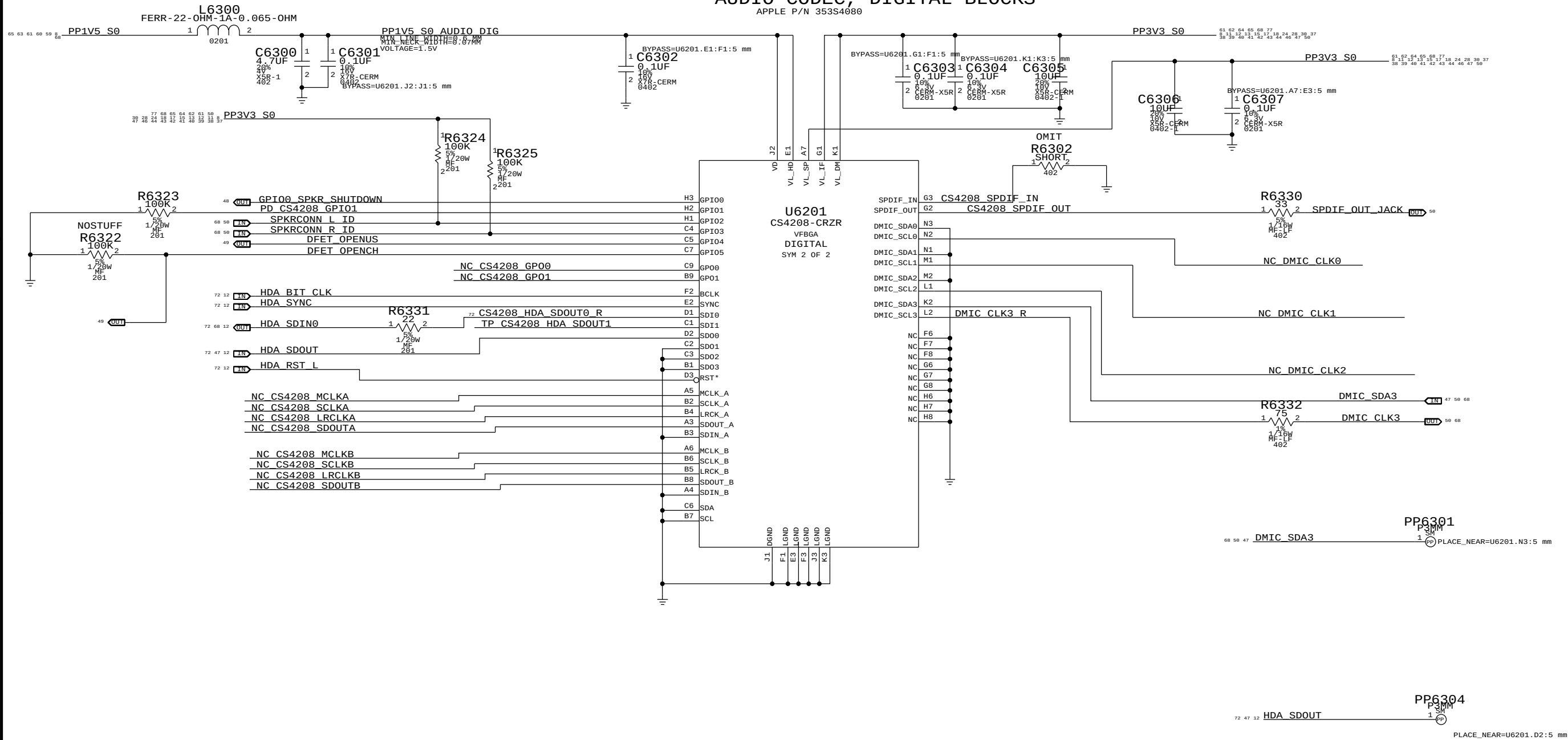
APPLE P/N 353S2456



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AUDIO CODEC, DIGITAL BLOCKS
APPLE P/N 353S4080



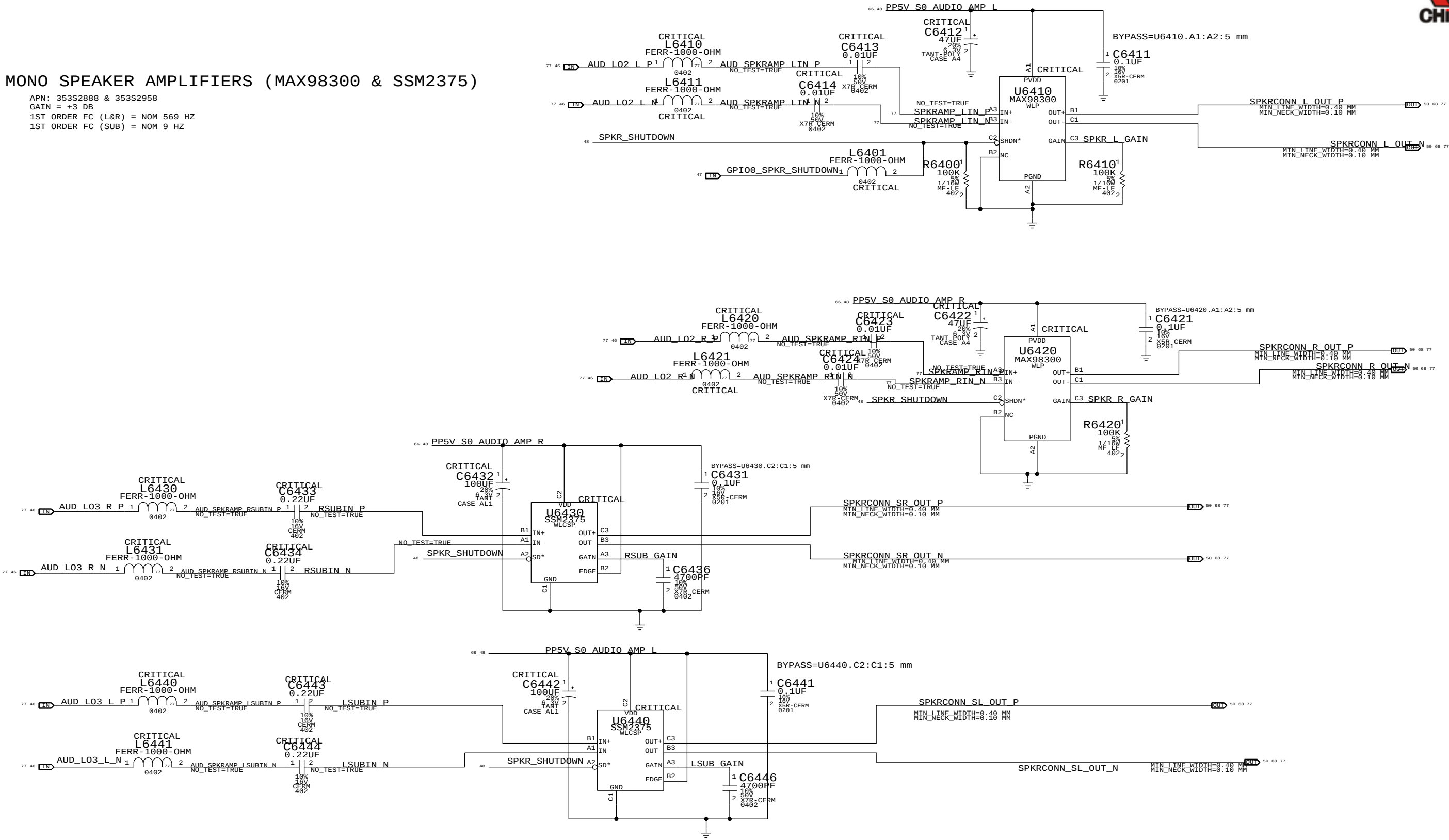
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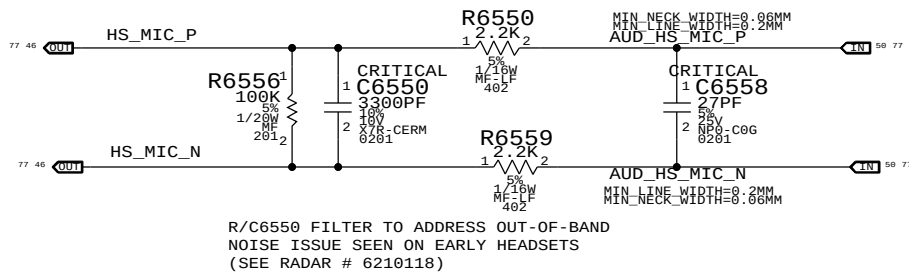
4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)

APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ

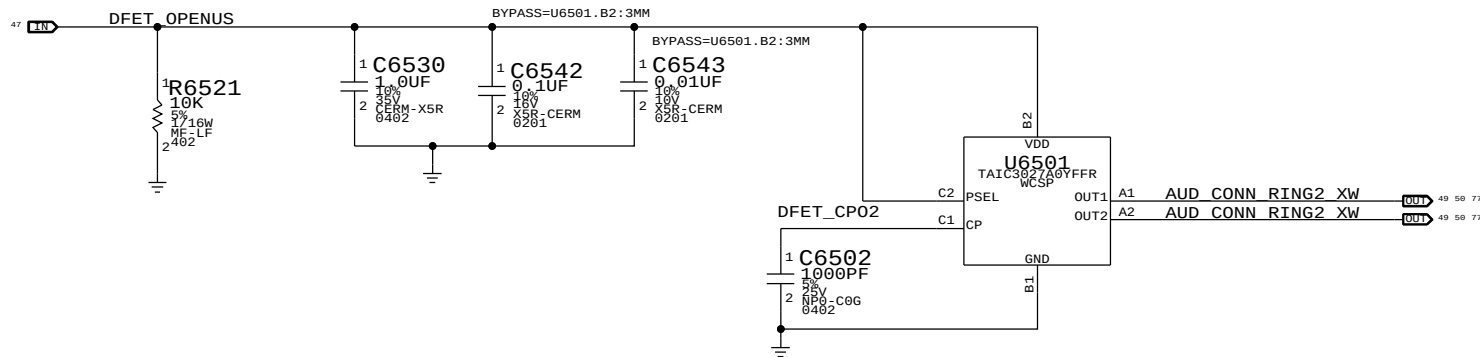
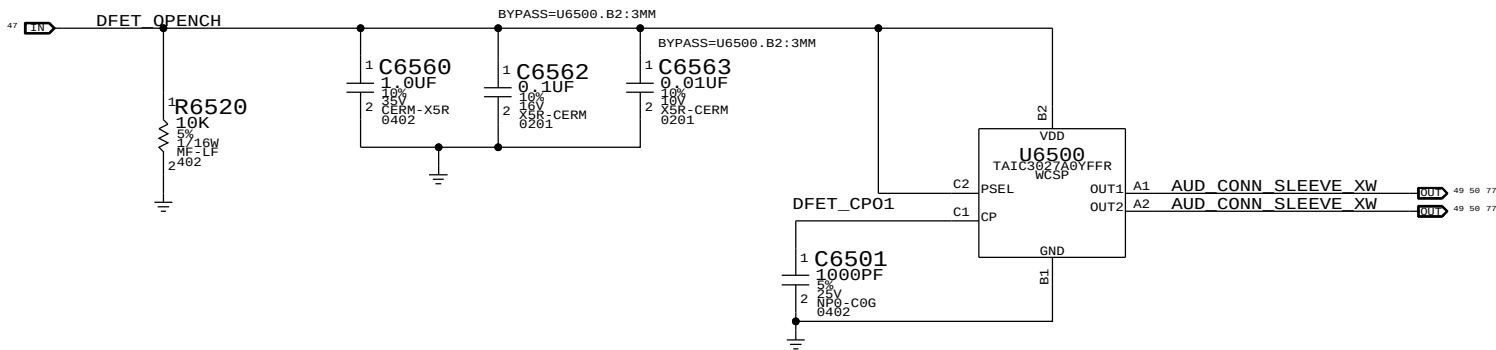


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R/C6550 FILTER TO ADDRESS OUT-OF-BAND
NOISE ISSUE SEEN ON EARLY HEADSETS
(SEE RADAR # 6210118)



SYNC_MASTER=J44		SYNC_DATE=08/12/2013	
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 Apple Inc.		DRAWING NUMBER	SIZE
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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL
HP/HS OUT	0X02 (2)	0X02 (2)	0X10 (16)	N/A
TWEETERS	0X03 (3)	0X03 (3)	0X12 (18)	CODEC GPIO0
SUB	0X04 (4)	0X04 (4)	0X13 (19)	CODEC GPIO0
SPDIF OUT	N/A	0X0E (14)	0X21 (33)	N/A

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF
DMIC 1	0X09 (9)	0X1C (28)	3.3V
DMIC 2	0X09 (9)	0X1C (28)	3.3V
HEADSET MIC	0X07 (7)	0X18 (24)	2.7V

OTHER CODEC GPIO LINES

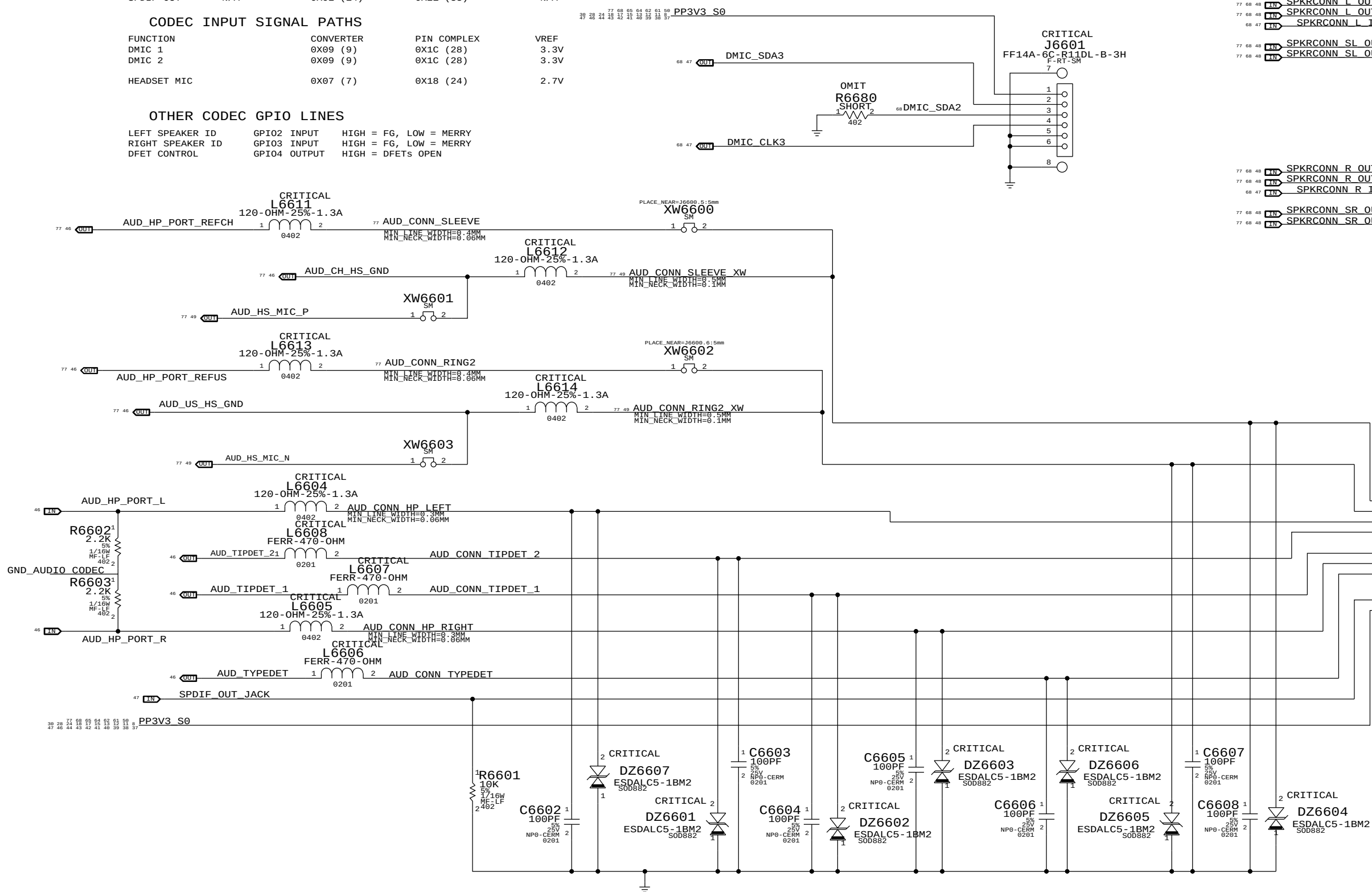
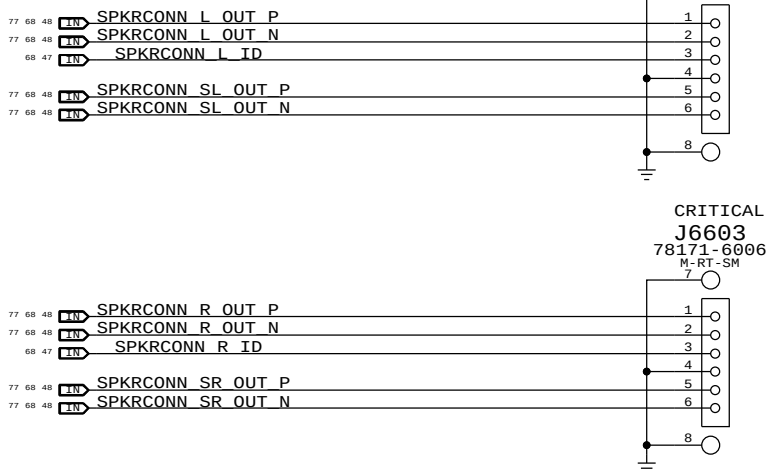
LEFT SPEAKER ID	GPIO2 INPUT	HIGH = FG, LOW = MERRY
RIGHT SPEAKER ID	GPIO3 INPUT	HIGH = FG, LOW = MERRY
DFET CONTROL	GPIO4 OUTPUT	HIGH = DFETS OPEN

SPEAKER CONNECTOR

HP=80HZ
APN: 518S0672

2-MIC CONNECTOR

APN: 518S0818



SYNC MASTER=J44 SYNC DATE=08/12/2013

AUDIO: JACK TRANSLATORS

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MagSafe DC Power Jack

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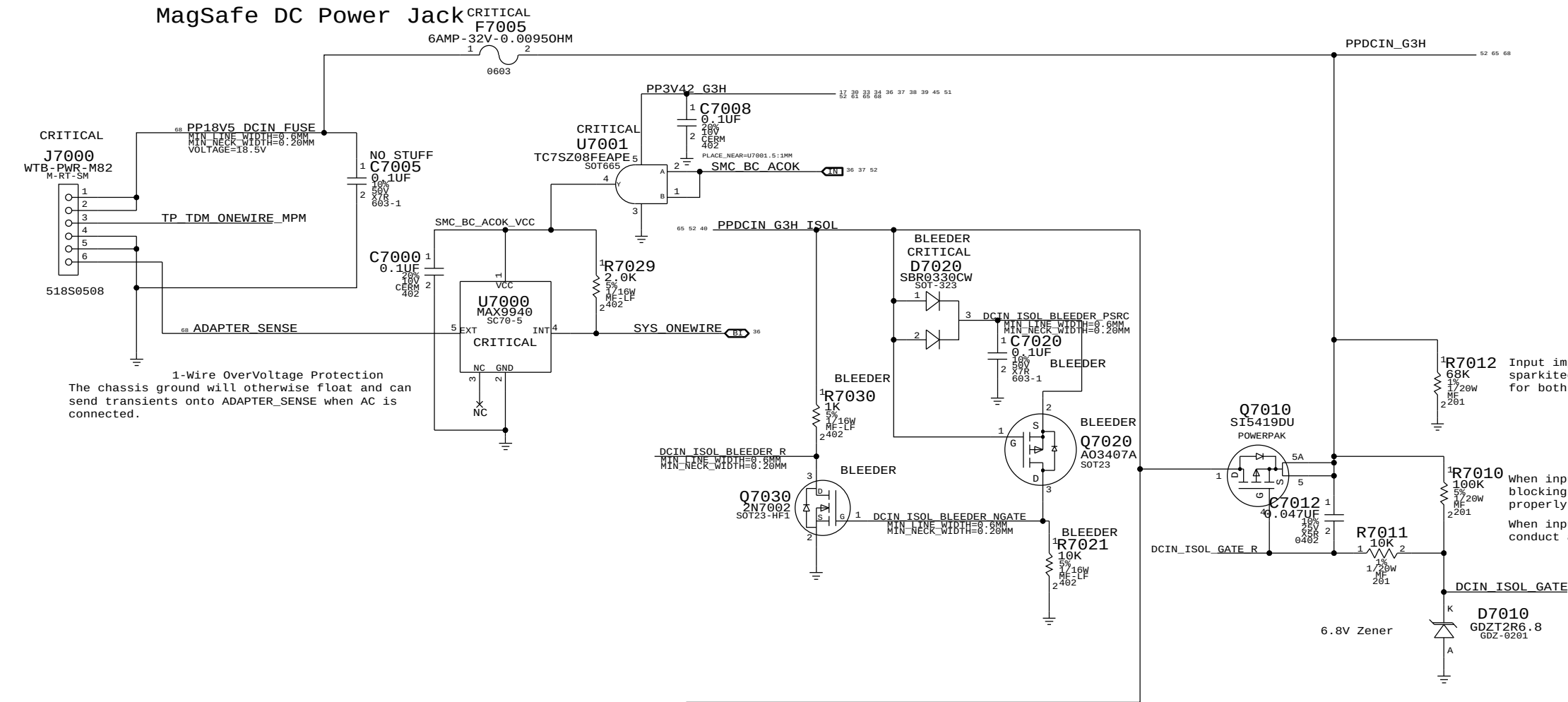
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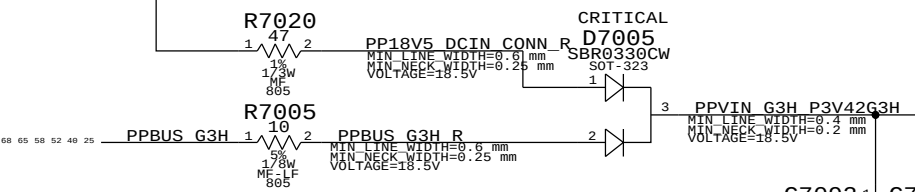
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Input impedance of 68K meets sparkitecture requirements for both MPM4 and MPM5.

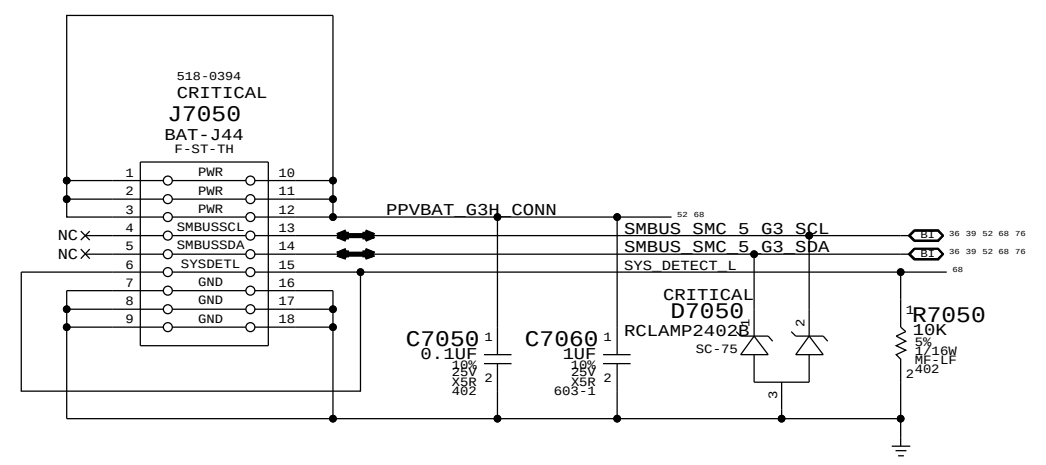
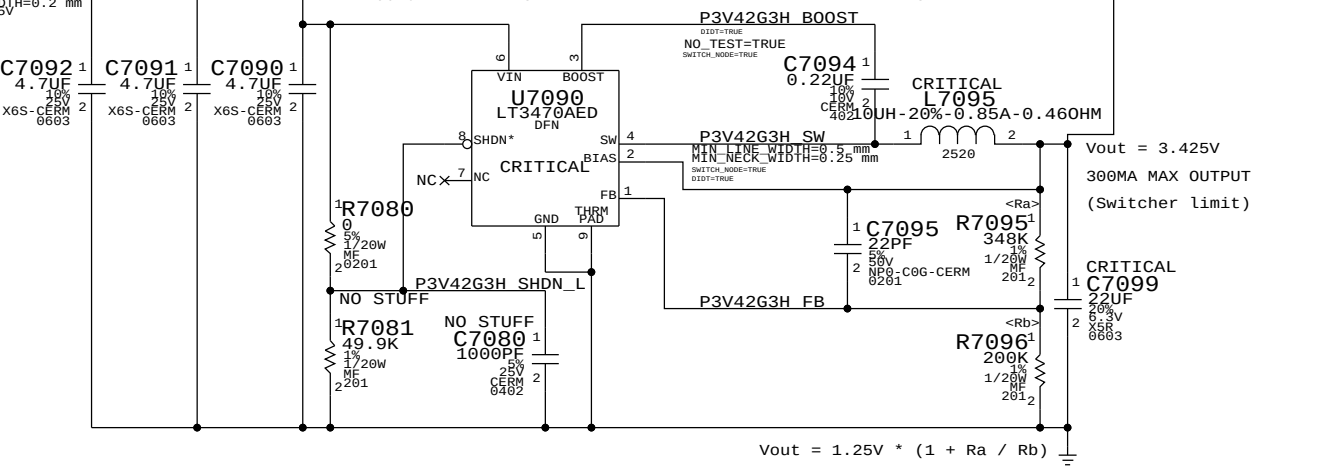
When input voltage is 2V the FET will be off blocking the leakage path and 22.1K can be properly detected.

When input voltage is at 16V+, FET will conduct and power charger and 3.42V reg

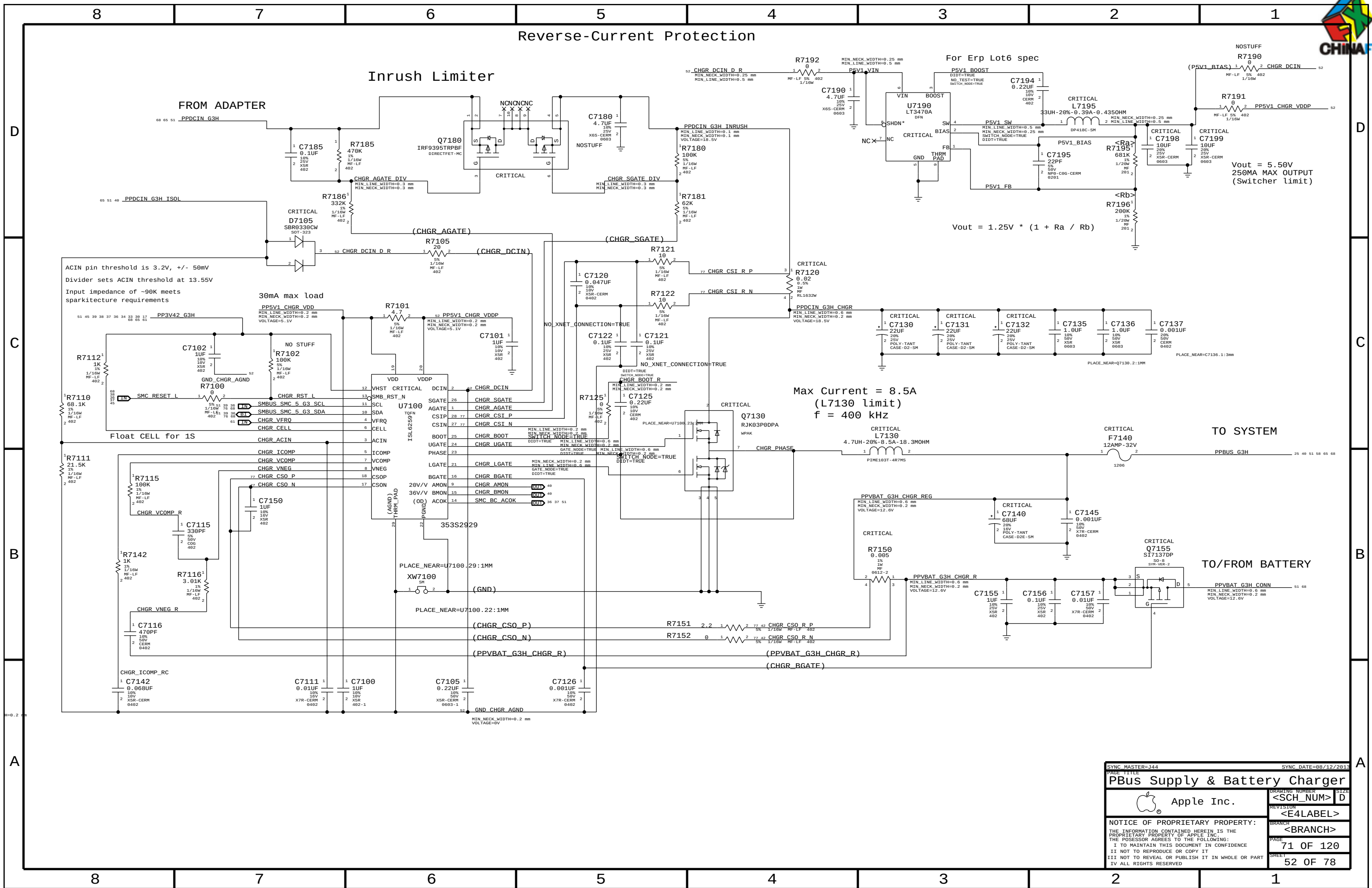


3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC Vref generator



DC-In & Battery Connectors	
Apple Inc.	<SCH_NUM> D
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C

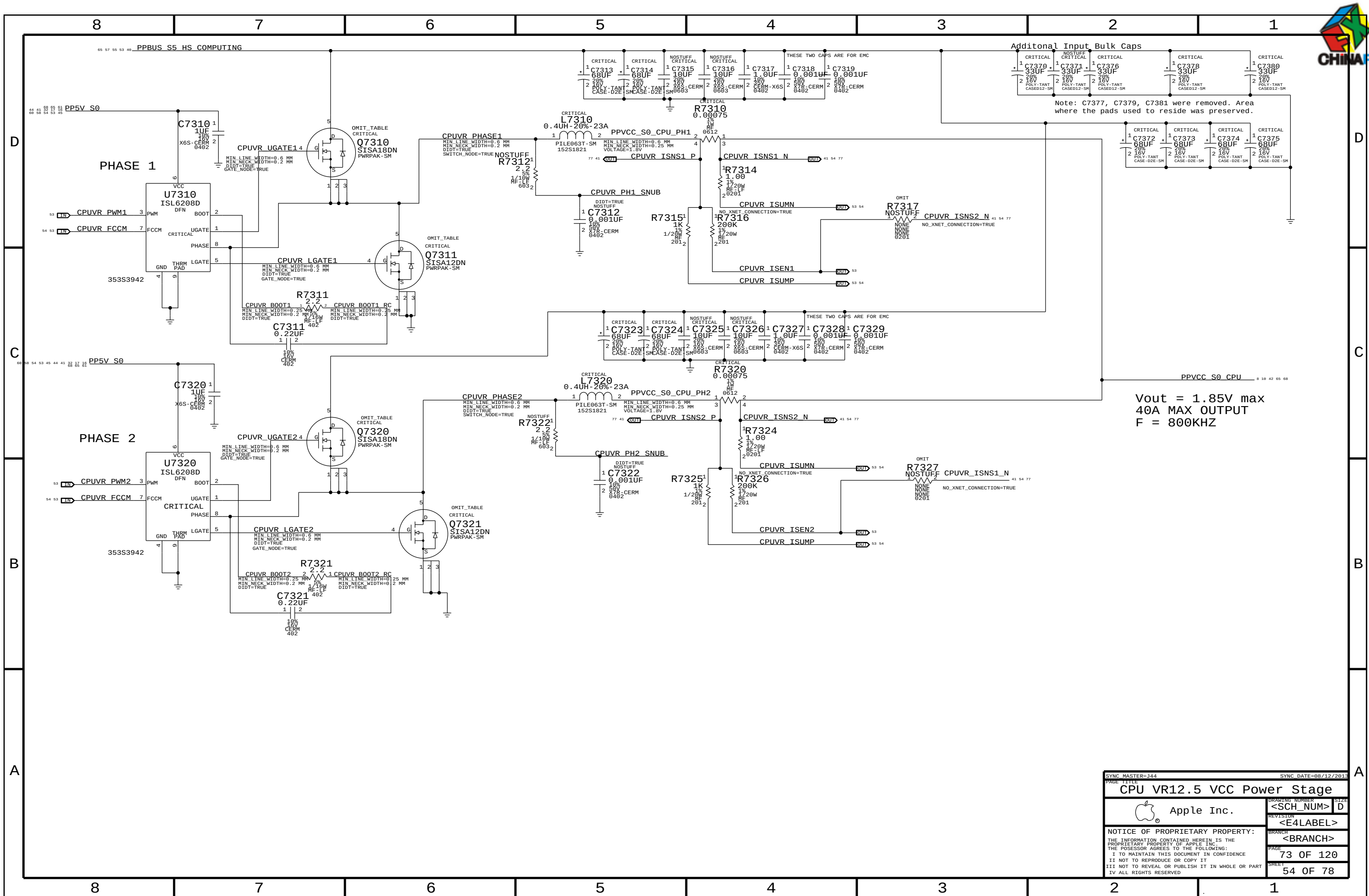
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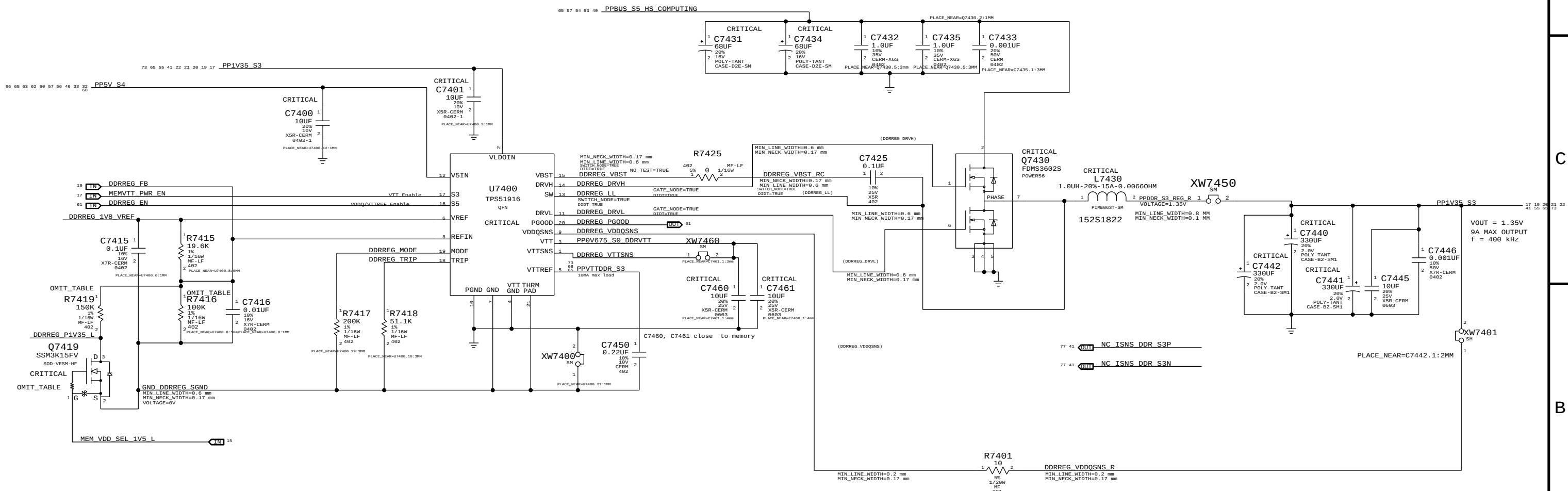
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CPU VR12.6 VCC Regulator IC			
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		PAGE 72 OF 120	
		SHEET 53 OF 78	

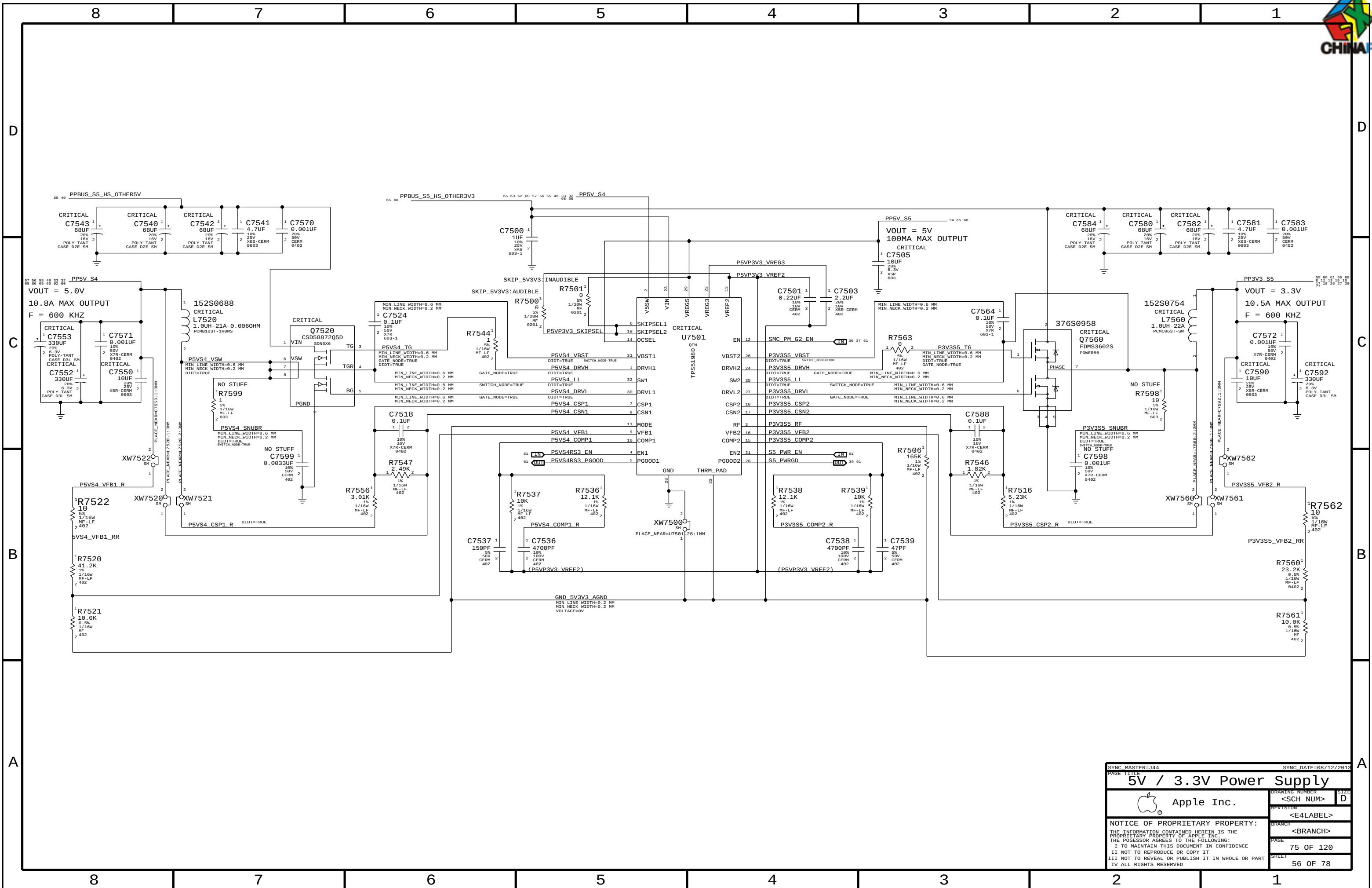


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CPU VR12.5 VCC Power Stage			
 Apple Inc.		DRAWING NUMBER	SIZE
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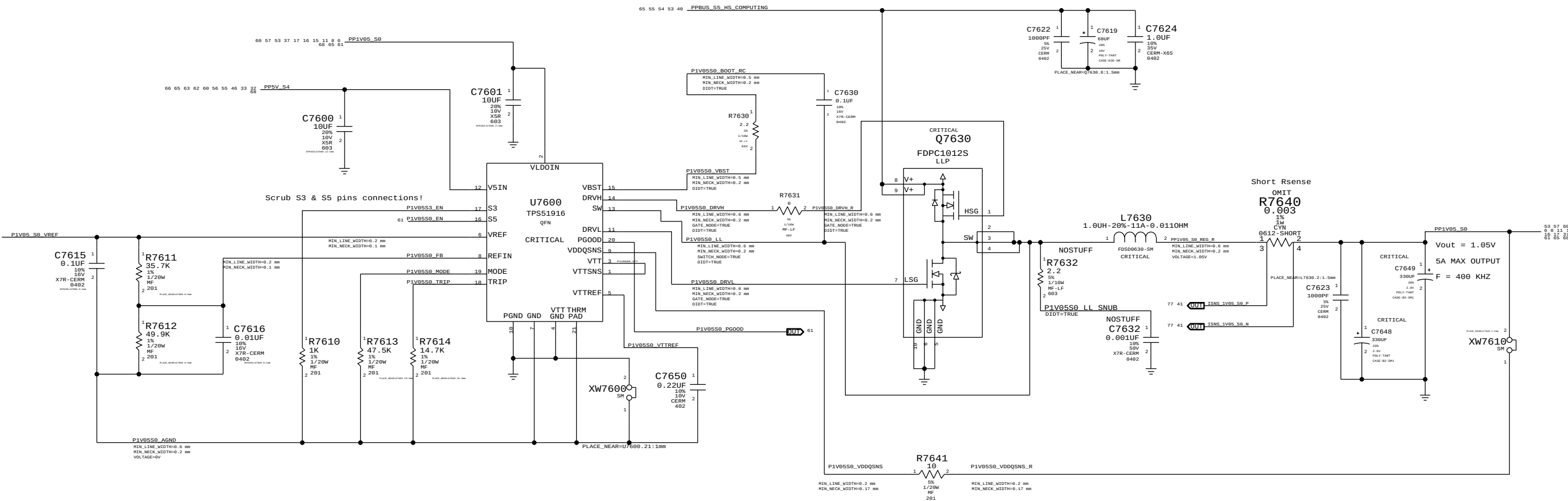
DDR3L (1V35 S3) REGULATOR



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0411	1	RES, MTL FILM, 1/16W, 100K, 1, 0402, SMD, LF	R7416	CRITICAL	PPDDR:1V5
114S0391	1	RES, MTL FILM, 1/16W, 60.4K, 1, 0402, SMD, LF	R7416	CRITICAL	PPDDR:1V35
376S0612	1	MOSFET, N-CH, 30V, 100MA, 7.00MM, SOT-723, HF	Q7419	CRITICAL	PPDDR:1V5
114S0428	1	RES, MTL FILM, 1/16W, 150k, 0402, SMD, LF	R7419		PPDDR:1V5



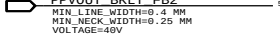
1.05V S0 Regulator



SYNC MASTER=344		SYNC DATE=08/12/2013	
PAGE TITLE			
1.05V S0 Power Supply			
		DRAWING NUMBER	SIZE
Apple Inc.		<SCH_NUM>	D
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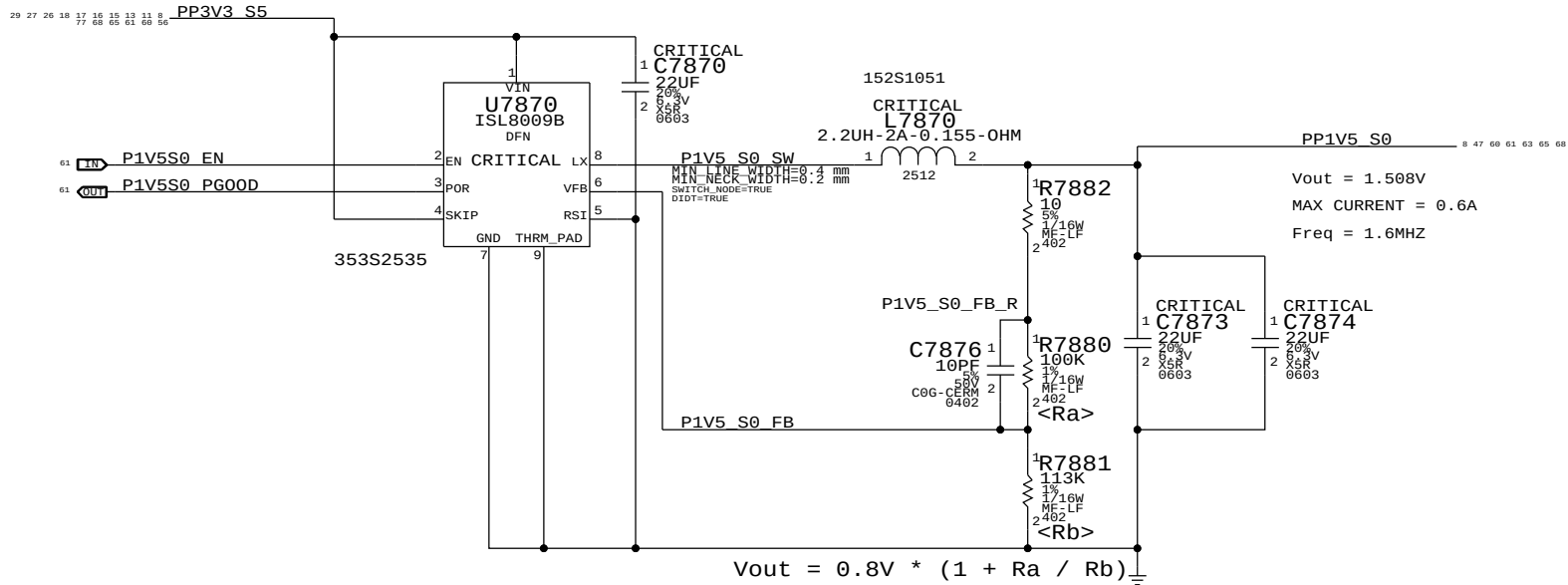
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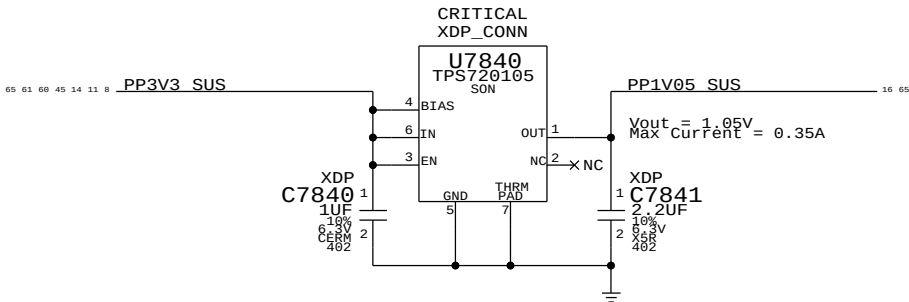


1.5V S0 Switcher



1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



SYNC MASTER=J44		SYNC DATE=08/12/2013	
PAGE TITLE			
Misc Power Supplies			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
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Loading specs per J41/43_PowerBudget_Riviera_rev0.99e

1.5V S0 Audio Switch (BYPASSED)

3.3V SUS Switch

3.3V S4 Switch

3.3V SSD Switch

1.05V PCH HSI0 Switch

3.3V S3 Switch

3.3V S0 Switch

3.3V Sensor Switch

5V S0 Switch

Part	TPS22924C
Type	Load Switch
R(on) @ 1.8V	19.6 mOhm Typ 21.8 mOhm Max
Current	2A Max

Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

Part	TPS22920
Type	Load Switch
R(on) @ 3.6V	5.5 MOHM TYP 8.8 MOHM MAX
Current	4A MAX

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 25C	7.8 mOhm Typ 8.5 mOhm Max
Current	5.3A Max

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

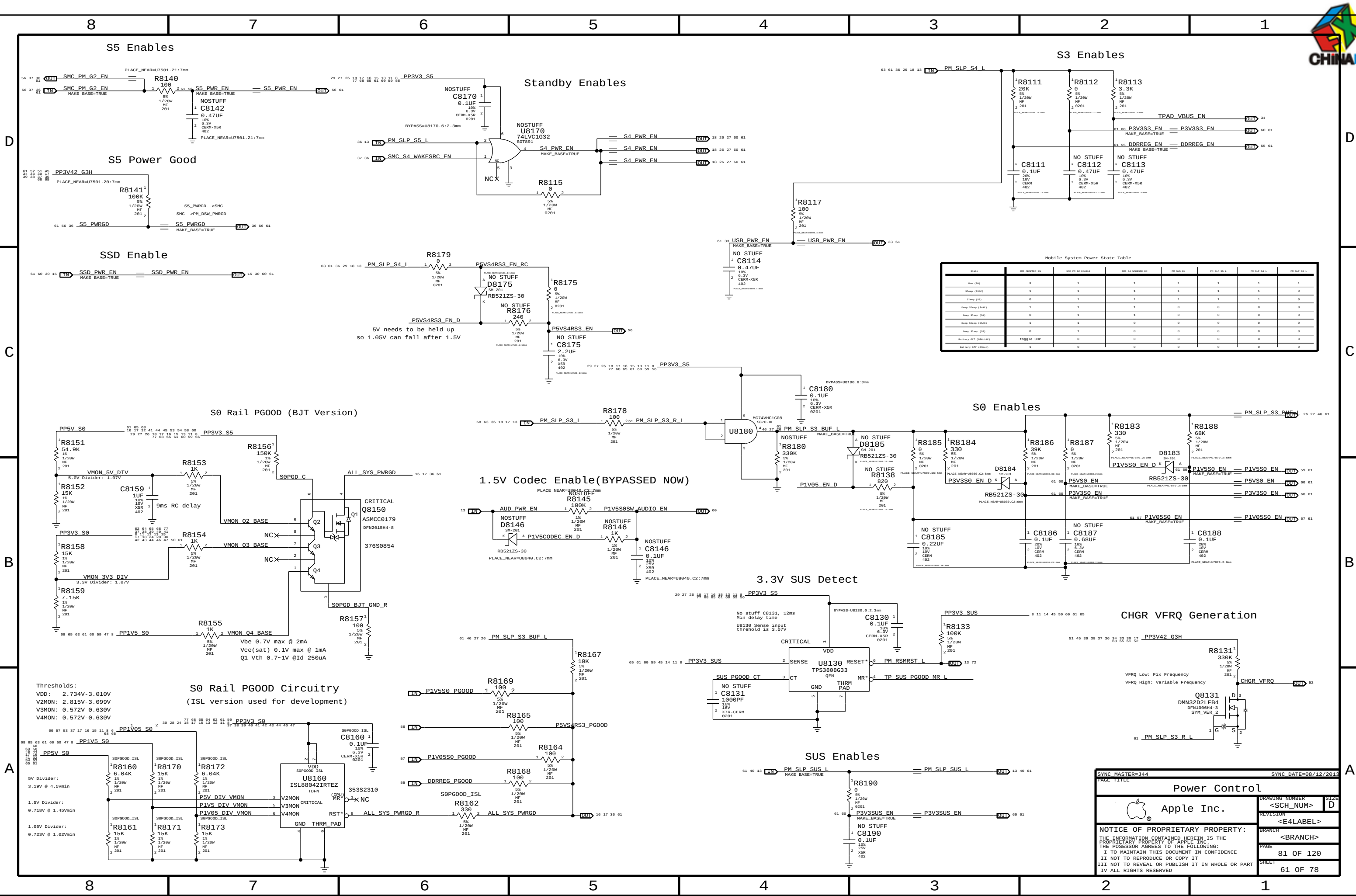
Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

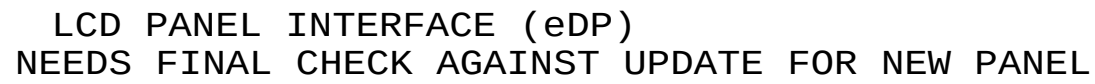
Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

Part	SLG5AP1443V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

REMOVED THE ANALOG POWER GATE AS SLG5AP1471 SHOULD BE AVAILABLE BY THEN

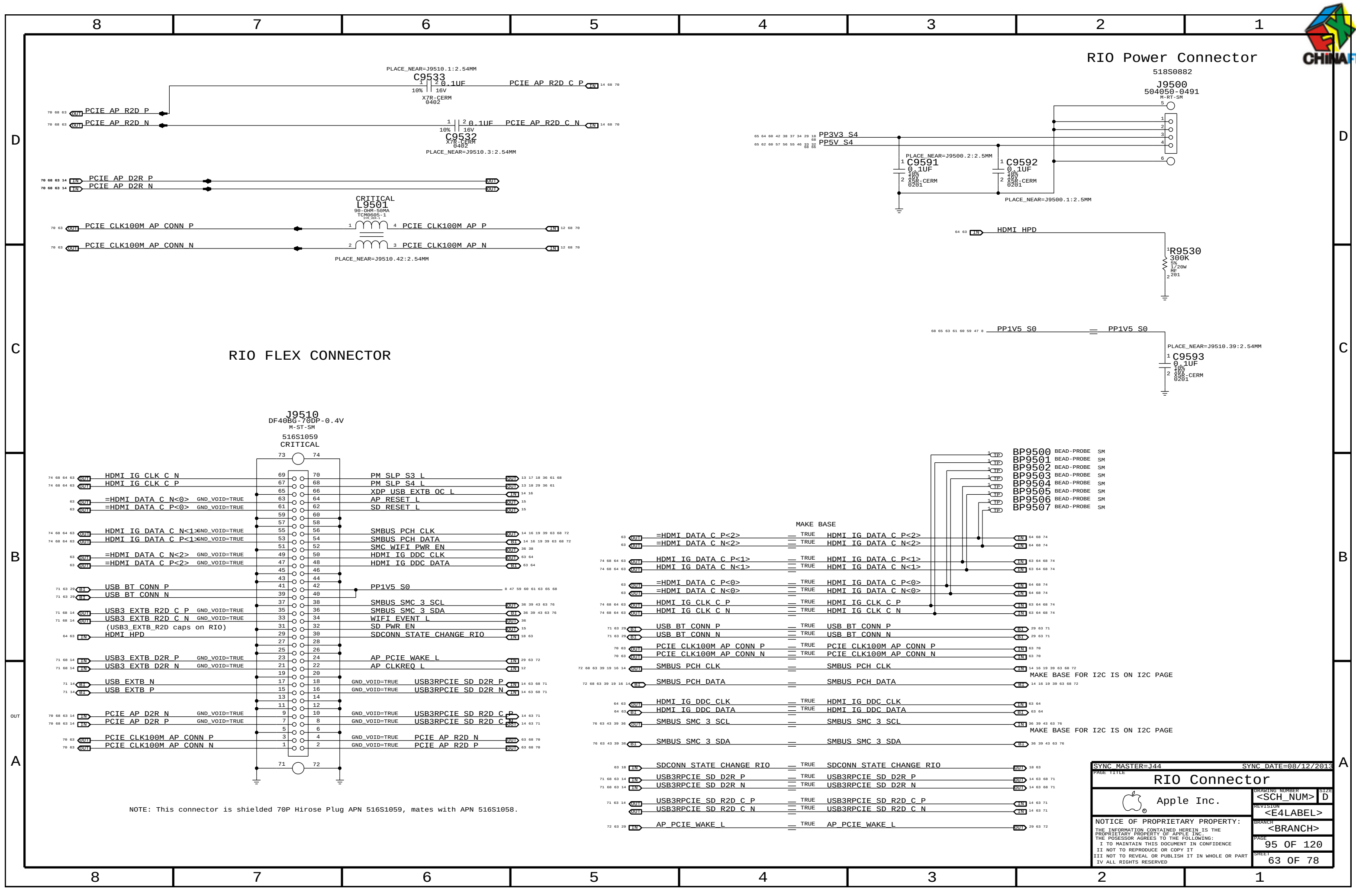
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Part	SLG5AP1443V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

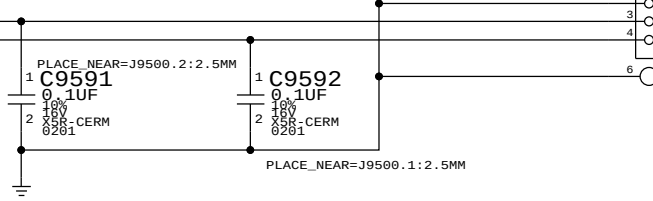
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eBP Display Connector			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	
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NOTE: This connector is shielded 70P Hirose Plug APN 516S1059, mates with APN 516S1058.

RIO Power Connector

518S0882
J9500
504050-0491
M-RT-SM



RIO FLEX CONNECTOR

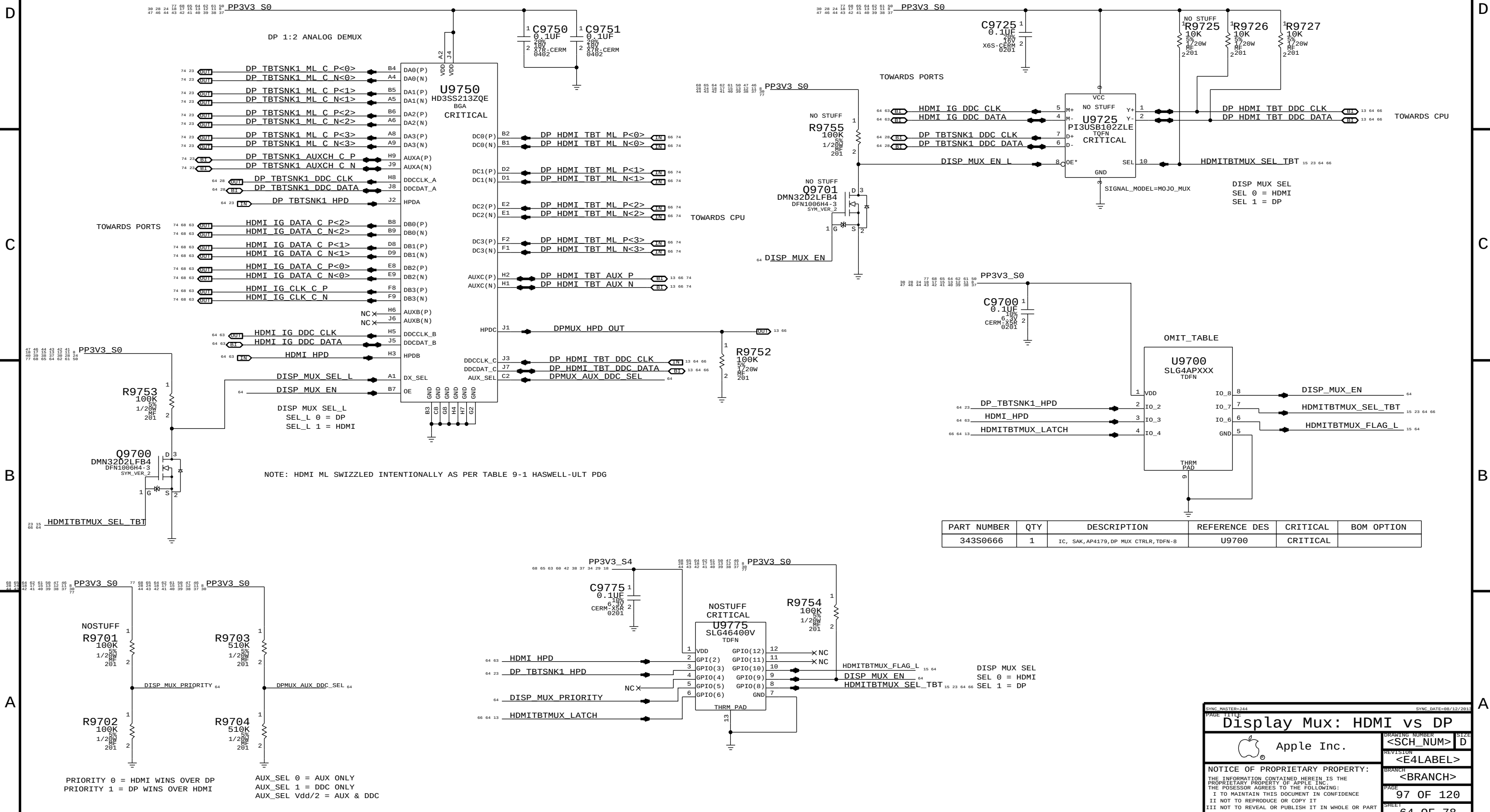
J9510
DF40BG-70DP-0.4V
M-ST-SM
516S1059
CRITICAL

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63	OUT	=HDMI_DATA_C_P<2>	TRUE HDMI_IG_DATA_C_P<2>
63	OUT	=HDMI_DATA_C_N<2>	TRUE HDMI_IG_DATA_C_N<2>
74	68	64	63
OUT	OUT	HDMI_IG_DATA_C_P<1>	TRUE HDMI_IG_DATA_C_P<1>
74	68	64	63
OUT	OUT	HDMI_IG_DATA_C_N<1>	TRUE HDMI_IG_DATA_C_N<1>
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74	68	64	63
OUT	OUT	HDMI_IG_CLK_C_P	TRUE HDMI_IG_CLK_C_P
74	68	64	63
OUT	OUT	HDMI_IG_CLK_C_N	TRUE HDMI_IG_CLK_C_N
71	63	29	63
BT	BT	USB_BT_CONN_P	TRUE USB_BT_CONN_P
71	63	29	63
BT	BT	USB_BT_CONN_N	TRUE USB_BT_CONN_N
70	63	OUT	PCIE_CLK100M_AP_CONN_P
70	63	OUT	PCIE_CLK100M_AP_CONN_N
74	68	63	39
16	14	16	14
BT	BT	SMBUS_PCH_CLK	TRUE SMBUS_PCH_CLK
64	63	OUT	HDMI_IG_DDC_CLK
64	63	OUT	HDMI_IG_DDC_DATA
76	63	43	39
36	36	43	39
OUT	OUT	SMBUS_SMC_3_SCL	TRUE SMBUS_SMC_3_SCL
76	63	43	39
BT	BT	SMBUS_SMC_3_SDA	TRUE SMBUS_SMC_3_SDA
63	18	OUT	SDCONN_STATE_CHANGE_RIO
71	68	63	14
BT	BT	USB3RPCIE_SD_D2R_P	TRUE USB3RPCIE_SD_D2R_P
71	68	63	14
BT	BT	USB3RPCIE_SD_D2R_N	TRUE USB3RPCIE_SD_D2R_N
71	68	63	14
BT	BT	USB3RPCIE_SD_R2D_C_P	TRUE USB3RPCIE_SD_R2D_C_P
71	68	63	14
BT	BT	USB3RPCIE_SD_R2D_C_N	TRUE USB3RPCIE_SD_R2D_C_N
72	68	63	29
BT	BT	AP_PCIE_WAKE_L	TRUE AP_PCIE_WAKE_L

SYNC_MASTER=J44		SYNC_DATE=08/12/2013			
PAGE TITLE					
RIO Connector					
Apple Inc.		DRAWING NUMBER	SIZE		
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DISPLAY MUX: DP OR HDMI

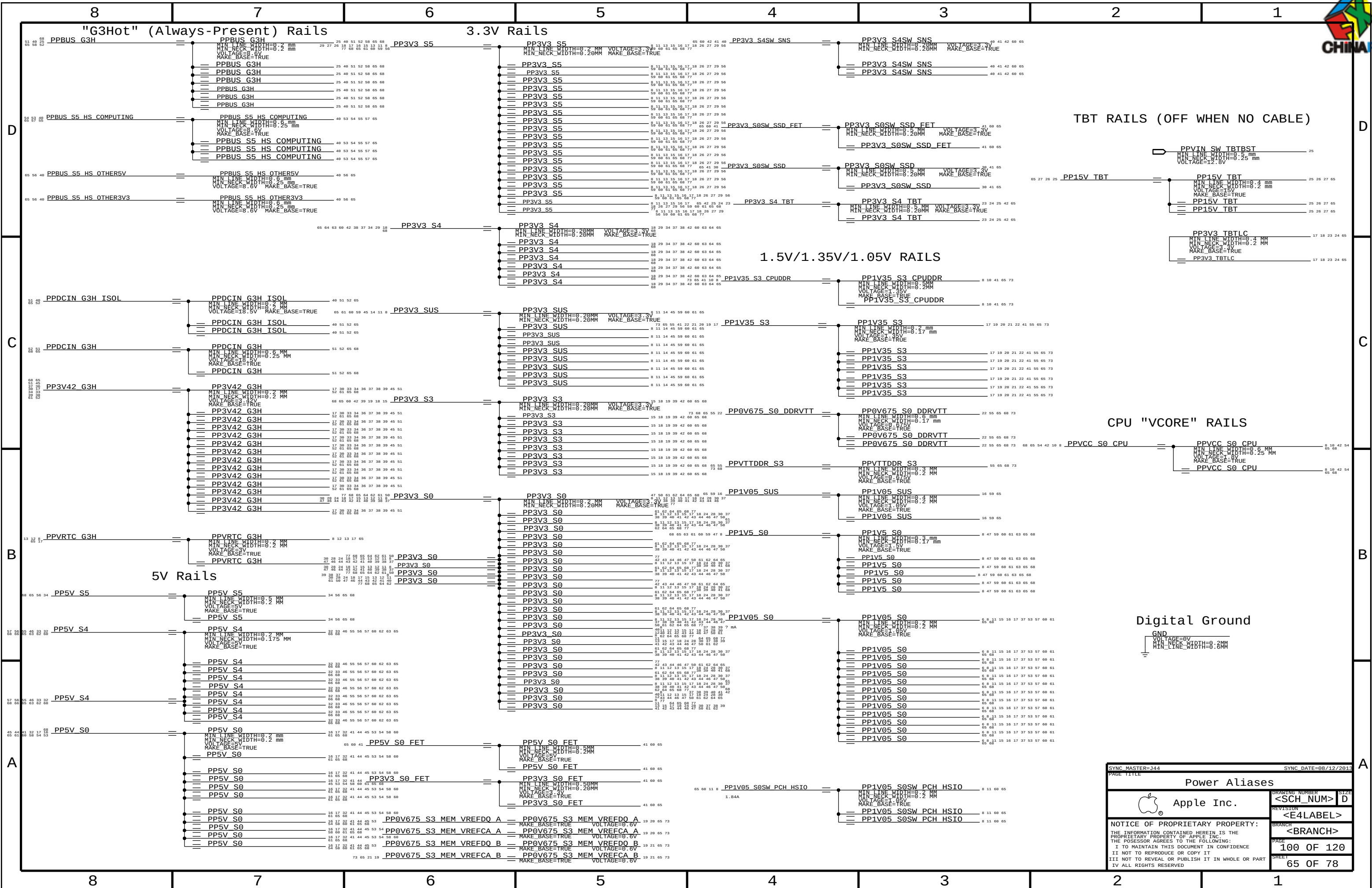


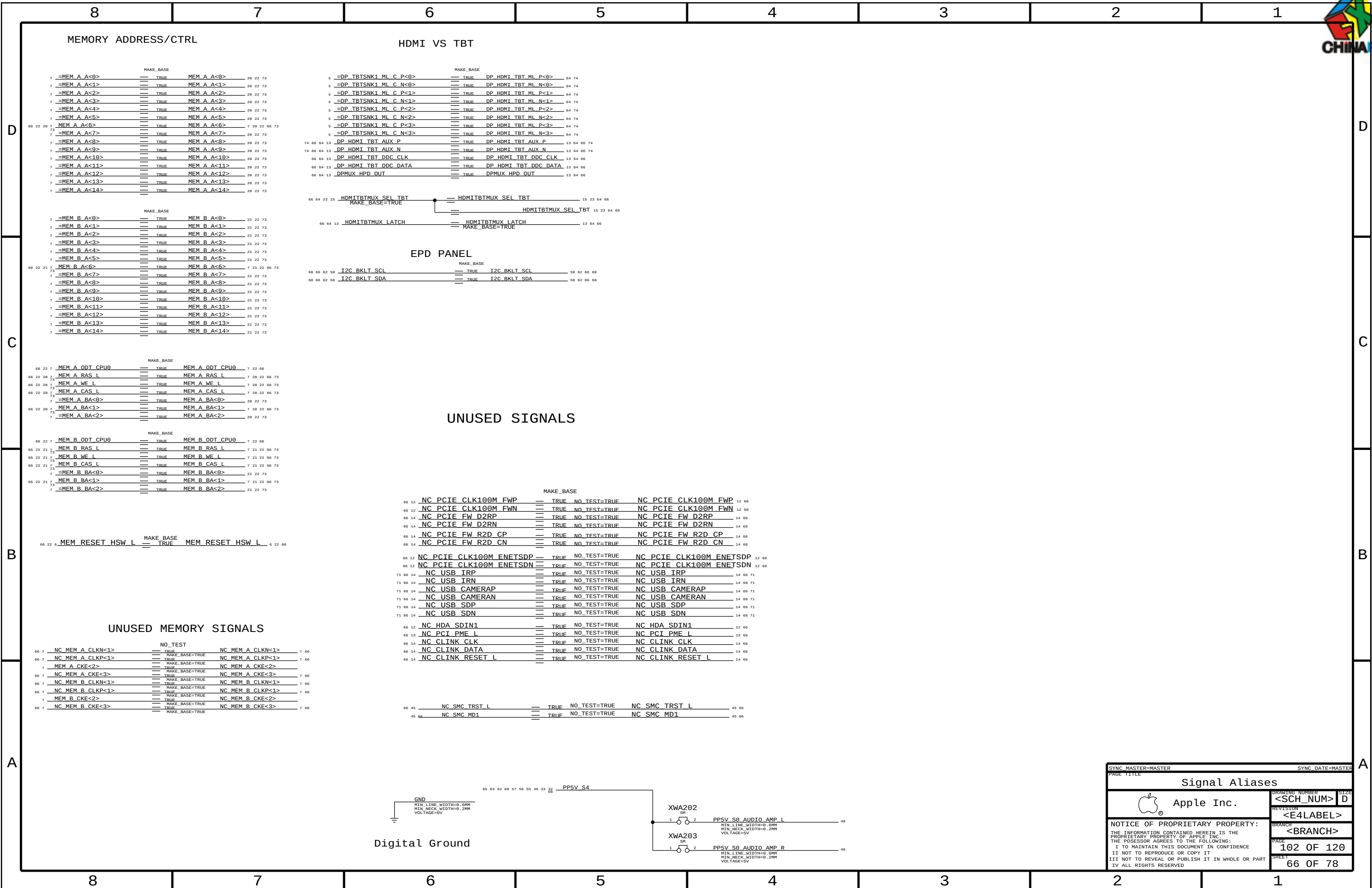
Display Mux: HDMI vs DP

Apple Inc.

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D

C

B

A

D

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B

A

Memory Bit/Byte Swizzle

MAKE_BASE			
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73 68 7	TRUE	MEM A DQ<1>	=MEM A DQ<56>
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SYNC MASTER=344		SYNC DATE=01/03/2013	
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Memory Bit/Byte Swizzle			
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D

D

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

C

Stackup-Defined Spacing Rules

3

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	.	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1X_DIELECTRIC	TOP, BOTTOM	0.058 MM	?
1X_DIELECTRIC	ISL3, ISL4, ISL9, ISL10	0.053 MM	?
1X_DIELECTRIC	ISL2, ISL5, ISL6, ISL7, ISL8, ISL11	0.101 MM	?

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

A

SYNC MASTER=J44		SYNC DATE=08/12/2013	
PAGE TITLE			
PCB Rule Definitions			
	Apple Inc.		
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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CPU_VCCSENSE	*	25 MIL	?	CPU_08MIL	*	0.203 MM	?
				CPU_12MIL	*	0.305 MM	?
				CPU_18MIL	*	0.457 MM	?
				CPU_25MIL	*	0.635 MM	?

PCI Express Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE_2SAME	*	=3X_DIELECTRIC	?	PCIE_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_TXRX	*	=6X_DIELECTRIC	?	PCIE_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIE_20THER	*	=4X_DIELECTRIC	?	PCIE_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
PCIE_2CLK	*	=7X_DIELECTRIC	?	PCIE_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIECLK_20THER	*	=7X_DIELECTRIC	?	PCIECLK_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_*	*	*	PCIE_20THER
PCIE_*	=SAME	*	PCIE_2SAME
PCIE_*	CLK_*	*	PCIE_2CLK
CLK_PCIE	*	*	PCIECLK_20THER
PCIE_TX	*_RX	*	PCIE_TXRX
PCIE_RX	*_TX	*	PCIE_TXRX

CPU Signal Properties

ELECTRICAL CONST SET	NET TYPE		
	PHYSICAL	SPACING	
XDP_TCK0	CPU_45S	CPU_18MTL	XDP_CPU_TCK 6 16
XDP_TCK0	CPU_45S	CPU_18MTL	PCH_JTAGX 12 16
XDP_TCK1	CPU_45S	CPU_18MTL	XDP_PCH_TCK 12 16
XDP_TDO	CPU_45S		XDP_CPU_TDO 6 16
XDP_TDO	CPU_45S		XDP_PCH_TDO 12 16
XDP_TDI	CPU_45S		XDP_CPU_TDI 6 16
XDP_TDI	CPU_45S		XDP_PCH_TDI 12 16
XDP_TMS	CPU_45S		XDP_CPU_TMS 6 16
XDP_TMS	CPU_45S		XDP_PCH_TMS 12 16
XDP_TRST_L	CPU_45S		XDP_TRST_L 16
XDP_TRST_L	CPU_45S		XDP_CPUPCH_TRST_L 6 12 16
XDP_PRDY_L	CPU_45S		XDP_CPU_PRDY_L 6 16
XDP_PREQ_L	CPU_45S		XDP_CPU_PREQ_L 6 16
CPU_VCCST_PWRGD	CPU_45S	CPU_08MTL	CPU_VCCST_PWRGD 8 16 17
CPU_VCCST_PWRGD	CPU_45S	CPU_08MTL	XDP_CPU_VCCST_PWRGD 16
CPU_BPM	CPU_45S	CPU_08MTL	XDP_BPM_L<1..0> 6 16
CPU_BPM_TP	CPU_45S		XDP_BPM_L<7..2> 6 16
CPU_RCOMP_SM	CPU_27P4S	CPU_25MTL	CPU_SM_RCOMP<2..0> 6
CPU_RCOMP_EDP	CPU_27P4S	CPU_25MTL	MCP_EDP_RCOMP 5
CPU_RCOMP_OPT	CPU_27P4S	CPU_12MTL	CPU_OPT_RCOMP 6
CPU_PROCHOT	CPU_45S	CPU_08MTL	CPU_PROCHOT_L 6 36 37 53
CPU_PROCHOT	CPU_45S	CPU_08MTL	CPU_PROCHOT_R_L 6
CPU_CATERR	CPU_45S	CPU_08MTL	CPU_CATERR_L 6 36
CPU_VIDALERT	CPU_45S	CPU_18MTL	CPU_VIDALERT_L 8 53
CPU_VIDALERT	CPU_45S	CPU_18MTL	CPU_VIDALERT_R_L 8
CPU_VIDSClk	CPU_45S	CPU_18MTL	CPU_VIDSClk 8 53
CPU_VIDSClk	CPU_45S	CPU_18MTL	CPU_VIDSClk_R 8
CPU_VIDSOUT	CPU_45S	CPU_18MTL	CPU_VIDSOUT 8 53
CPU_VIDSOUT	CPU_45S	CPU_18MTL	CPU_VIDSOUT_R 8
CPU_PECI	CPU_45S	CPU_18MTL	CPU_PECI 6 37
CPU_PECI	CPU_45S	CPU_18MTL	CPU_PECI_R 36 37
CPU_PECI_SMC	CPU_45S	CPU_18MTL	SMC_PECI_L 36 37
CPU_PECI_SMC	CPU_45S	CPU_18MTL	SMC_PECI_L_R 37
CPU_CFG	CPU_45S		CPU_CFG<19..11> 6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<10..8> 6 16
CPU_CFG	CPU_45S		CPU_CFG<7..5> 6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<4> 6 16
CPU_CFG_3	CPU_45S		CPU_CFG<3> 6 16
CPU_CFG	CPU_45S		CPU_CFG<2> 6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<1..0> 6 16
CPU_MEM_RESET	CPU_45S	CPU_08MTL	MEM_RESET_L 20 21 22
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P 8 53
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N 9 53

PCI Express Properties

ELECTRICAL CONST SET	NET TYPE		
	PHYSICAL	SPACING	
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<3..1> 12 30 68
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<3..1> 12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<0> 12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<0> 12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_P<3..0> 12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_N<3..0> 12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_P<3..0> 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_N<3..0> 30 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<0> 14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<0> 14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<0> 23
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<0> 23
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<3..1> 14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<3..1> 14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<3..1> 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<3..1> 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_P<3..0> 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_N<3..0> 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_P<3..0> 14 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_N<3..0> 14 23 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_P 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_N 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_P 14 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_N 14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_P 14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_N 14 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P 63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N 63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_P 12 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_N 12 63 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_P 12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_N 12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P 31 32
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N 31 32
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_P 12 30 68
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_N 12 30 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_P 12 23 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_N 12 23 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_P 14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_N 14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_P 31 32
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_N 31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_P 31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_N 31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_P 14 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_N 14 32

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CPU & PCIe Constraints			
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USB 2 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAIS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?	USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIAIS	*	=6X_DIELECTRIC	?	USB_RBIAIS	TOP,BOTTOM	=10X_DIELECTRIC	?

USB 3 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?	USB3_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?	USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_20THER	*	=4X_DIELECTRIC	?	USB3_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	*	*	USB3_20THER
USB3_*	=SAME	*	USB3_2SAME
USB3_TX	*_RX	*	USB3_TXRX
USB3_RX	*_TX	*	USB3_TXRX

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5X_DIELECTRIC	?

SATA Interface Constraints (Not Used)

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?	SATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?	SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_20THER	*	=4X_DIELECTRIC	?	SATA_20THER	TOP,BOTTOM	=6X_DIELECTRIC	

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	*	*	SATA_20THER
SATA_*	=SAME	*	SATA_2SAME
SATA_TX	*_RX	*	SATA_TXRX
SATA_RX	*_TX	*	SATA_TXRX

USB Constraints

ELECTRICAL CONST SET	NET TYPE		
	PHYSICAL	SPACING	
USB_BT	USB_85D	USB	USB BT P
USB_BT	USB_85D	USB	USB BT N
USB_BT	USB_85D	USB	USB BT CONN P
USB_BT	USB_85D	USB	USB BT CONN N
USB_EXT_A	USB_85D	USB	USB EXT_A P
USB_EXT_A	USB_85D	USB	USB EXT_A N
DEFAULT	DEFAULT	DEFAULT	SMC DEBUGPRT_RX_L
DEFAULT	DEFAULT	DEFAULT	SMC DEBUGPRT_TX_L
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_P
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_N
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_F_P
USB_EXT_A	USB_85D	USB	USB2_EXT_A MUXED_F_N
USB_EXT_A	USB_85D	USB	USB_LT1_P
USB_EXT_A	USB_85D	USB	USB_LT1_N
USB_EXT_B	USB_85D	USB	USB_EXTB_P
USB_EXT_B	USB_85D	USB	USB_EXTB_N
USB_TPAD	USB_85D	USB	USB_TPAD_P
USB_TPAD	USB_85D	USB	USB_TPAD_N
USB_TPAD	USB_85D	USB	USB_TPAD_R_P
USB_TPAD	USB_85D	USB	USB_TPAD_R_N
USB3_EXT_A_D2R	USB_85D	USB3_RX	USB3_EXT_A D2R_P
USB3_EXT_A_D2R	USB_85D	USB3_RX	USB3_EXT_A D2R_N
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_P
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_N
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_C_P
USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A R2D_C_N
USB3_EXT_B_D2R	USB_85D	USB3_RX	USB3_EXTB D2R_P
USB3_EXT_B_D2R	USB_85D	USB3_RX	USB3_EXTB D2R_N
USB3_EXT_B_R2D	USB_85D	USB3_TX	USB3_EXTB R2D_C_P
USB3_EXT_B_R2D	USB_85D	USB3_TX	USB3_EXTB R2D_C_N
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD D2R_P
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD D2R_N
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD R2D_C_P
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD R2D_C_N
USB_NC	USB_85D	USB	NC_USB_IRP
USB_NC	USB_85D	USB	NC_USB_IRN
USB_NC	USB_85D	USB	TP_USB_5P
USB_NC	USB_85D	USB	TP_USB_5N
USB_NC	USB_85D	USB	NC_USB_SDP
USB_NC	USB_85D	USB	NC_USB_SDN
USB_NC	USB_85D	USB	NC_USB_CAMERAP
USB_NC	USB_85D	USB	NC_USB_CAMERAN
PCH_USB_RBIAIS	PCH_USB_RBIAIS	USB_RBIAIS	PCH_USB_RBIAIS
SATA_85D	SATA_85D	SATA_RX	DUMMY SATA D2R_P
SATA_85D	SATA_85D	SATA_RX	DUMMY SATA D2R_N
SATA_85D	SATA_85D	SATA_TX	DUMMY SATA R2D_P
SATA_85D	SATA_85D	SATA_TX	DUMMY SATA R2D_N
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R

Notes:
This is here to keep the SATA rules.

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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2X_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2X_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPT_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
PCH_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_12MIL	*	0.305 MM	?
PCH_15MIL	*	0.381 MM	?
PCH_18MIL	*	0.457 MM	?
PCH_20MIL	*	0.508 MM	?

PCH Net Properties

ELECTRICAL CONST SET	NET TYPE		
	PHYSICAL	SPACING	
LPC_AD	LPC_45S	LPC	LPC_AD<3...0> 14 36 45 68
LPC_AD	LPC_45S	LPC	LPC_FRAME_L 14 36 45 68
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R 12 17
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC 17 36 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS 17 45 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS_R 12 17
SMBUS_PCH	SMB_45S	SMR	SMBUS_PCH_CLK 14 16 19 39 63 68
SMBUS_PCH	SMB_45S	SMR	SMBUS_PCH_DATA 14 16 19 39 63 68
SML_PCH_0	SMB_45S	SMR	SML_PCH_0_CLK 14 39
SML_PCH_0	SMB_45S	SMR	SML_PCH_0_DATA 14 39
	SMB_45S	SMR	SMBUS_SMC_1_S0_SCL 14 32 36 39 43 68 76
	SMB_45S	SMR	SMBUS_SMC_1_S0_SDA 14 32 36 39 43 68 76
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK 12 47
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK_R 12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC 12 47
HDA_SYNC	HDA_45S	HDA	HDA_SYNC_R 12
HDA_RST	HDA_45S	HDA	HDA_RST_R_L 12
HDA_RST	HDA_45S	HDA	HDA_RST_L 12 47
HDA_SDIN	HDA_45S	HDA	HDA_SDIN0 12 47 68
HDA_SDIN	HDA_45S	HDA	CS4208_HDA_SDOUT0_R 47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT 12 47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT_R 12 17
SPT_MLB	SPT_45S	SPT	SPI_ALT_CLK 45
SPT_MLB	SPT_45S	SPT	SPI_CLK 45
SPT_MLB	SPT_45S	SPT	SPI_CLK_R 14 45
SPT_MLB	SPT_45S	SPT	SPI_MLB_CLK 45
SPT_MLB	SPT_45S	SPT	SPI_SMC_CLK 36 45
SPT_MLB	SPT_45S	SPT	SPI_ALT_CS_L 45
SPT_MLB	SPT_45S	SPT	SPI_CS0_L 45
SPT_MLB	SPT_45S	SPT	SPI_CS0_R_L 14 45
SPT_MLB	SPT_45S	SPT	SPI_MLB_CS_L 45
SPT_MLB	SPT_45S	SPT	SPI_SMC_CS_L 36 45
SPT_MLB	SPT_45S	SPT	SPI_ALT_MISO 45
SPT_MLB	SPT_45S	SPT	SPI_MISO 14 45
SPT_MLB	SPT_45S	SPT	SPI_MISO_R 45
SPT_MLB	SPT_45S	SPT	SPI_MLB_MISO 45
SPT_MLB	SPT_45S	SPT	SPI_SMC_MISO 36 45
SPT_MLB	SPT_45S	SPT	SPI_ALT_MOSI 45
SPT_MLB	SPT_45S	SPT	SPI_MOSI 45
SPT_MLB	SPT_45S	SPT	SPI_MOSI_R 14 45
SPT_MLB	SPT_45S	SPT	SPI_MLB_MOSI 45
SPT_MLB	SPT_45S	SPT	SPI_SMC_MOSI 36 45
SPT_MLB_I02	SPT_45S	SPT	SPI_I0<2> 14 45
SPT_MLB_I02	SPT_45S	SPT	SPIROM_WP_L 45
SPT_MLB_I03	SPT_45S	SPT	SPI_I0<3> 14 45
SPT_MLB_I03	SPT_45S	SPT	SPIROM_HOLD_L 45
SPT_MLB_I03	SPT_45S	SPT	SPIROM_USE_MLB 15 45 68
PCH_RTCX	PCH_45S	PCH_15MTI	PCH_CLK32K_RTCX1 12 17
PCH_SRTCST	PCH_45S	PCH_15MTI	PCH_SRTCST_L 12
PCH_RTCRST	PCH_45S	PCH_15MTI	RTC_RESET_L 12
PCH_THRMTRIP	PCH_45S	PCH_18MTI	PM_THRMTRIP_L 15 37
PCH_THRMTRIP	PCH_45S	PCH_18MTI	PM_THRMTRIP_R_L 37
	PCH_45S	PCH_15MTI	PCH_INTRUDER_L 12
	PCH_45S	PCH_15MTI	PCH_INTVRMEN 12
	PCH_45S	PCH_15MTI	PCH_DSWVRMEN 13
	PCH_45S	PCH_15MTI	PM_RSMRST_L 13 61
	PCH_45S	PCH_15MTI	PM_SYSRST_L 13 17 36 68
	PCH_45S	PCH_15MTI	XDP_DBRESET_L 16 17
	PCH_45S	PCH_15MTI	PM_PCH_SYS_PWROK 13 16 17 36
	PCH_45S	PCH_15MTI	XDP_SYS_PWROK 16
	PCH_45S	PCH_15MTI	SYS_PWROK_R 17
	PCH_45S	PCH_15MTI	PM_PCH_PWROK 13 17
	PCH_45S	PCH_15MTI	PM_S0_PG00D 17
	PCH_45S	PCH_15MTI	SMC_DELAYED_PWRGD 17 24 25 36 37
	PCH_45S	PCH_15MTI	PM_DSW_PWRGD 13 36
	PCH_45S	PCH_15MTI	PM_PWRBTN_L 13 16 36
	PCH_45S	PCH_15MTI	XDP_CPU_PWRBTN_L 16
	PCH_45S	PCH_15MTI	PCIE_WAKE_L 13 29 31
	PCH_45S	PCH_15MTI	AP_PCIE_WAKE_L 29 63
	PCH_45S	PCH_15MTI	CAM_PCIE_WAKE_L 31
	PCH_45S	PCH_15MTI	TBT_CIO_PLUG_EVENT_L 15 18 23
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALIN 12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALOUT 12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALOUT_R 17
PCH_RCOMP_PCIE	PCH_27P4S	PCH_12MTI	PCH_PCIE_RCOMP 14
PCH_RCOMP_OPT	PCH_27P4S	PCH_12MTI	PCH_OPI_COMP 15
PCH_RCOMP_SATA	PCH_27P4S	PCH_12MTI	PCH_SATA_RCOMP 12

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PCH Constraints			
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTL	*	=2x_DIELECTRIC	?
MEM_CTL2CTL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_20THERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_2OTHER	*	=6x_DIELECTRIC	?
MEM_CMD2CMD_BM	*	=2x_DIELECTRIC	?
MEM_CMD2CTL_BM	*	=2x_DIELECTRIC	?
MEM_CTL2CTL_BM	*	=2x_DIELECTRIC	?
MEM_12MIL	*	0.305 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTL2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_20THERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CMD2CMD_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CMD2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CTL2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	*	*	MEM_20THER
MEM_*_DQS_*	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_CTL	*	*	MEM_20THER
MEM_CLK	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTL	*	MEM_CMD2CTL
MEM_CTL	MEM_CTL	*	MEM_CTL2CTL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_20THERMEM
MEM_CMD	MEM_CMD	BGA_MEM	MEM_CMD2CMD_BM
MEM_CMD	MEM_CTL	BGA_MEM	MEM_CMD2CTL_BM
MEM_CTL	MEM_CTL	BGA_MEM	MEM_CTL2CTL_BM

Haswell ULT Memory Down DDR3L 1x8 Length Matching

DDR3 Signal Group	Unit	Min Length	Max Length
CTLmax - CTLmin	mils	0	100
CTL to CLK	mils	CLK - 500	CLK + 500
CMDi to CMDj	mils	CMDj - 100	CMDj + 100
CMD to CLK	mils	CLK - 500	CLK + 500
(DQmax - DQmin) per byte	mils	0	250
(DQS - DQmax) per byte	mils	-100	150
DQS to DQS#	mils	-5	5
DQS to CLK (Rule 1)	mils	CLK - 6500	CLK + 500
Max(CLK-DQS) - Min(CLK-DQS)	mils	0	5500
CLK to CLK#	mils	-5	5

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<0>	7 20 22
MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM A CKE<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM A CS L<0>	7 20 22
MEM_A_ODT0	MEM_40S	MEM_CTL	MEM A ODT<0>	20 22
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A A<15..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A BA<2..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A RAS L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A CAS L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM A WE L	7 20 22 66
MEM_A_DQBYTE0	MEM_45S	MEM_A_DQBYTE_0	MEM A DQ<7..0>	7 67 68
MEM_A_DQBYTE1	MEM_45S	MEM_A_DQBYTE_1	MEM A DQ<15..8>	7 67 68
MEM_A_DQBYTE2	MEM_45S	MEM_A_DQBYTE_2	MEM A DQ<23..16>	7 67 68
MEM_A_DQBYTE3	MEM_45S	MEM_A_DQBYTE_3	MEM A DQ<31..24>	7 67 68
MEM_A_DQBYTE4	MEM_45S	MEM_A_DQBYTE_4	MEM A DQ<39..32>	7 20 67 68
MEM_A_DQBYTE5	MEM_45S	MEM_A_DQBYTE_5	MEM A DQ<47..40>	7 67 68
MEM_A_DQBYTE6	MEM_45S	MEM_A_DQBYTE_6	MEM A DQ<55..48>	7 67 68
MEM_A_DQBYTE7	MEM_45S	MEM_A_DQBYTE_7	MEM A DQ<63..56>	7 67 68
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM A DQS P<0>	7 67
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM A DQS N<0>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM A DQS P<1>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM A DQS N<1>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM A DQS P<2>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM A DQS N<2>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM A DQS P<3>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM A DQS N<3>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM A DQS P<4>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM A DQS N<4>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM A DQS P<5>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM A DQS N<5>	7 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM A DQS P<6>	7 20 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM A DQS N<6>	7 20 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM A DQS P<7>	7 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM A DQS N<7>	7 67
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK P<0>	7 21 22
MEM_B_CLK	MEM_72D	MEM_CLK	MEM B CLK N<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM B CKE<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM B CS L<0>	7 21 22
MEM_B_ODT0	MEM_40S	MEM_CTL	MEM B ODT<0>	21 22
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B A<15..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B BA<2..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B RAS L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B CAS L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM B WE L	7 21 22 66
MEM_B_DQBYTE0	MEM_45S	MEM_B_DQBYTE_0	MEM B DQ<7..0>	7 67 68
MEM_B_DQBYTE1	MEM_45S	MEM_B_DQBYTE_1	MEM B DQ<15..8>	7 67 68
MEM_B_DQBYTE2	MEM_45S	MEM_B_DQBYTE_2	MEM B DQ<23..16>	7 67 68
MEM_B_DQBYTE3	MEM_45S	MEM_B_DQBYTE_3	MEM B DQ<31..24>	7 67 68
MEM_B_DQBYTE4	MEM_45S	MEM_B_DQBYTE_4	MEM B DQ<39..32>	7 21 67 68
MEM_B_DQBYTE5	MEM_45S	MEM_B_DQBYTE_5	MEM B DQ<47..40>	7 67 68
MEM_B_DQBYTE6	MEM_45S	MEM_B_DQBYTE_6	MEM B DQ<55..48>	7 67 68
MEM_B_DQBYTE7	MEM_45S	MEM_B_DQBYTE_7	MEM B DQ<63..56>	7 67 68
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM B DQS P<0>	7 67
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM B DQS N<0>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM B DQS P<1>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM B DQS N<1>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM B DQS P<2>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM B DQS N<2>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM B DQS P<3>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM B DQS N<3>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM B DQS P<4>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM B DQS N<4>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM B DQS P<5>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM B DQS N<5>	7 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM B DQS P<6>	7 21 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM B DQS N<6>	7 21 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM B DQS P<7>	7 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM B DQS N<7>	7 67
		MEM_PWR	PP1V35 S3	17 19 20 21 22 41 55 65
		MEM_PWR	PP1V35 S3 CPUDDR	8 10 41 65
		MEM_PWR	PP0V675 S0 DDRVTT	22 55 65 68
		MEM_PWR	PPVTTDDR S3	55 65 68
		MEM_12MIL	CPU DIMMA VREFD0	7 19
		MEM_12MIL	CPU DIMMA VREFD0 A ISOL	19
		MEM_12MIL	CPU DIMMB VREFD0	7 19
		MEM_12MIL	CPU DIMMB VREFD0 B ISOL	19
		MEM_12MIL	CPU DIMM VREFCA	7 19
		MEM_12MIL	CPU DIMM VREFCA A ISOL	19
		MEM_12MIL	CPU DIMM VREFCA B ISOL	19
		MEM_12MIL	PP0V675 S3 MEM VREFD0 A	19 20 65
		MEM_12MIL	PP0V675 S3 MEM VREFD0 B	19 21 65
		MEM_12MIL	PP0V675 S3 MEM VREFCA A	19 20 65
		MEM_12MIL	PP0V675 S3 MEM VREFCA B	19 21 65

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Thunderbolt, DP, HDMI Constraints

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt & DisplayPort Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTD_P_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTD_P_2SAME	*	=3X_DIELECTRIC	?
TBTD_P_TXRX	*	=6X_DIELECTRIC	?
TBTD_P_20THER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTD_P_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
TBTD_P_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
TBTD_P_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTD_P_*	*	*	TBTD_P_20THER
TBTD_P_*	=SAME	*	TBTD_P_2SAME
TBTD_P_TX	*_RX	*	TBTD_P_TXRX
TBTD_P_RX	*_TX	*	TBTD_P_TXRX

DisplayPort & HDMI Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
HDMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3x_DIELECTRIC	?
DP_20THER	*	=4x_DIELECTRIC	?
HDMICLK_20THER	*	=7x_DIELECTRIC	?
HDMICLK_2DPHDMI	*	=4x_DIELECTRIC	?
HDMIDATA_2SAME	*	=3x_DIELECTRIC	?
HDMIDATA_20THER	*	=4x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_20THER	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICLK_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?
HDMICLK_2DPHDMI	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMIDATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
HDMIDATA_20THER	TOP,BOTTOM	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDMI_DATA	*	*	HDMIDATA_20THER
HDMI_DATA	=SAME	*	HDMIDATA_2SAME
HDMI_DATA	TBTD_P_TX	*	HDMIDATA_2SAME
HDMI_DATA	TBTD_P_RX	*	DP_2SAME
HDMI_CLK	*	*	HDMICLK_20THER
HDMI_CLK	HDMI_DATA	*	HDMICLK_2DPHDMI
HDMI_CLK	DISPLAYPORT	*	HDMICLK_2DPHDMI
HDMI_CLK	TBTD_P_TX	*	HDMICLK_2DPHDMI

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_20THER
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	HDMI_DATA	*	DP_2SAME
DISPLAYPORT	TBTD_P_TX	*	DP_2SAME
DISPLAYPORT	TBTD_P_RX	*	TBTD_P_TXRX

DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.
MAX LENGTH OF DISPLAYPORT/TMDS TRACES: 13 INCHES.

ELECTRICAL CONST SET	PHYSICAL	SPACING	NET TYPE
DP_85D	DISPLAYPORT	DP TBTSRC ML C P<3..0>	
DP_85D	DISPLAYPORT	DP TBTSRC ML C N<3..0>	
DP_85D	DISPLAYPORT	DP TBTSRC AUXCH C P	
DP_85D	DISPLAYPORT	DP TBTSRC AUXCH C N	
SPI_TBT_CLK	TBT_SPI_45S	TBT_SPI	TBT SPI CLK
SPI_TBT_MOST	TBT_SPI_45S	TBT_SPI	TBT SPI MOSI
SPI_TBT_MISO	TBT_SPI_45S	TBT_SPI	TBT SPI MISO
SPI_TBT_CS_L	TBT_SPI_45S	TBT_SPI	TBT SPI CS_L
DP_HDMI_TBT_ML	DP_85D	DISPLAYPORT	DP HDMI TBT ML P<3..0>
DP_HDMI_TBT_MI	DP_85D	DISPLAYPORT	DP HDMI TBT ML N<3..0>
DP_HDMI_TBT_AUX	DP_85D		DP HDMI TBT AUX P
DP_HDMI_TBT_AUX	DP_85D		DP HDMI TBT AUX N
HDMI_CLOCK	HDMI_85D	HDMI_CLK	HDMI IG CLK C P
HDMI_CLOCK	HDMI_85D	HDMI_CLK	HDMI IG CLK C N
HDMI_DATA	HDMI_85D	HDMI_DATA	HDMI IG DATA C P<2..0>
HDMI_DATA	HDMI_85D	HDMI_DATA	HDMI IG DATA C N<2..0>

Only used on hosts supporting Thunderbolt video-in

Thunderbolt, DP, HDMI Net Properties

ELECTRICAL CONST SET	PHYSICAL	SPACING	NET TYPE
TBT_A_R2D	TBTD_P_85D	TBTD_P_TX	TBT A R2D C P<1..0>
TBT_A_R2D	TBTD_P_85D	TBTD_P_TX	TBT A R2D C N<1..0>
TBT_A_R2D	TBTD_P_85D	TBTD_P_TX	TBT A R2D P<1..0>
TBT_A_R2D	TBTD_P_85D	TBTD_P_TX	TBT A R2D N<1..0>
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPA ML C P<1>
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPA ML C N<1>
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPA ML P<1>
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPA ML N<1>
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP A LSX_ML P<1>
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP A LSX_ML N<1>
DP_TBTPA_MI	DP_85D	DISPLAYPORT	DP TBTPA ML C P<3>
DP_TBTPA_MI	DP_85D	DISPLAYPORT	DP TBTPA ML C N<3>
DP_TBTPA_MI	DP_85D	DISPLAYPORT	DP TBTPA ML P<3>
DP_TBTPA_MI	DP_85D	DISPLAYPORT	DP TBTPA ML N<3>
TBT_A_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT A D2R C P<0>
TBT_A_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT A D2R C N<0>
TBT_A_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT A D2R P<0>
TBT_A_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT A D2R N<0>
TBT_A_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT A D2R C P<1>
TBT_A_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT A D2R C N<1>
TBT_A_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT A D2R P<1>
TBT_A_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT A D2R N<1>
TBT_A_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT A D2R1 AUXDDC_P
TBT_A_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT A D2R1 AUXDDC_N
DP_TBTPA_AUXCH	DP_85D		DP TBTPA AUXCH C P
DP_TBTPA_AUXCH	DP_85D		DP TBTPA AUXCH C N
DP_TBTPA_AUXCH	DP_85D		DP TBTPA AUXCH P
DP_TBTPA_AUXCH	DP_85D		DP TBTPA AUXCH N
TBT_B_R2D	TBTD_P_85D	TBTD_P_TX	TBT B R2D C P<1..0>
TBT_B_R2D	TBTD_P_85D	TBTD_P_TX	TBT B R2D C N<1..0>
TBT_B_R2D	TBTD_P_85D	TBTD_P_TX	TBT B R2D P<1..0>
TBT_B_R2D	TBTD_P_85D	TBTD_P_TX	TBT B R2D N<1..0>
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPB ML C P<1>
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPB ML C N<1>
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPB ML P<1>
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP TBTPB ML N<1>
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP B LSX_ML P<1>
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP B LSX_ML N<1>
DP_TBTPB_MI	DP_85D	DISPLAYPORT	DP TBTPB ML C P<3>
DP_TBTPB_MI	DP_85D	DISPLAYPORT	DP TBTPB ML C N<3>
DP_TBTPB_MI	DP_85D	DISPLAYPORT	DP TBTPB ML P<3>
DP_TBTPB_MI	DP_85D	DISPLAYPORT	DP TBTPB ML N<3>
TBT_B_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT B D2R C P<0>
TBT_B_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT B D2R C N<0>
TBT_B_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT B D2R P<0>
TBT_B_D2R_0	TBTD_P_85D	TBTD_P_RX	TBT B D2R N<0>
TBT_B_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT B D2R C P<1>
TBT_B_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT B D2R C N<1>
TBT_B_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT B D2R P<1>
TBT_B_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT B D2R N<1>
TBT_B_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT B D2R1 AUXDDC_P
TBT_B_D2R_1	TBTD_P_85D	TBTD_P_RX	TBT B D2R1 AUXDDC_N
DP_TBTPB_AUXCH	DP_85D		DP TBTPB AUXCH C P
DP_TBTPB_AUXCH	DP_85D		DP TBTPB AUXCH C N
DP_TBTPB_AUXCH	DP_85D		DP TBTPB AUXCH P
DP_TBTPB_AUXCH	DP_85D		DP TBTPB AUXCH N
DP_TBTSNK0_MI	DP_85D	DISPLAYPORT	DP TBTSNK0 ML C P<3..0>
DP_TBTSNK0_MI	DP_85D	DISPLAYPORT	DP TBTSNK0 ML C N<3..0>
DP_TBTSNK0_MI	DP_85D	DISPLAYPORT	DP TBTSNK0 ML P<3..0>
DP_TBTSNK0_MI	DP_85D	DISPLAYPORT	DP TBTSNK0 ML N<3..0>
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK0 AUXCH C P
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK0 AUXCH C N
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK0 AUXCH P
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK0 AUXCH N
DP_TBTSNK1_MI	DP_85D	DISPLAYPORT	DP TBTSNK1 ML C P<3..0>
DP_TBTSNK1_MI	DP_85D	DISPLAYPORT	DP TBTSNK1 ML C N<3..0>
DP_TBTSNK1_MI	DP_85D	DISPLAYPORT	DP TBTSNK1 ML P<3..0>
DP_TBTSNK1_MI	DP_85D	DISPLAYPORT	DP TBTSNK1 ML N<3..0>
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK1 AUXCH C P
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK1 AUXCH C N
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK1 AUXCH P
DP_TBTSNK_AUXCH	DP_85D		DP TBTSNK1 AUXCH N
DP_INT_MI	DP_85D	DISPLAYPORT	DP INT ML C P<3..0>
DP_INT_MI	DP_85D	DISPLAYPORT	DP INT ML C N<3..0>
DP_INT_MI	DP_85D	DISPLAYPORT	DP INT ML P<3..0>
DP_INT_MI	DP_85D	DISPLAYPORT	DP INT ML N<3..0>
DP_INT_AUXCH	DP_85D		DP INT AUXCH C P
DP_INT_AUXCH	DP_85D		DP INT AUXCH C N
DP_INT_AUXCH	DP_85D		DP INT AUXCH P
DP_INT_AUXCH	DP_85D		DP INT AUXCH N

Notes:
AUX and DDC was removed from DISPLAYPORT or TBTD_P_RX/TX because it's not high speed, and to save routing space.

Only used on dual-port hosts.

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TBT,DP,HDMI Constraints			
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Camera Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	31 32
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	31 32
S2_MEM_CKE	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	31 32
S2_MEM_CS	S2_MFM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	31 32
S2_MEM_CMD	S2_MFM_45S	S2_MEM_CTRL	MEM_CAM_ODT	32
S2_MEM_CMD	S2_MFM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	31 32
S2_MEM_CMD	S2_MFM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	31 32
S2_MEM_CMD	S2_MFM_45S	S2_MEM_CMD	MEM_CAM_WE_L	31 32
S2_MEM_CMD	S2_MFM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	31 32
S2_MEM_CMD	S2_MFM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	31 32
S2_MEM_CMD	S2_MFM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	31 32
S2_MEM_DQS0	S2_MFM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	31 32
S2_MEM_DQS0	S2_MFM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	31 32
S2_MEM_DQS1	S2_MFM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	31 32
S2_MEM_DQS1	S2_MFM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	31 32
S2_MEM_DATA_0	S2_MFM_45S	S2_MFM_DATA0	MEM_CAM_DM<0>	31 32
S2_MEM_DATA_1	S2_MFM_45S	S2_MFM_DATA1	MEM_CAM_DM<1>	31 32
S2_MEM_A	S2_MFM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	31 32
S2_MEM_DATA_0	S2_MFM_45S	S2_MFM_DATA0	MEM_CAM_DQ<7..0>	31 32
S2_MEM_DATA_1	S2_MFM_45S	S2_MFM_DATA1	MEM_CAM_DQ<15..8>	31 32
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	32 68
		S2_MFM_PWR	PP1V35_CAM	31 32
		S2_MFM_PWR	PP0V675_CAM_VREF	31 32
		S2_MFM_PWR	PP0V675_MEM_CAM_VREFCA	32
		S2_MFM_PWR	PP0V675_MEM_CAM_VREFDQ	32

MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MIPI_20THER	*	=4X_DIELECTRIC	?	MIPI_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?	MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_20THER	*	=7X_DIELECTRIC	?	MIPICLK_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?	S2_DATA2SELF	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_DQS20WNDATA	*	=2X_DIELECTRIC	?	S2_DQS20WNDATA	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?	S2_CMD2CMD	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?	S2_CMD2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?	S2_CTRL2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?	S2_20THERMEM	TOP,BOTTOM	=6X_DIELECTRIC	?
S2MEM_2PWR	*	=2X_DIELECTRIC	?	S2MEM_2PWR	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?	S2MEM_2GND	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?	S2MEM_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER	S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WNDATA
S2_MEM_DQS*	*	*	S2MEM_20THER	S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WNDATA
S2_MEM_CMD	*	*	S2MEM_20THER				
S2_MEM_CTRL	*	*	S2MEM_20THER				
S2_MEM_CLK	*	*	S2MEM_20THER				
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF				
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD				
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL				
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL				
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM				

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

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Camera Constraints			
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SMC SMBus & Charger Net Properties

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING
SMBUS_SMC_2	SMB_45S	SMB
SMBUS_SMC_2	SMB_45S	SMB
SMBUS_SMC_1	SMB_45S	SMB
SMBUS_SMC_1	SMB_45S	SMB
SMBUS_SMC_0	SMB_45S	SMB
SMBUS_SMC_0	SMB_45S	SMB
SMBUS_SMC_5	SMB_45S	SMB
SMBUS_SMC_5	SMB_45S	SMB
SMBUS_SMC_3	SMB_45S	SMB
SMBUS_SMC_3	SMB_45S	SMB

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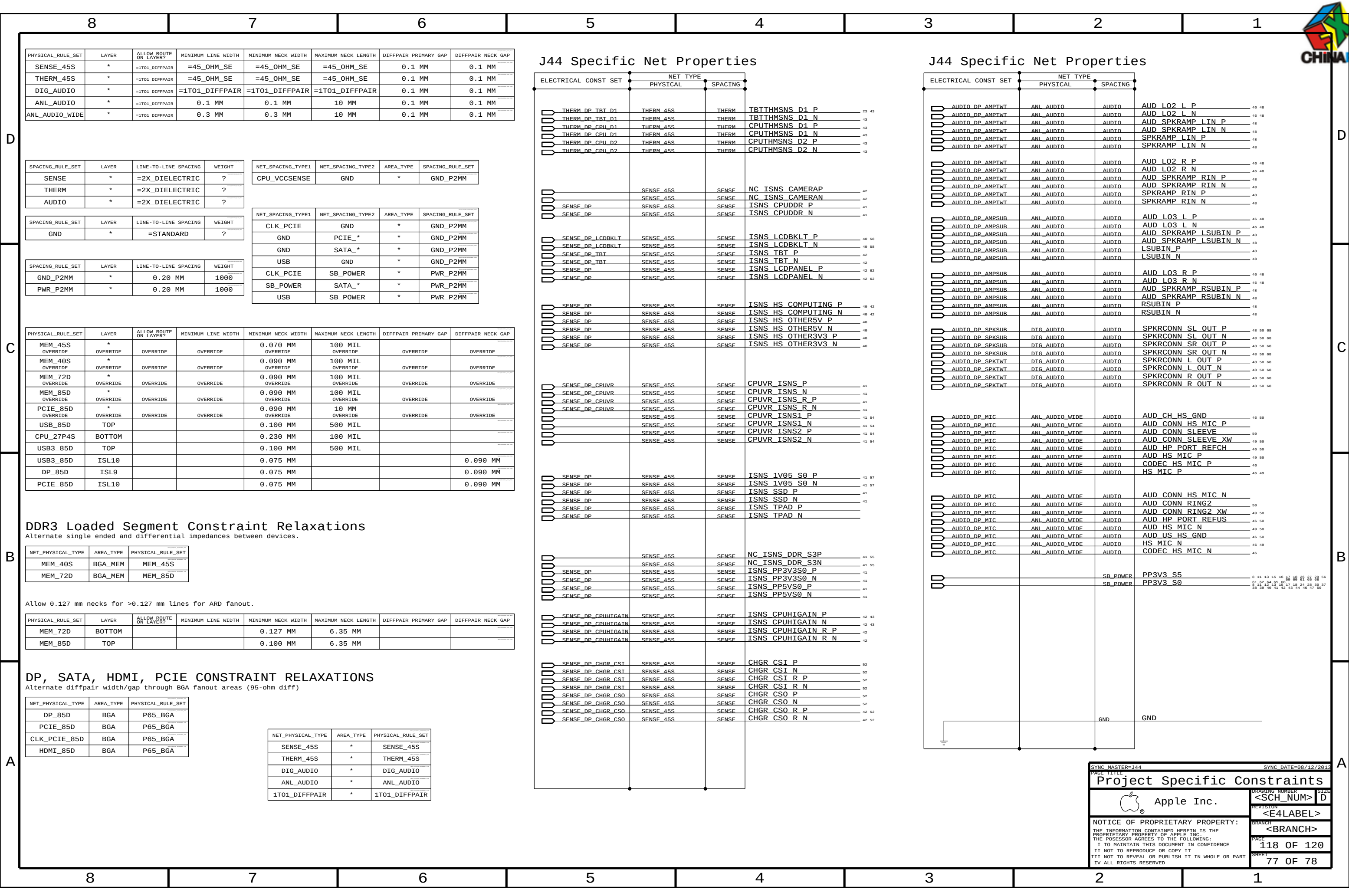
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